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(54) **SYSTEMS AND METHODS FOR SINGLE CELL PROCESSING**

(71) Applicant: **10X GENOMICS, INC.**, Pleasanton, CA (US)

(72) Inventors: **Michael Schnall-Levin**, San Francisco, CA (US); **Ali R. Alemozafar**, San Francisco, CA (US)

(73) Assignee: **10X GENOMICS, INC.**, Pleasanton, CA (US)

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Primary Examiner — Sahana S Kaup

(74) *Attorney, Agent, or Firm* — WILSON SONSINI
GOODRICH & ROSATI

(57) **ABSTRACT**

The present disclosure provides systems and methods for measuring one or more analytes at the single cell level. In some instances, sequencing of a composite barcode sequence may identify a type of analyte, identify a cell, and determine that the type of analyte originated from the cell. In some instances, a probability that a type of analyte originated from the cell may be determined. Multiple types of analytes may be identified to have originated from the cell, and/or the respective probabilities determined. An analyte may be a protein, such as a surface-bound protein or an internal protein. An analyte may be a metabolite or other small molecule. An analyte may be any constituent of a cell.

20 Claims, 11 Drawing Sheets

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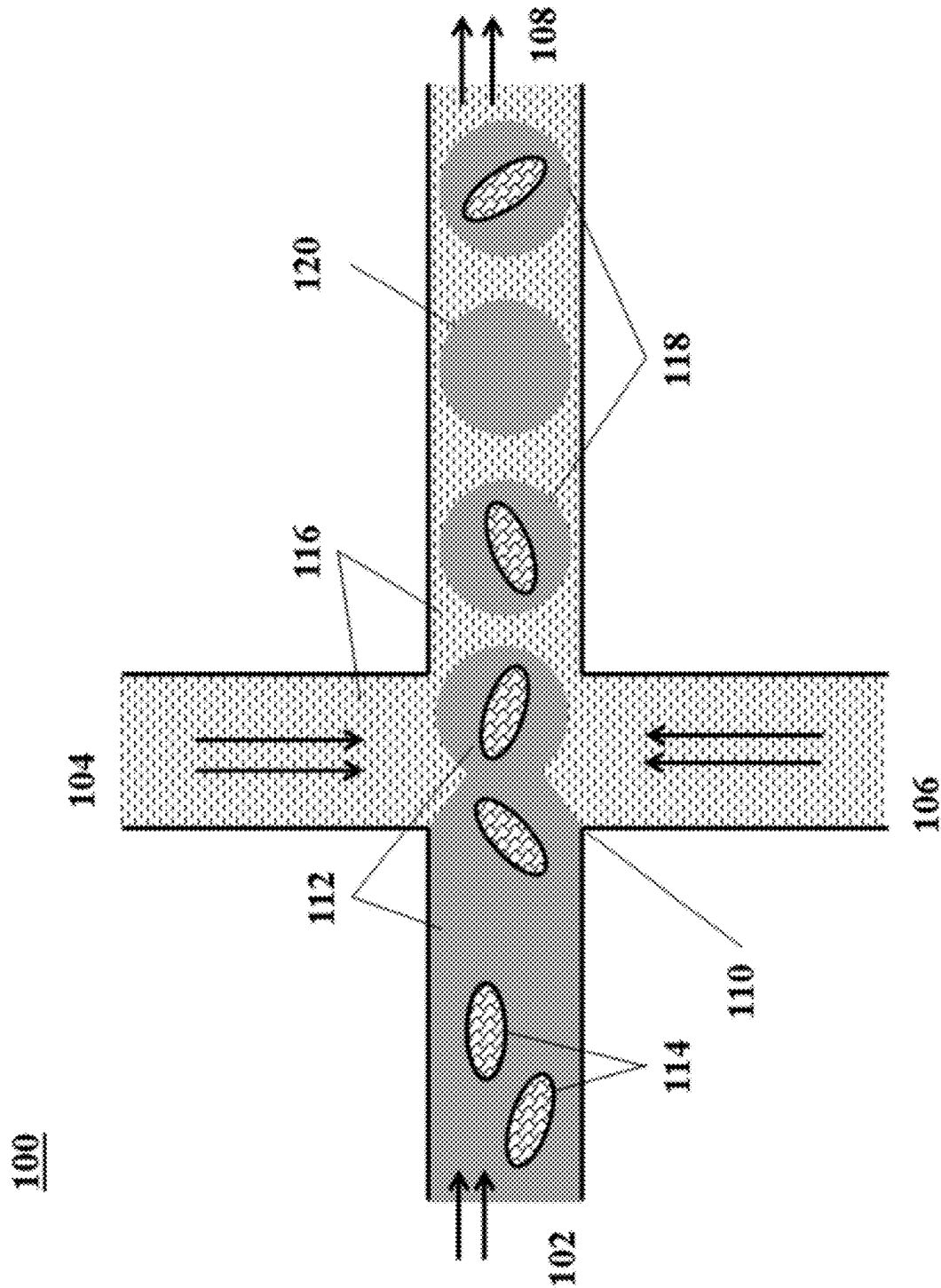


FIG. 1

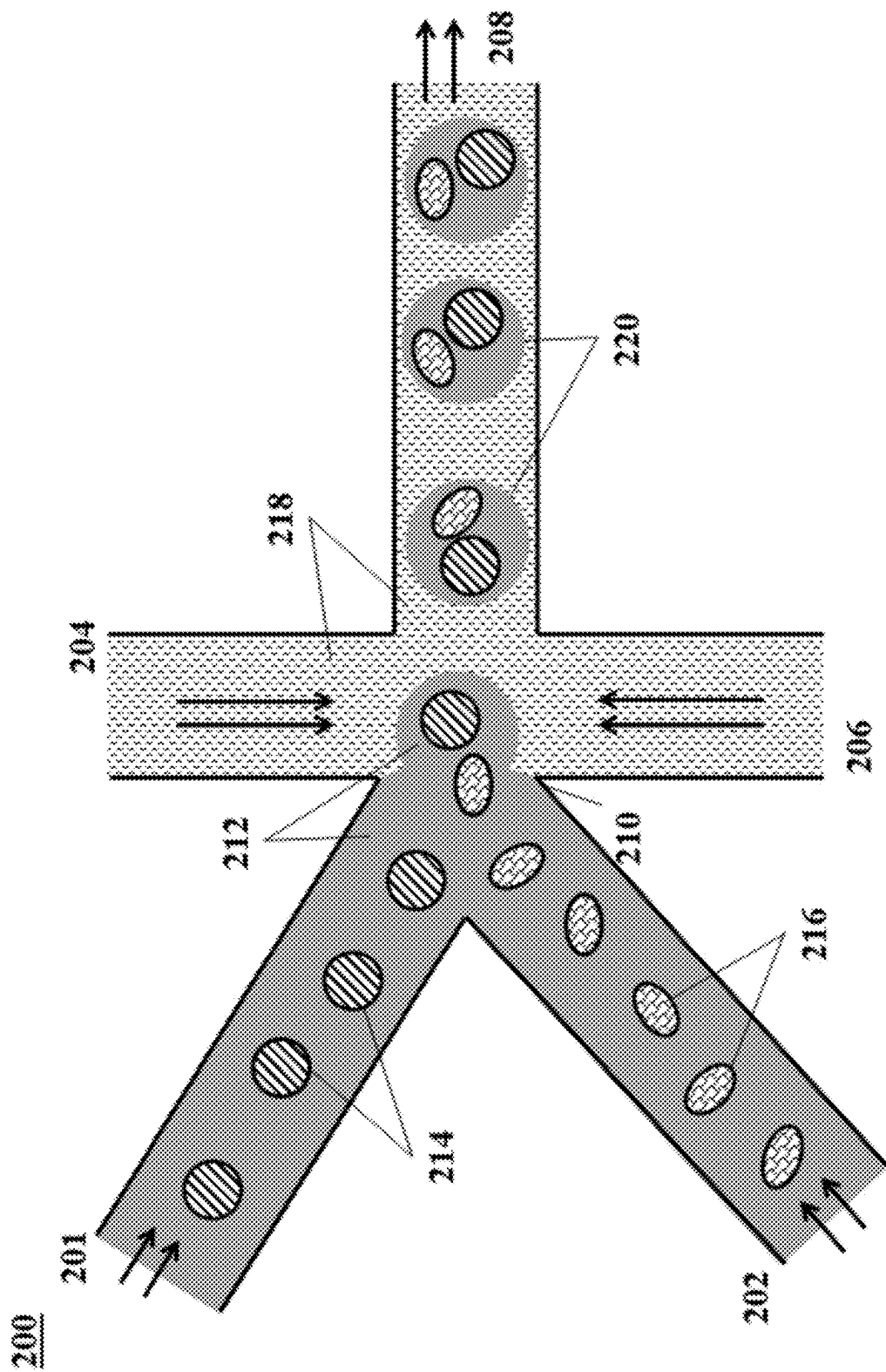


FIG. 2

300

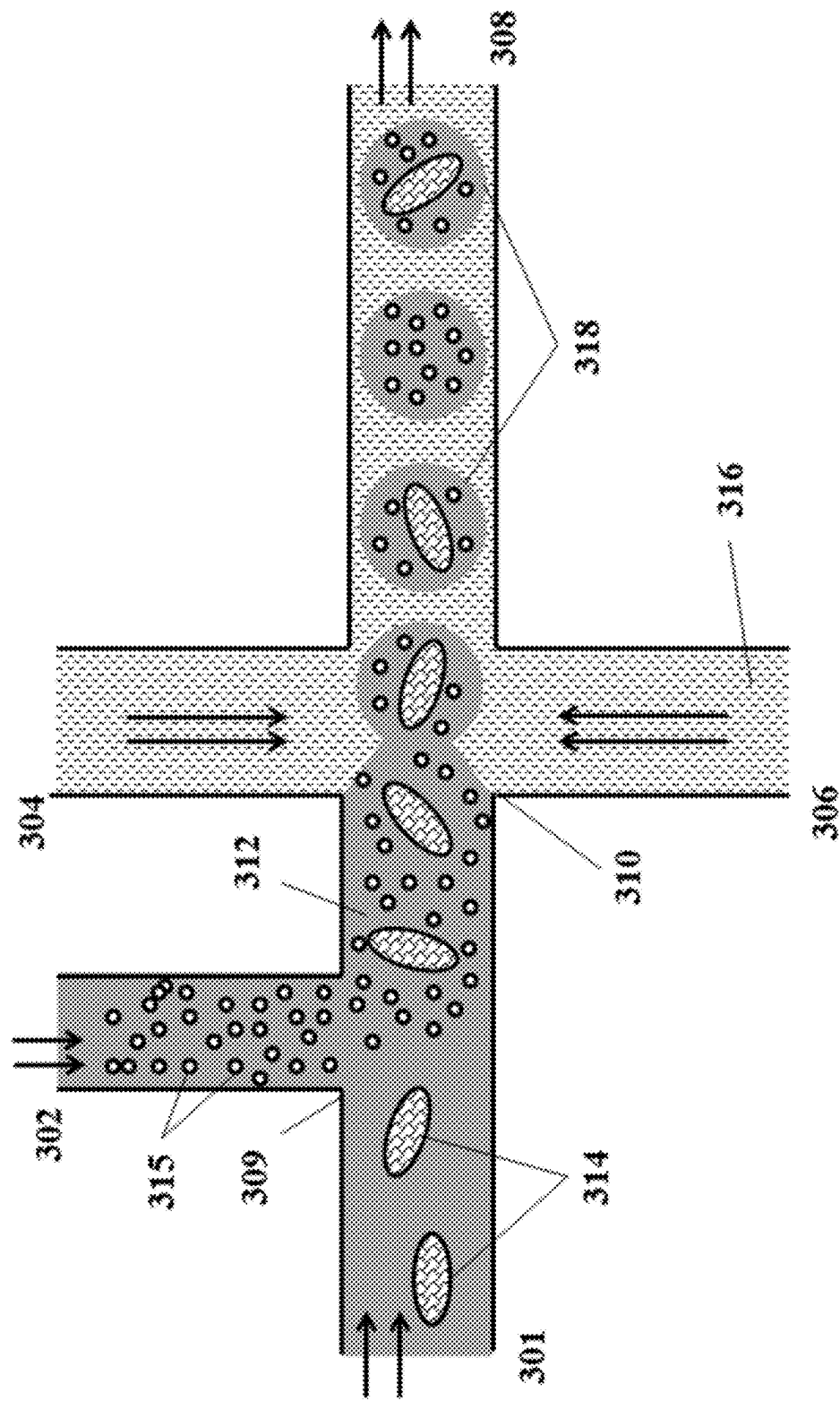


FIG. 3

400

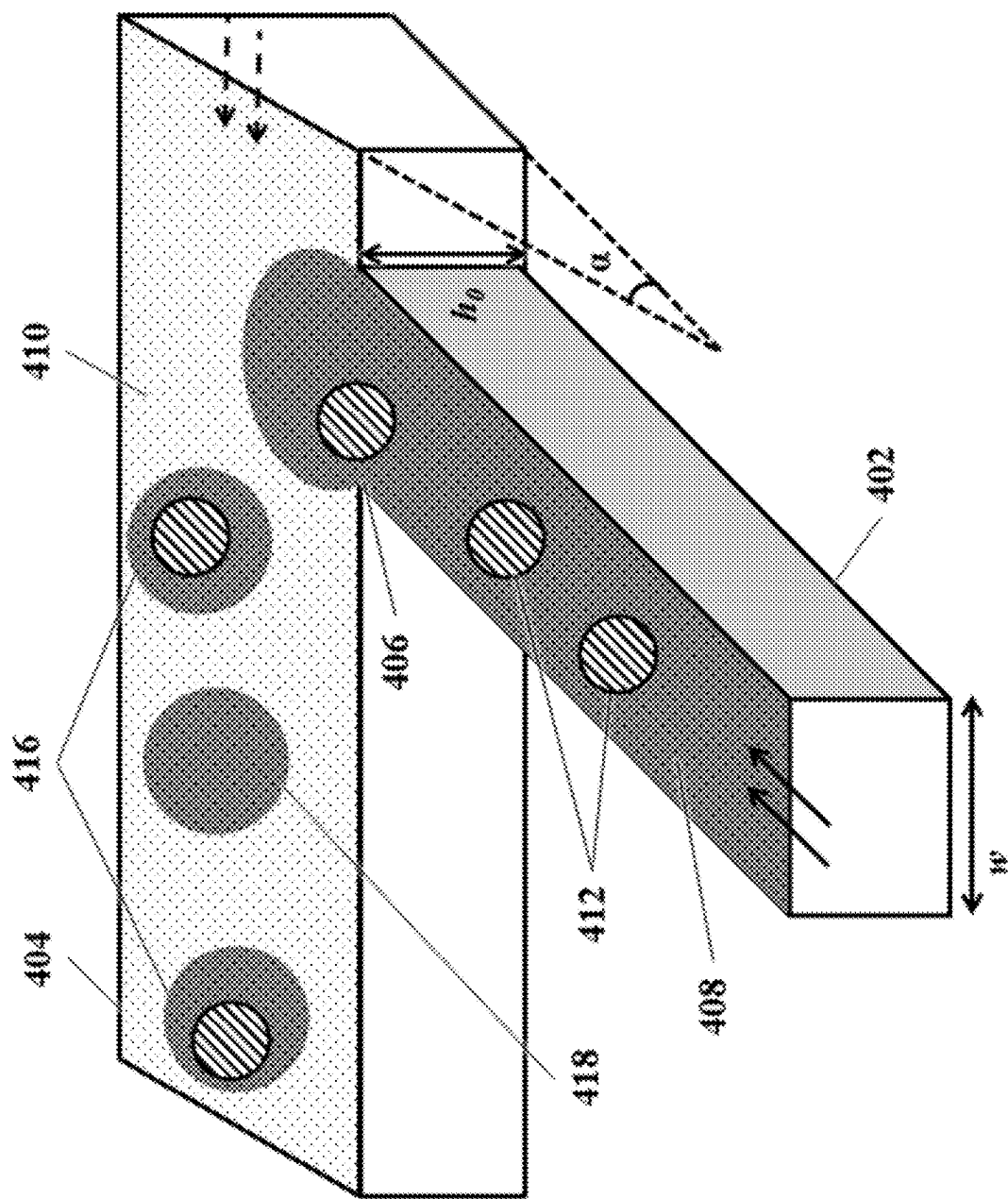


FIG. 4

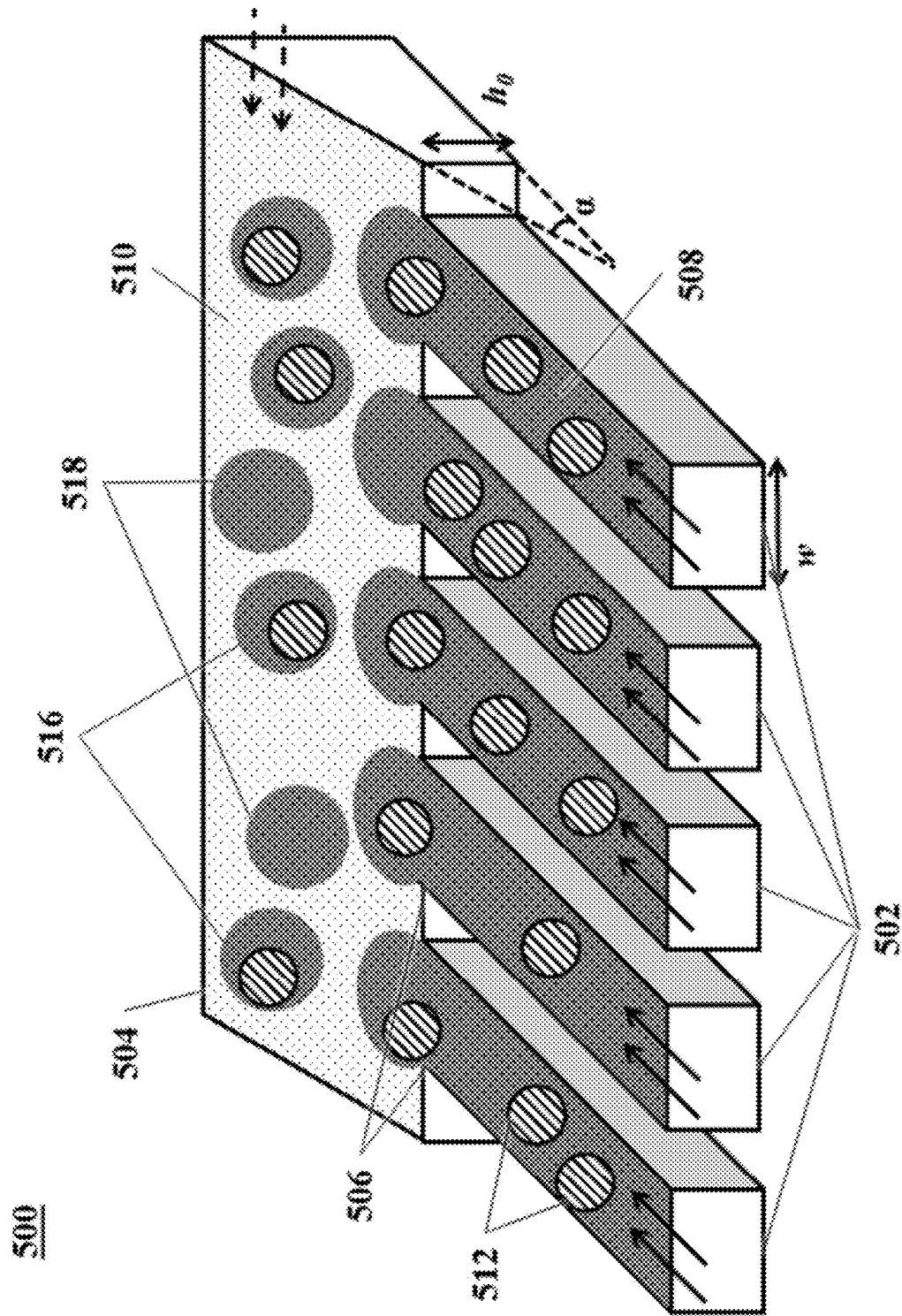


FIG. 5

600

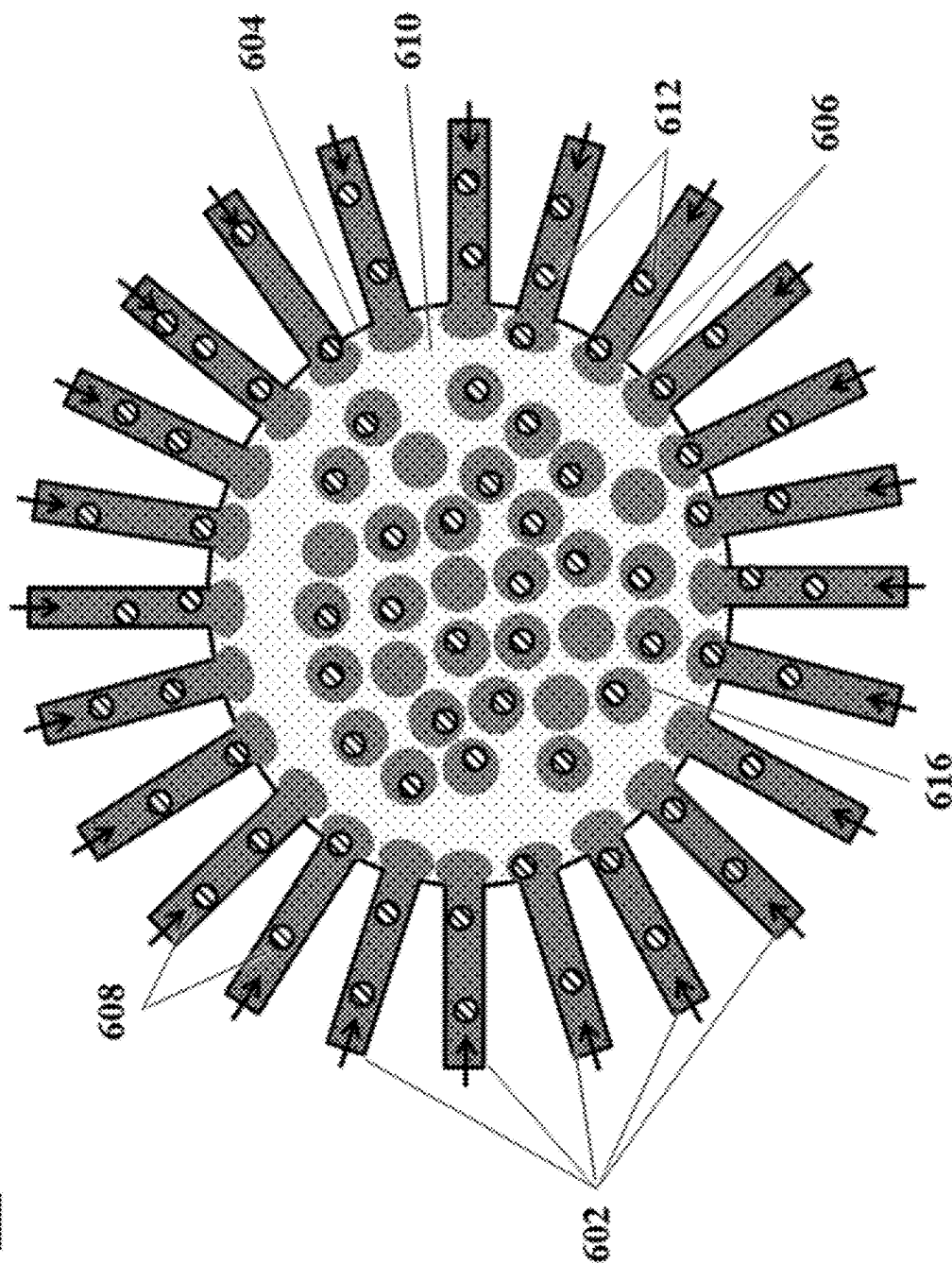


FIG. 6

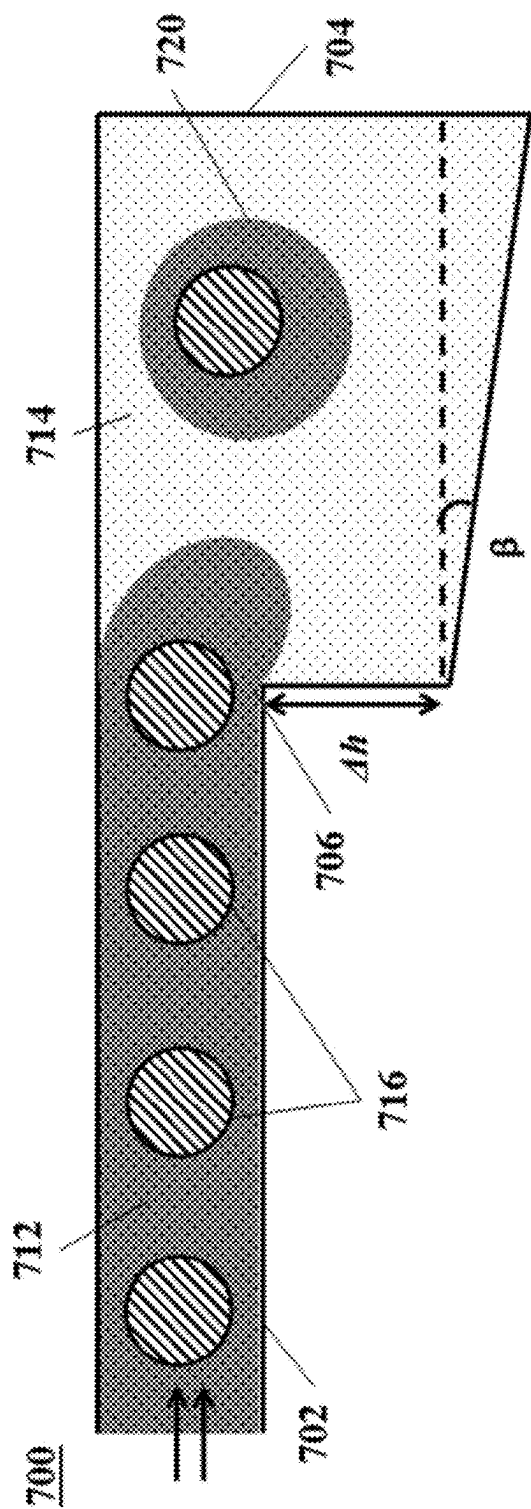


FIG. 7A

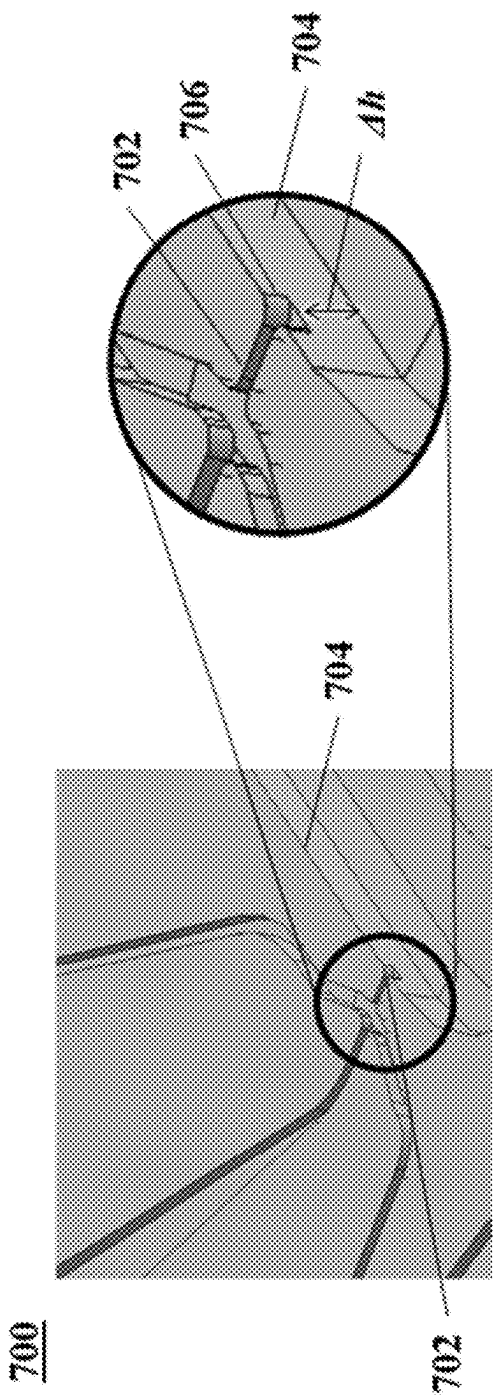
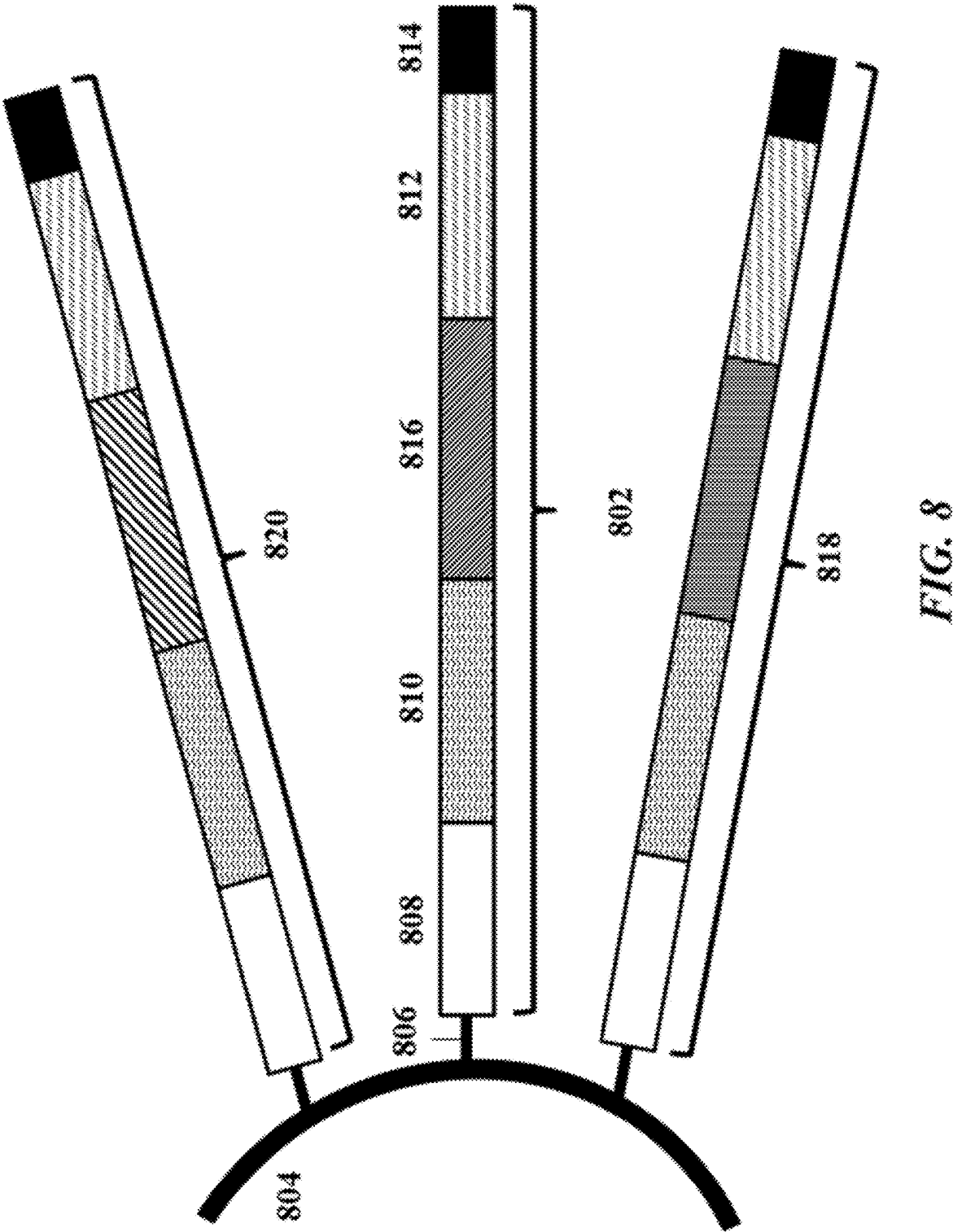


FIG. 7B



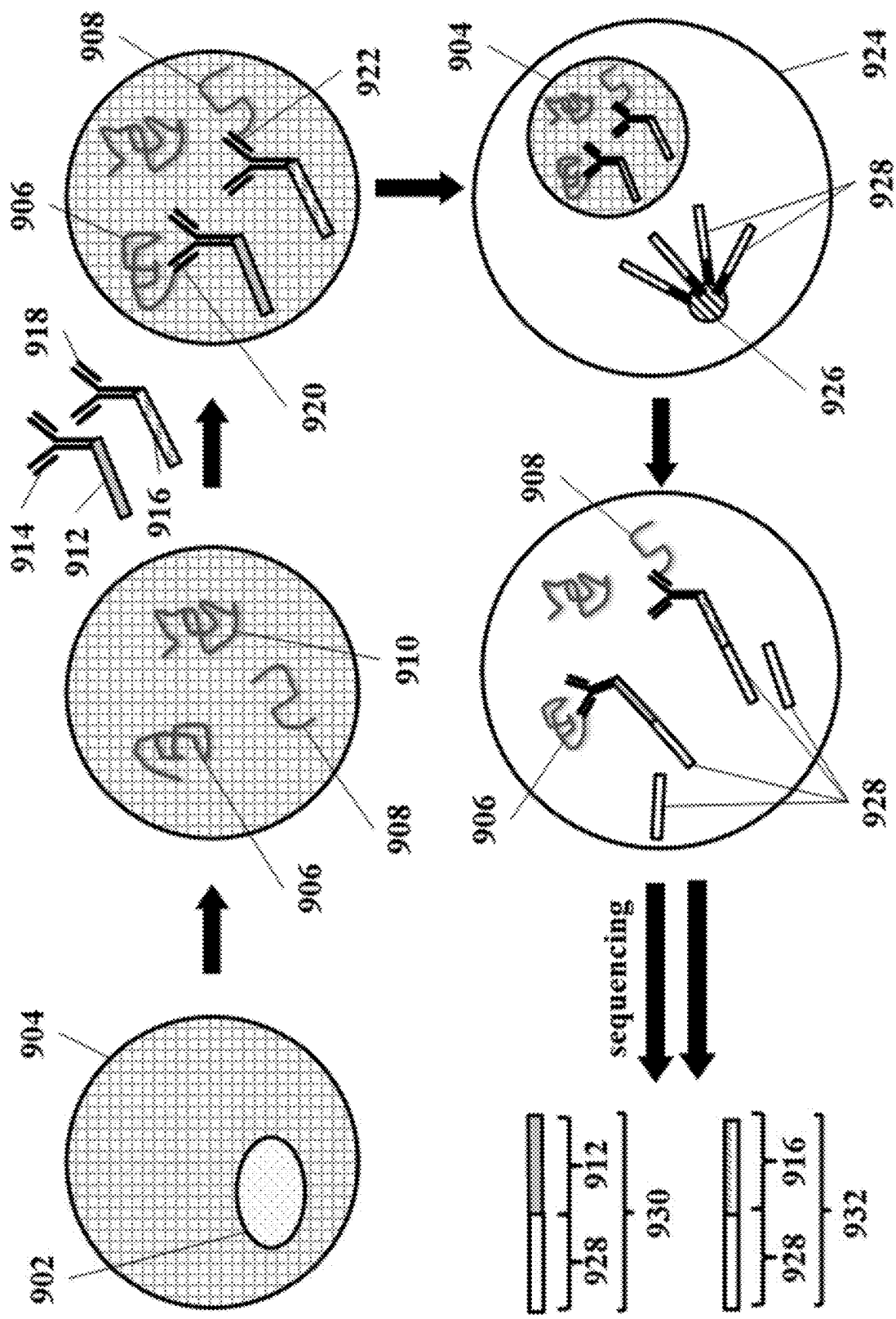


FIG. 9

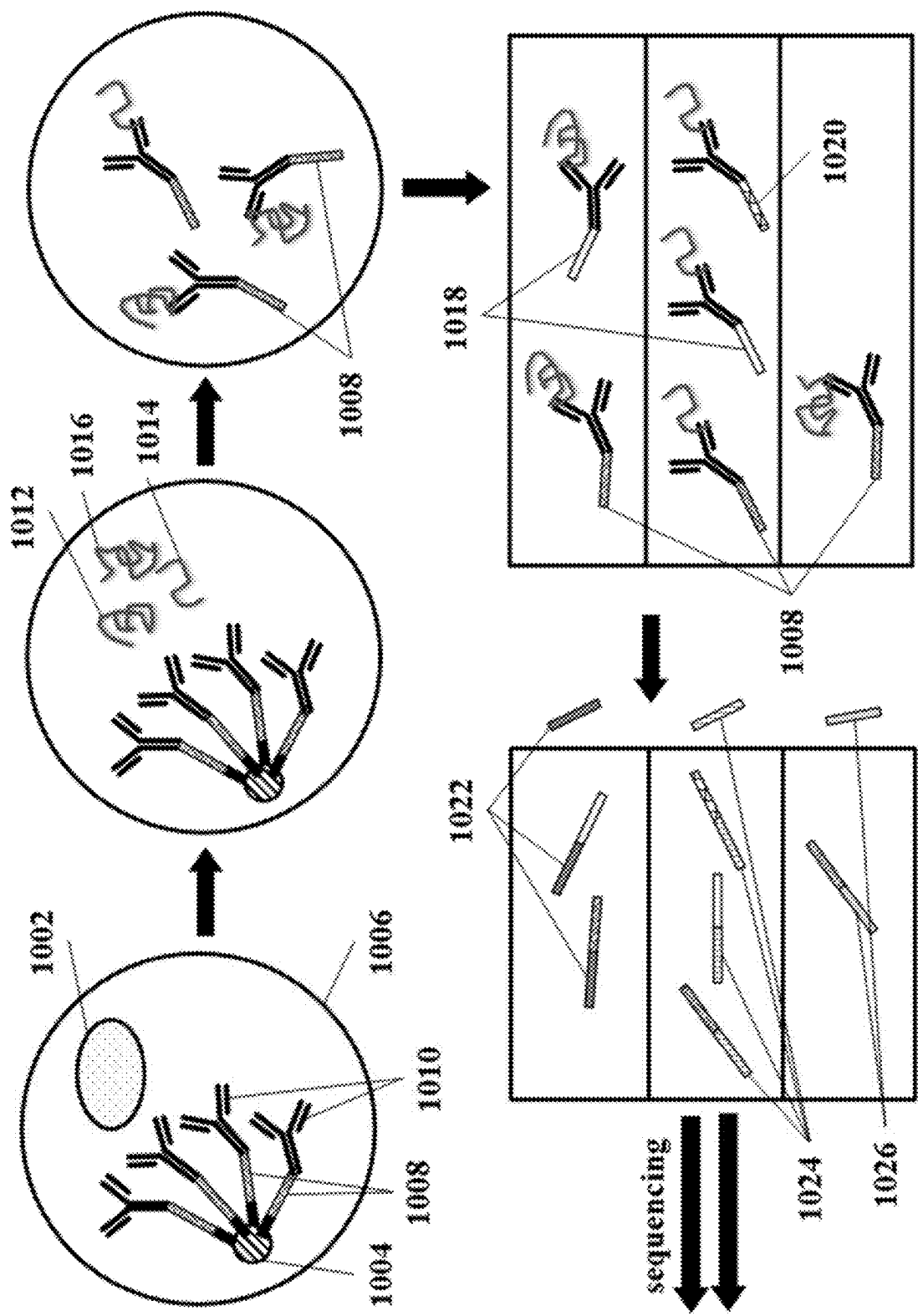


FIG. 10

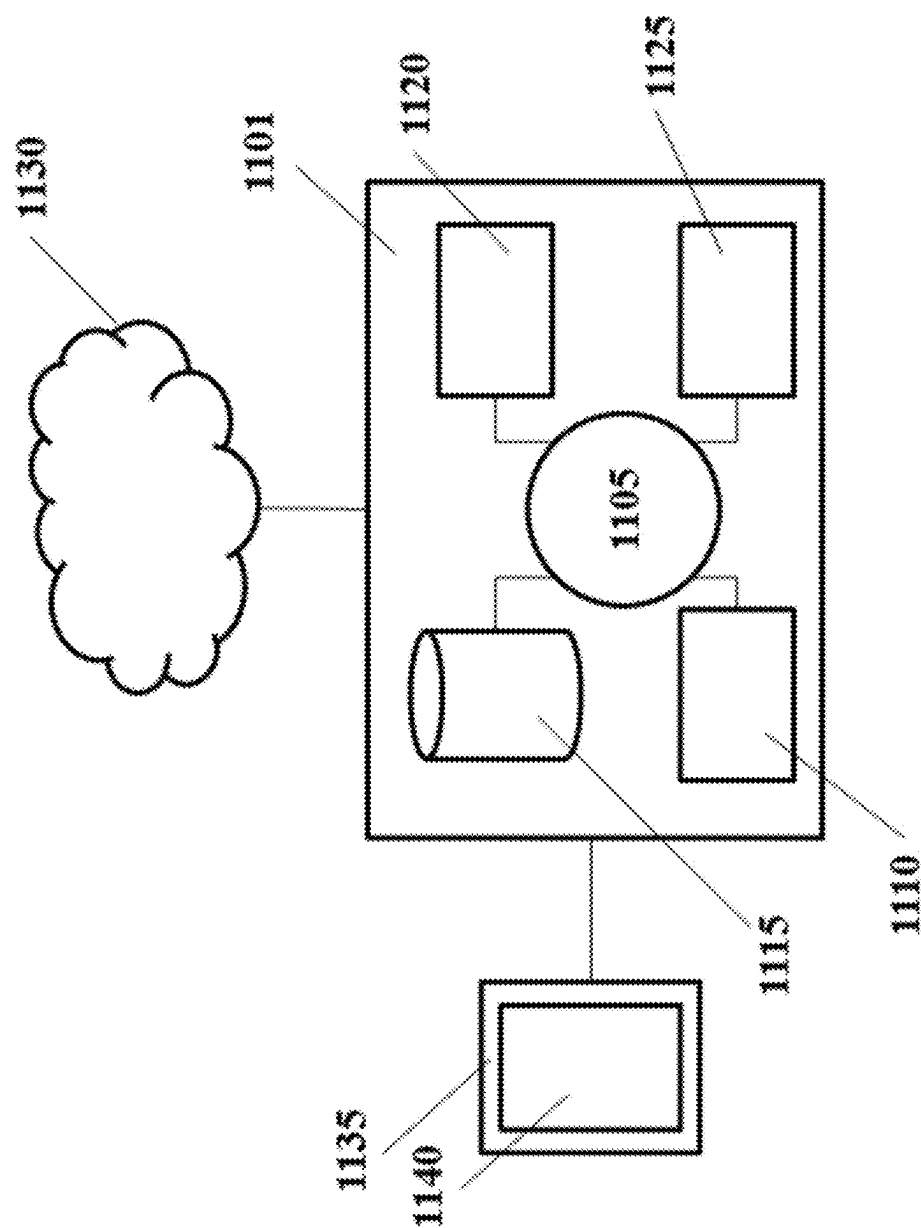


FIG. 11

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SYSTEMS AND METHODS FOR SINGLE CELL PROCESSING

CROSS-REFERENCE

This application is a continuation of International Application No. PCT/US2018/064763, filed Dec. 10, 2018, which claims the benefit of U.S. Provisional Patent Application No. 62/597,578, filed Dec. 12, 2017 which applications are entirely incorporated herein by reference.

BACKGROUND

Samples may be processed for various purposes, such as identification of a type of moiety or analyte within the sample. The sample may be a biological sample. The biological samples may be processed for various purposes, such as detection of a disease (e.g., cancer) or identification of a particular species. There are various approaches for processing samples, such as polymerase chain reaction (PCR) and sequencing.

Biological samples may be processed within various reaction environments, such as partitions. Partitions may be wells or droplets. Droplets or wells may be employed to process biological samples in a manner that enables the biological samples to be partitioned and processed separately. For example, such droplets may be fluidically isolated from other droplets, enabling accurate control of respective environments in the droplets.

Biological samples in partitions may be subjected to various processes, such as chemical processes or physical processes. Samples in partitions may be subjected to heating or cooling, or chemical reactions, such as to yield species that may be qualitatively or quantitatively processed.

SUMMARY

During processing, it may be difficult to identify the origin (e.g., cell) of an isolated analyte (e.g., protein, metabolite, nucleic acid molecule, etc.). Recognized herein is a need for systems and methods for processing that efficiently and accurately measure one or more analytes at the single cell level. The systems and methods may identify the origin(s) of one or more analytes. For example, the analyte can be a protein (e.g., surface-bound protein, internal protein, etc.). The analyte can be a metabolite (e.g., alcohol, amino acid, nucleotide, antioxidant, organic acid, polyol, vitamin, cofactor, etc.). The analyte can be any constituent of a cell. The analyte can be a targeted analyte, such as having binding specificity to an antibody that is specifically captured and analyzed. The analyte can be a non-targeted analyte that is non-specifically captured and analyzed.

In an aspect, provided is a method for identifying a protein within a cell containing or suspected of containing the protein, comprising: (a) bringing a cell bead generated from the cell in contact with an antibody having binding specificity with the protein, under conditions sufficient to permit the antibody to flow into the cell bead and couple to the protein, wherein the antibody is coupled to a first nucleic acid barcode molecule comprising a first nucleic acid sequence; (b) co-partitioning the cell bead and a bead in a partition, wherein the bead comprises a second nucleic acid barcode molecule comprising a second nucleic acid sequence; (c) using the first nucleic acid barcode molecule and the second nucleic acid barcode molecule to generate a nucleic acid molecule comprising the first nucleic acid sequence and the second nucleic acid sequence; (d) sequenc-

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ing the nucleic acid molecule or derivative thereof to identify the first nucleic acid sequence and the second nucleic acid sequence, wherein the first nucleic acid sequence identifies the protein and the second nucleic acid sequence identifies the cell bead; and (e) determining that the protein originated from within the cell.

In some embodiments, the method further comprises, subsequent to (a), washing out unbound antibodies from the cell bead.

In some embodiments, the cell is lysed in the cell bead.

In some embodiments, the bead is a gel bead.

In some embodiments, the partition is a droplet in an emulsion.

In some embodiments, the second nucleic acid molecule is releasably coupled to the bead. In some embodiments, the method further comprises releasing the second nucleic acid barcode molecule from the bead.

In some embodiments, the bead comprises a plurality of nucleic acid barcode molecules, each of the plurality of nucleic acid barcode molecules comprising the second nucleic acid sequence. In some embodiments, each of the plurality of nucleic acid barcode molecules comprises a unique nucleic acid sequence that is unique to each of the plurality of nucleic acid barcode molecules.

In some embodiments, (a) comprises bringing the cell bead in contact with a plurality of different antibodies each having binding specificity with a different protein, wherein a given antibody of the plurality of different antibodies is coupled to a nucleic acid barcode molecule comprising a nucleic acid sequence that identifies the given antibody. In some embodiments, the cell bead comprises a plurality of different proteins from the cell, wherein (d) comprises identifying the different proteins, and wherein (e) comprises determining that the different proteins originated from within the cell.

In some embodiments, the determining comprises determining a probability that the protein originated from within the cell.

In another aspect, provided is a method for identifying a protein from a cell containing or suspected of containing the protein, comprising: (a) co-partitioning the cell and a bead in a first partition, wherein the bead comprises a plurality of antibodies configured to couple to proteins, wherein each antibody of the plurality of antibodies is coupled to a first nucleic acid barcode molecule comprising a first nucleic acid sequence; (b) subjecting the cell to conditions sufficient to permit an antibody of the plurality of antibodies to couple to the protein to generate a protein-antibody complex comprising the first nucleic acid barcode molecule; (c) co-partitioning the protein-antibody complex in a second partition with a second nucleic acid barcode molecule comprising a second nucleic acid sequence under conditions sufficient to permit the first nucleic acid barcode molecule of the protein-antibody complex and the second nucleic acid barcode molecule to generate a nucleic acid molecule comprising the first nucleic acid sequence and the second nucleic acid sequence; (d) sequencing the nucleic acid molecule or derivative thereof to identify the first nucleic acid sequence and the second nucleic acid sequence, wherein the first nucleic acid sequence identifies the cell and the second nucleic acid sequence identifies the protein; and (e) determining that the protein originated from the cell.

In some embodiments, the plurality of antibodies is configured to couple non-specifically to proteins.

In some embodiments, the first partition is a droplet.

In some embodiments, the second partition is a well.

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In some embodiments, (b) comprises lysing the cell in the first partition.

In some embodiments, a given antibody of the plurality of antibodies is coupled to a nucleic acid barcode molecule comprising a nucleic acid sequence that is unique to the given antibody.

In some embodiments, the bead is a gel bead.

In some embodiments, the plurality of antibodies is releasably coupled to the bead. In some embodiments, the method further comprises releasing the plurality of antibodies from the bead.

In some embodiments, the method further comprises, subsequent to (b), separating a first set of protein-antibody complexes comprising a type of the protein from other protein-antibody complexes comprising other types of proteins, wherein the first set of protein-antibody complexes comprises the protein-antibody complex. In some embodiments, the first set of protein-antibody complexes comprises proteins from a plurality of cells, wherein the plurality of cells comprises the cell.

In some embodiments, the protein is from within the cell.

In some embodiments, the protein is a surface bound protein.

In some embodiments, the determining comprises determining a probability that the protein originated from within the cell.

In another aspect, provided is a method for identifying a metabolite within a cell containing or suspected of containing the metabolite, comprising: (a) bringing a cell bead generated from the cell in contact with an antibody having binding specificity with the metabolite, under conditions sufficient to permit the antibody to flow into the cell bead and couple to the metabolite, wherein the antibody is coupled to a first nucleic acid barcode molecule comprising a first nucleic acid sequence; (b) co-partitioning the cell bead and a bead in a partition, wherein the bead comprises a second nucleic acid barcode molecule comprising a second nucleic acid; (c) using the first nucleic acid barcode molecule and the second nucleic acid barcode molecule to generate a nucleic acid molecule comprising the first nucleic acid sequence and the second nucleic acid sequence; (d) sequencing the nucleic acid molecule to identify the first nucleic acid sequence and the second nucleic acid sequence, wherein the first nucleic acid sequence identifies the metabolite and the second nucleic acid sequence identifies the cell bead; and (e) determining that the metabolite originated from within the cell.

In some embodiments, the metabolite comprises a structural unit having binding specificity to the antibody.

In some embodiments, the metabolite is selected from the group consisting of an alcohol, amino acid, nucleotide, antioxidant, organic acid, polyol, and vitamin.

In some embodiments, the metabolite is from a class of metabolites.

In some embodiments, the metabolite is a cofactor.

In some embodiments, the method further comprises, subsequent to (a), washing out unbound antibodies from the cell bead.

In some embodiments, the cell is lysed in the cell bead.

In some embodiments, the bead is a gel bead.

In some embodiments, the partition is a droplet in an emulsion.

In some embodiments, the second nucleic acid molecule is releasably coupled to the bead. In some embodiments, the method further comprises releasing the second nucleic acid barcode molecule from the bead.

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In some embodiments, the bead comprises a plurality of nucleic acid barcode molecules, each of the plurality of nucleic acid barcode molecules comprising the second nucleic acid sequence. In some embodiments, each of the plurality of nucleic acid barcode molecules comprises a unique nucleic acid sequence that is unique to each of the plurality of nucleic acid barcode molecules.

In some embodiments, the determining comprises determining a probability that the metabolite originated from within the cell.

In another aspect, provided is a method for determining that a first analyte and a second analyte originated from a cell, wherein the first analyte is from a surface of the cell and the second analyte is from within the cell, comprising: (a) processing the cell or cell derivative with a first antibody under conditions sufficient to permit the first antibody to couple to the first analyte, wherein the first antibody comprises a first nucleic acid barcode molecule comprising a first nucleic acid sequence; (b) processing the cell or cell derivative with a second antibody under conditions sufficient to permit the second antibody to couple to the second analyte, wherein the second antibody comprises a second nucleic acid barcode molecule comprising a second nucleic acid sequence; (c) using the first nucleic acid barcode molecule and the second nucleic acid barcode molecule to generate a first nucleic acid molecule and a second nucleic acid molecule, respectively, wherein the first nucleic acid molecule comprises the first nucleic acid sequence and a common barcode sequence and the second nucleic acid molecule comprises the second nucleic acid sequence and the common barcode sequence, wherein the common barcode sequence identifies the cell or cell derivative; (d) sequencing the first nucleic acid molecule and the second nucleic acid molecule, or derivatives thereof, to identify the first nucleic acid sequence, the second nucleic acid sequence, and the common barcode sequence, wherein the first nucleic acid sequence identifies the first analyte, the second nucleic acid sequence identifies the second analyte, and the common barcode sequence identifies the cell or cell derivative; and (e) determining that the first analyte and the second analyte originated from the cell.

In some embodiments, the first analyte and the second analyte are different types of analytes.

In some embodiments, the first analyte and the second analyte are the same type of analyte.

In some embodiments, the first analyte and/or the second analyte is a protein.

In some embodiments, the second analyte is a metabolite.

In some embodiments, (a) comprises processing the cell and (b) comprise processing the cell derivative. In some embodiments, the cell derivative is a cell bead.

In some embodiments, (c) is performed by co-partitioning the cell or cell derivative with a plurality of barcode molecules each comprising the common barcode sequence. In some embodiments, the plurality of barcode molecules is coupled to a bead.

In some embodiments, the determining comprises determining a probability that the first analyte or the second analyte originated from the cell.

In some embodiments, the determining comprises determining a first probability that the first analyte originated from the cell and determining a second probability that the second analyte originated from within the cell.

Additional aspects and advantages of the present disclosure will become readily apparent to those skilled in this art from the following detailed description, wherein only illustrative embodiments of the present disclosure are shown and

described. As will be realized, the present disclosure is capable of other and different embodiments, and its several details are capable of modifications in various obvious respects, all without departing from the disclosure. Accordingly, the drawings and description are to be regarded as illustrative in nature, and not as restrictive.

INCORPORATION BY REFERENCE

All publications, patents, and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated to be incorporated by reference. To the extent publications and patents or patent applications incorporated by reference contradict the disclosure contained in the specification, the specification is intended to supersede and/or take precedence over any such contradictory material.

BRIEF DESCRIPTION OF THE DRAWINGS

The novel features of the invention are set forth with particularity in the appended claims. A better understanding of the features and advantages of the present invention will be obtained by reference to the following detailed description that sets forth illustrative embodiments, in which the principles of the invention are utilized, and the accompanying drawings (also "Figure" and "FIG." herein), of which:

FIG. 1 shows an example of a microfluidic channel structure for partitioning individual biological particles.

FIG. 2 shows an example of a microfluidic channel structure for delivering barcode carrying beads to droplets.

FIG. 3 shows an example of a microfluidic channel structure for co-partitioning biological particles and reagents.

FIG. 4 shows an example of a microfluidic channel structure for the controlled partitioning of beads into discrete droplets.

FIG. 5 shows an example of a microfluidic channel structure for increased droplet generation throughput.

FIG. 6 shows another example of a microfluidic channel structure for increased droplet generation throughput.

FIG. 7A shows a cross-section view of another example of a microfluidic channel structure with a geometric feature for controlled partitioning. FIG. 7B shows a perspective view of the channel structure of FIG. 7A.

FIG. 8 illustrates an example of a barcode carrying bead.

FIG. 9 illustrates a method for identifying and measuring an analyte in a cell.

FIG. 10 illustrates another method for identifying and measuring an analyte in a cell.

FIG. 11 shows a computer system that is programmed or otherwise configured to implement methods provided herein.

DETAILED DESCRIPTION

While various embodiments of the invention have been shown and described herein, it will be obvious to those skilled in the art that such embodiments are provided by way of example only. Numerous variations, changes, and substitutions may occur to those skilled in the art without departing from the invention. It should be understood that various alternatives to the embodiments of the invention described herein may be employed.

Where values are described as ranges, it will be understood that such disclosure includes the disclosure of all possible sub-ranges within such ranges, as well as specific numerical values that fall within such ranges irrespective of whether a specific numerical value or specific sub-range is expressly stated.

The term "barcode," as used herein, generally refers to a label, or identifier, that conveys or is capable of conveying information about an analyte. A barcode can be part of an analyte. A barcode can be independent of an analyte. A barcode can be a tag attached to an analyte (e.g., nucleic acid molecule) or a combination of the tag in addition to an endogenous characteristic of the analyte (e.g., size of the analyte or end sequence(s)). A barcode may comprise a plurality of barcode units or elements. A barcode may be unique. Barcodes can have a variety of different formats. For example, barcodes can include: polynucleotide barcodes; random nucleic acid and/or amino acid sequences; and synthetic nucleic acid and/or amino acid sequences. A barcode can be attached to an analyte in a reversible or irreversible manner. A barcode can be added to, for example, a fragment of a deoxyribonucleic acid (DNA) or ribonucleic acid (RNA) sample before, during, and/or after sequencing of the sample. Barcodes can allow for identification and/or quantification of individual sequencing-reads.

The term "real time," as used herein, can refer to a response time of less than about 1 second, a tenth of a second, a hundredth of a second, a millisecond, or less. The response time may be greater than 1 second. In some instances, real time can refer to simultaneous or substantially simultaneous processing, detection or identification.

The term "subject," as used herein, generally refers to an animal, such as a mammal (e.g., human) or avian (e.g., bird), or other organism, such as a plant. For example, the subject can be a vertebrate, a mammal, a rodent (e.g., a mouse), a primate, a simian or a human. Animals may include, but are not limited to, farm animals, sport animals, and pets. A subject can be a healthy or asymptomatic individual, an individual that has or is suspected of having a disease (e.g., cancer) or a pre-disposition to the disease, and/or an individual that is in need of therapy or suspected of needing therapy. A subject can be a patient. A subject can be a microorganism or microbe (e.g., bacteria, fungi, archaea, viruses).

The term "genome," as used herein, generally refers to genomic information from a subject, which may be, for example, at least a portion or an entirety of a subject's hereditary information. A genome can be encoded either in DNA or in RNA. A genome can comprise coding regions (e.g., that code for proteins) as well as non-coding regions. A genome can include the sequence of all chromosomes together in an organism. For example, the human genome ordinarily has a total of 46 chromosomes. The sequence of all of these together may constitute a human genome.

The terms "adaptor(s)", "adapter(s)" and "tag(s)" may be used synonymously. An adaptor or tag can be coupled to a polynucleotide sequence to be "tagged" by any approach, including ligation, hybridization, or other approaches.

The term "sequencing," as used herein, generally refers to methods and technologies for determining the sequence of nucleotide bases in one or more polynucleotides. The polynucleotides can be, for example, nucleic acid molecules such as deoxyribonucleic acid (DNA) or ribonucleic acid (RNA), including variants or derivatives thereof (e.g., single stranded DNA). Sequencing can be performed by various systems currently available, such as, without limitation, a sequencing system by Illumina®, Pacific Biosciences

(PacBio®), Oxford Nanopore®, or Life Technologies (Ion Torrent®). Alternatively or in addition, sequencing may be performed using nucleic acid amplification, polymerase chain reaction (PCR) (e.g., digital PCR, quantitative PCR, or real time PCR), or isothermal amplification. Such systems may provide a plurality of raw genetic data corresponding to the genetic information of a subject (e.g., human), as generated by the systems from a sample provided by the subject. In some examples, such systems provide sequencing reads (also “reads” herein). A read may include a string of nucleic acid bases corresponding to a sequence of a nucleic acid molecule that has been sequenced. In some situations, systems and methods provided herein may be used with proteomic information.

The term “bead,” as used herein, generally refers to a particle. The bead may be a solid or semi-solid particle. The bead may be a gel bead. The gel bead may include a polymer matrix (e.g., matrix formed by polymerization or cross-linking). The polymer matrix may include one or more polymers (e.g., polymers having different functional groups or repeat units). Polymers in the polymer matrix may be randomly arranged, such as in random copolymers, and/or have ordered structures, such as in block copolymers. Cross-linking can be via covalent, ionic, or inductive, interactions, or physical entanglement. The bead may be a macromolecule. The bead may be formed of nucleic acid molecules bound together. The bead may be formed via covalent or non-covalent assembly of molecules (e.g., macromolecules), such as monomers or polymers. Such polymers or monomers may be natural or synthetic. Such polymers or monomers may be or include, for example, nucleic acid molecules (e.g., DNA or RNA). The bead may be formed of a polymeric material. The bead may be magnetic or non-magnetic. The bead may be rigid. The bead may be flexible and/or compressible. The bead may be disruptable or dissolvable. The bead may be a solid particle (e.g., a metal-based particle including but not limited to iron oxide, gold or silver) covered with a coating comprising one or more polymers. Such coating may be disruptable or dissolvable.

The term “sample,” as used herein, generally refers to a biological sample of a subject. The biological sample may comprise any number of macromolecules, for example, cellular macromolecules. The sample may be a cell sample. The sample may be a cell line or cell culture sample. The sample can include one or more microbes. The biological sample may be a nucleic acid sample or protein sample. The biological sample may also be a carbohydrate sample or a lipid sample. The biological sample may be derived from another sample. The sample may be a tissue sample, such as a biopsy, core biopsy, needle aspirate, or fine needle aspirate. The sample may be a fluid sample, such as a blood sample, urine sample, or saliva sample. The sample may be a skin sample. The sample may be a cheek swab. The sample may be a plasma or serum sample. The sample may be a cell-free or cell free sample. A cell-free sample may include extracellular polynucleotides. Extracellular polynucleotides may be isolated from a bodily sample that may be selected from the group consisting of blood, plasma, serum, urine, saliva, mucosal excretions, sputum, stool and tears.

The term “biological particle,” as used herein, generally refers to a discrete biological system derived from a biological sample. The biological particle may be a macromolecule. The biological particle may be a small molecule. The biological particle may be a virus. The biological particle may be a cell or derivative of a cell. The biological particle may be an organelle. The biological particle may be a rare

cell from a population of cells. The biological particle may be any type of cell, including without limitation prokaryotic cells, eukaryotic cells, bacterial, fungal, plant, mammalian, or other animal cell type, mycoplasmas, normal tissue cells, tumor cells, or any other cell type, whether derived from single cell or multicellular organisms. The biological particle may be a constituent of a cell. The biological particle may be or may include DNA, RNA, organelles, proteins, metabolites, or any combination thereof. The biological particle may be or may include a matrix (e.g., a gel or polymer matrix) comprising a cell or one or more constituents from a cell (e.g., cell bead), such as DNA, RNA, organelles, proteins, metabolites, or any combination thereof, from the cell. The biological particle may be obtained from a tissue of a subject. The biological particle may be a hardened cell. Such hardened cell may or may not include a cell wall or cell membrane. The biological particle may include one or more constituents of a cell, but may not include other constituents of the cell. An example of such constituents is a nucleus or an organelle. A cell may be a live cell. The live cell may be capable of being cultured, for example, being cultured when enclosed in a gel or polymer matrix, or cultured when comprising a gel or polymer matrix.

The term “macromolecular constituent,” as used herein, generally refers to a macromolecule contained within or from a biological particle. The macromolecular constituent may comprise a nucleic acid. In some cases, the biological particle may be a macromolecule. The macromolecular constituent may comprise DNA. The macromolecular constituent may comprise RNA. The RNA may be coding or non-coding. The RNA may be messenger RNA (mRNA), ribosomal RNA (rRNA) or transfer RNA (tRNA), for example. The RNA may be a transcript. The RNA may be small RNA that are less than 200 nucleic acid bases in length, or large RNA that are greater than 200 nucleic acid bases in length. Small RNAs may include 5.8S ribosomal RNA (rRNA), 5S rRNA, transfer RNA (tRNA), microRNA (miRNA), small interfering RNA (siRNA), small nucleolar RNA (snoRNAs), Piwi-interacting RNA (piRNA), tRNA-derived small RNA (tsRNA) and small rDNA-derived RNA (srRNA). The RNA may be double-stranded RNA or single-stranded RNA. The RNA may be circular RNA. The macromolecular constituent may comprise a protein. The macromolecular constituent may comprise a peptide. The macromolecular constituent may comprise a polypeptide.

The term “molecular tag,” as used herein, generally refers to a molecule capable of binding to a macromolecular constituent. The molecular tag may bind to the macromolecular constituent with high affinity. The molecular tag may bind to the macromolecular constituent with high specificity. The molecular tag may comprise a nucleotide sequence. The molecular tag may comprise a nucleic acid sequence. The nucleic acid sequence may be at least a portion or an entirety of the molecular tag. The molecular tag may be a nucleic acid molecule or may be part of a nucleic acid molecule. The molecular tag may be an oligonucleotide or a polypeptide. The molecular tag may comprise a DNA aptamer. The molecular tag may be or comprise a primer. The molecular tag may be, or comprise, a protein. The molecular tag may comprise a polypeptide. The molecular tag may be a barcode.

The term “partition,” as used herein, generally, refers to a space or volume that may be suitable to contain one or more species or conduct one or more reactions. A partition may be a physical compartment, such as a droplet or well. The partition may isolate space or volume from another space or

volume. The droplet may be a first phase (e.g., aqueous phase) in a second phase (e.g., oil) immiscible with the first phase. The droplet may be a first phase in a second phase that does not phase separate from the first phase, such as, for example, a capsule or liposome in an aqueous phase. A partition may comprise one or more other (inner) partitions. In some cases, a partition may be a virtual compartment that can be defined and identified by an index (e.g., indexed libraries) across multiple and/or remote physical compartments. For example, a physical compartment may comprise a plurality of virtual compartments.

Provided herein are systems and methods for processing that efficiently and accurately measure one or more analytes at the single cell level. The systems and methods may identify the cell that the one or more analytes originated from. An analyte can be a protein (e.g., surface-bound protein, internal protein, etc.). An analyte can be a metabolite (e.g., alcohol, amino acid, nucleotide, antioxidant, organic acid, polyol, vitamin, cofactor, etc.). An analyte can be any constituent of a cell, such as a small molecule, large molecule, or macromolecule. An analyte can be a biological particle. The analyte can be a targeted analyte, such as having binding specificity to an antibody that is specifically captured and analyzed. The analyte can be a non-targeted analyte that is non-specifically captured and analyzed. Identification and Measurement of Analytes at the Single Cell Level

A method for identifying a targeted analyte within a cell may comprise generating a cell bead from the cell and washing in an antibody into the cell bead. The antibody may have binding specificity to the targeted analyte and be coupled to a first barcode molecule. The first barcode molecule may comprise a first barcode sequence. Inside the cell bead, the antibody can bind to or otherwise couple to the targeted analyte, forming an analyte-antibody complex and tagging the targeted analyte with the first barcode sequence. The cell bead may be co-partitioned with a bead in a partition, wherein the bead is coupled to a second barcode molecule. The partition may be a droplet in an emulsion (e.g., water-in-oil, oil-in-water, etc.). The second barcode molecule may comprise a second barcode sequence. In the partition, the second barcode molecule may be released from the bead and attached to the analyte-antibody complex, thereby tagging the targeted analyte with both the first barcode sequence and the second barcode sequence. The first barcode molecule and the second barcode molecule may form a composite barcode molecule that comprises a composite sequence comprising the first barcode sequence and the second barcode sequence. The first barcode sequence may be indicative of the targeted analyte and the second barcode sequence may be indicative of the cell. The composite barcode molecule may be sequenced to identify the targeted analyte and the cell, and determine that the targeted analyte originated from within the cell.

The first barcode molecule may be a nucleic acid molecule, such as an oligonucleotide or polynucleotide. The first barcode sequence may uniquely identify the type of analyte having binding specificity to the antibody tagged by the first barcode sequence. For example, a first antibody having binding specificity to a first type of analyte and a second antibody having binding specificity to a second type of analyte can be tagged with different barcode sequences. A plurality of antibodies, each having a binding specificity to a same type of analyte (e.g., first type of protein), may each be tagged with the same barcode sequence.

The second barcode molecule may be a nucleic acid molecule, such as an oligonucleotide or polynucleotide. The

second barcode sequence may uniquely identify the bead that the second barcode molecule is coupled thereto. For example, a first bead and a second bead can be coupled to barcode molecules that comprise different barcode sequences. A plurality of barcode molecules, each coupled to the same bead, may comprise the same barcode sequence. A bead, associated with the unique barcode sequence (e.g., the second barcode sequence), may be co-partitioned with a cell. The unique barcode sequence of the bead may be used to uniquely identify the cell co-partitioned with the bead.

The composite barcode molecule may be a nucleic acid molecule, such as an oligonucleotide or polynucleotide. The composite barcode sequence may comprise the first barcode sequence and the second barcode sequence. In some instances, the composite barcode sequence may comprise one or more additional barcode sequences (e.g., unique molecular identifier, anchoring element, etc.) that are described elsewhere herein.

In some instances, the composite barcode molecule may be sequenced to determine a probability that the targeted analyte originated from within the cell. For example, in some instances, two or more beads of a plurality of beads may have the identical barcode sequence (e.g., second barcode sequence), which may or may not be intentional. In such cases, identifying the second barcode sequence may be indicative that the targeted analyte originated from one of two or more cells (each co-partitioned with a different bead). For example, where five beads have the identical second barcode sequence, identifying the second barcode sequence may be indicative that the targeted analyte originated from one of five cells (each co-partitioned with a different bead). In some instances, the probability that a targeted analyte originated from a cell may be based at least in part on the probability that the different beads in the system comprise unique second barcode sequences (e.g., if there is substantially a 100% probability that a first bead has a unique second barcode sequence among a plurality of other beads, identifying the second barcode sequence may be indicative that there is substantially a 100% probability the targeted analyte originated from the cell co-partitioned with the first bead). In some instances, a probability that the targeted analyte originated from within a cell may be at least about 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100%. Alternatively, the probability may be less than about 50%.

The targeted analyte can be an internal protein and the antibody contacting the cell bead may have a binding specificity to the internal protein. In another example, the targeted analyte can be a metabolite and the antibody contacting the cell bead may have a binding specificity to the metabolite. In some instances, a metabolite may be an alcohol, amino acid, nucleotide, antioxidant, organic acid, polyol, or vitamin. A metabolite may be a cofactor. The targeted analyte can be any constituent of a cell, such as any small molecule, large molecule, or macromolecule (e.g., macromolecular constituent). In yet another example, the targeted analyte can be from a class, set, or subset of analytes (e.g., proteins, metabolites, small molecules, etc.) sharing a structural similarity or homology (e.g., moiety, functional group, etc.), and the antibody contacting the cell bead may have a binding specificity to the class, set, or subset of analytes via the structural similarity. In such cases, the first barcode sequence may uniquely identify the class, set, or subset of analytes. Upon binding (or otherwise coupling) to the antibody, the targeted analyte may be classified by the first barcode sequence as a member of the class, set, or subset of analytes.

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Systems and methods for partitioning and using cell beads and barcode-coupled beads are described in further detail elsewhere herein, such as with reference to FIGS. 1-8.

FIG. 9 illustrates a method for identifying and measuring an analyte in a cell. A cell 902 may be processed to generate a cell bead 904. This may be performed by providing the cell 902 in a partition, such as a droplet. Systems and methods for generating a cell bead, such as by encapsulating and polymerizing a cell, are described elsewhere herein. The cell may be lysed to release one or more analytes in the cell bead 904, such as a first analyte 906, a second analyte 908, and a third analyte 910. Alternatively, the cell 902 may be lysed prior to formation of the cell bead 904, and a cell bead generated with the lysed contents thereafter. For example, the cell is provided in a droplet and lysed, and the droplet is subsequently subjected to polymerization to generate the cell bead 904. A plurality of antibodies may be washed into the cell bead. The plurality of antibodies may comprise a first set of antibodies comprising a first type of antibody 914 and a second set of antibodies comprising a second type of antibody 918. The first type of antibody can have a binding specificity to the first analyte 906 and have coupled thereto a first barcode molecule comprising a first barcode sequence 912. The second type of antibody 918 can have a binding specificity to the second analyte 908 and have coupled thereto a second barcode molecule comprising a second barcode sequence 916. Antibodies having binding specificities to different types of analytes may each have a different barcode sequence. A barcode molecule may be a nucleic acid molecule, such as an oligonucleotide or polynucleotide.

Washed into the cell bead 904, the plurality of antibodies may bind or otherwise couple to one or more analytes in accordance with the respective binding specificity. For example, a first antibody of the first type of antibody 914 may bind to (or otherwise couple to) the first analyte 906, thereby forming a first analyte-antibody complex 920 and tagging the first analyte 906 with the first barcode sequence 912. A second antibody of the second type of antibody 918 may bind to (or otherwise couple to) the second analyte 908, thereby forming a second analyte-antibody complex 922 and tagging the second analyte 908 with the second barcode sequence 916. Unbound antibodies, such as having binding specificities to analytes that are not present or those unable to find a free analyte, may be washed out of the cell bead.

Alternative to, or in addition to, washing in a plurality of types of antibodies in bulk to the cell bead 904, antibodies may be washed into the cell bead 904 one type at a time (e.g., sequentially). In some instances, unbound antibodies may be washed out between each washing in. In some instances, unbound antibodies of one or more types may be washed out together.

The cell bead 904 may be co-partitioned with a bead 926 in a partition 924. The partition 924 may be a droplet in an emulsion. The bead 926 may be coupled to a plurality of barcode molecules, each comprising a common third barcode sequence 928. For example, FIG. 8 illustrates a bead coupled to a plurality of barcode molecules. The third barcode sequence 928 may be unique to the bead 926. The plurality of barcode molecules may be coupled by releasable linkage to the bead 926, as described elsewhere herein. In the partition 924, the analytes from the cell bead 904 may be tagged with the plurality of barcode molecules (e.g., after release from the bead 926).

In some instances, only the analytes bound by (or coupled to) an antibody which is coupled to another barcode molecule (e.g., the first barcode molecule coupled to the first type of antibody 914, the second barcode molecule coupled

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to the second type of antibody 916) may be tagged by the third barcode sequence 928. For example, the plurality of barcode molecules coupled to the bead 926 may comprise a specific priming sequence that is configured to couple to the another barcode molecule. Alternatively or in addition, other analytes (e.g., unbound analytes) may be tagged by the third barcode sequence 928.

An analyte may thus be tagged by a composite barcode sequence, the composite barcode sequencing comprising the third barcode sequence 928 and another barcode sequence (e.g., the first barcode sequence 912, the second barcode sequence 914) tagged by an antibody. For example, the first analyte 906 may be tagged by a first composite barcode sequence 930 comprising the first barcode sequence 912 and the third barcode sequence 930. The second analyte 908 may be tagged by a second composite barcode sequence 932 comprising the second barcode sequence 916 and the third barcode sequence 932. Upon pooling the composite sequences, such as from different partitions (e.g., comprising different cells and beads), the composite sequences may be sequenced to measure one or more analytes in a cell by attributing the one or more analytes to the cell.

For example, from the first composite barcode sequence 930, identification of the first barcode sequence 912 may be indicative of the first type of analyte 906 and identification of the third barcode sequence 928 may be indicative of the cell 904, permitting the determination that the first type of analyte 906 originated from the cell 904. In another example, from the second composite barcode sequence 932, identification of the second barcode sequence 916 may be indicative of the second type of analyte 908 and identification of the third barcode sequence 928 may be indicative of the cell 904, permitting the determination that the second type of analyte 906 originated from the cell 904. Alternatively or in addition, such as when there is a probability that two or more beads may share their common barcode sequence (e.g., third barcode sequence 932), identification of the third barcode sequence 928 may be indicative of a probability that a particular analyte originated from the cell 904. In some cases, the respective probabilities that multiple analytes originated from the cell may be determined.

While not illustrated in FIG. 9, in other partitions which co-partition a different cell with a different bead comprising a different common barcode sequence, identification of the different common barcode sequence in a composite barcode sequence may be indicative of the different cell, allowing the attribution (e.g., origin, probability of origin) of one or more analytes to the different cell.

Beneficially, the systems and methods may attribute internal proteins (e.g., not surface-bound) and metabolites to a single cell, and allow for their accurate and efficient measurement.

Methods of the present disclosure may permit identification of surface bound and internal analytes (e.g., proteins). For example, with reference to FIG. 9, the cell 902 may be brought in contact with a first set of antibodies prior to lysing, such as prior to providing the cell in the partition to generate the cell bead 904. In some cases, the cell 902 is in a collection of cells that may be brought in contact with the first set of antibodies in bulk. Each antibody of the first set of antibodies may comprise a barcode molecule with a first barcode sequence that is indicative of surface-bound analytes. In some instances, an antibody in the first set of antibodies may have binding specificity to a given analyte or a given type of analytes. For such instances, in some cases, the barcode molecule on the antibody may comprise additional barcode sequence(s) that are selected to be specific

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for, and indicative of, the given analyte or the given type of analytes (e.g., type of protein). For example, between two antibodies in the first set of antibodies which each has a binding specificity to a different type of analyte, both antibodies may comprise the same first barcode sequence (indicative of surface-bound analytes), but different additional sequence(s) (indicative of different types of analytes). When the cell **902** (prior to lysing) is brought in contact with the first set of antibodies, the antibodies in the first set of antibodies may couple to surface-bound analytes (e.g., surface-bound proteins) of the cell **902** to form antibody-analyte complexes, thereby tagging each antibody-bound, surface-bound analyte with the first barcode sequence (indicative of surface-bound analytes).

Subsequent to processing the cell **902** with the first set of antibodies, the cell **902** may be used to generate the cell bead **904**, as described above in the context of FIG. **9**. The cell may be lysed prior to or after generation of the cell bead **904**. Analytes from within the cell **902** may then be processed with a second set of antibodies (e.g., antibody **914** and antibody **918** used to assay for proteins, metabolites, etc.) by washing in the second set of antibodies. Each antibody of the second set of antibodies may comprise a barcode molecule with a second barcode sequence that is indicative of internal analytes. In some instances, an antibody in the second set of antibodies may have binding specificity to a given analyte or a given type of analytes. For such instances, in some cases, the barcode molecule on the antibody may comprise additional barcode sequence(s) that are selected to be specific for, and indicative of, the given analyte or the given type of analytes. For example, between two antibodies in the second set of antibodies which each has a binding specificity to a different type of analyte, both antibodies may comprise the same second barcode sequence (indicative of internal analytes), but with different additional sequence(s) (indicative of different types of analytes). When the analytes from within the cell **902** are brought in contact with the second set of antibodies, the antibodies in the second set of antibodies may couple to the internal analytes (from within the cell **902**) to form antibody-analyte complexes, thereby tagging each antibody-bound, internal analyte with the second barcode sequence (indicative of internal analytes).

In some instances, as described above in the context of FIG. **9**, the cell bead (processed by the first set and the second set of antibodies) may be co-partitioned with a bead comprising a plurality of barcode molecules comprising a third barcode sequence (indicative of the cell), and each analyte, including surface-bound analytes (tagged by the first barcode sequence) and internal analytes (tagged by the second barcode sequence), may be tagged by the third barcode sequence, such as to form composite barcode sequences. For example, a first composite sequence may comprise the first barcode sequence and the third barcode sequence. In another example, a second composite sequence may comprise the second barcode sequence and the third barcode sequence.

Upon pooling the composite sequences, such as from different partitions (e.g., comprising different cells and beads), the composite sequences may be sequenced to measure one or more analytes in a cell by attributing the one or more analytes to the cell. Following sequencing, the first and second barcode sequences may identify surface-bound and internal analytes, respectively, and the third barcode sequence may identify the cell (e.g., **902**). In some instances, additional sequence(s) may identify a given analyte or a given type of analyte. Beneficially, the systems and methods of the present disclosure may identify, for an analyte iden-

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tified, not only the originating cell, but the originating location (e.g., surface-bound, internal, etc.) in the cell.

Another method for measuring an analyte in a cell may comprise co-partitioning the cell and a bead in a first partition. The first partition may be a droplet in an emulsion. The bead may comprise a plurality of antibodies configured to couple to analytes (e.g., proteins). The plurality of antibodies may be releasably attached to the bead, such as described elsewhere herein. In some instances, a given antibody of the plurality of antibodies may have non-binding specificity, such as to be capable of binding or otherwise coupling to different analytes. Alternatively or in addition, a given antibody may have binding specificity to a particular type of analyte (e.g., a particular type of protein). Alternatively or in addition, a given antibody may have binding specificity to a particular class, set, or subset of analytes. Each antibody of the plurality of antibodies may be coupled to a common first barcode molecule comprising a common first barcode sequence. The common first barcode sequence may be unique to the bead. The cell may be subject to conditions sufficient to permit the given antibody to couple with an analyte to generate a first analyte-antibody complex, thereby tagging the analyte with the first barcode sequence. For example, the cell may be lysed and the antibodies released from the bead.

The first analyte-antibody complex may be pooled with other analyte-antibody complexes, such as from the same partition and/or different partitions (e.g., different droplets). The pool of analyte-antibody complexes may be sorted by type of analyte in each analyte-antibody complex into different partitions. For example, the partitions may be wells. The first analyte-antibody complex may be co-partitioned in a second partition (e.g., well) along with other analyte-antibody complexes of the same type as the first analyte-antibody complex, if any. The first analyte-antibody complex may be co-partitioned with a second barcode molecule comprising a second barcode sequence under conditions sufficient to permit the first barcode molecule of the first analyte-antibody complex and the second barcode molecule to generate a composite barcode molecule comprising a composite barcode sequence comprising the first barcode sequence and the second barcode sequence, thereby tagging the analyte with both the first barcode sequence and the second barcode sequence. The second barcode molecule may be delivered to the second partition prior to, during, or subsequent to, partitioning of the first analyte-antibody complex in the second partition. The second barcode molecule may be delivered on a support, such as a bead. The second barcode molecule may be delivered as part of a plurality of barcode molecules each comprising the second barcode sequence. The second barcode sequence may be unique to the second partition. Each well (each comprising different types of analyte-antibody complexes) may comprise a different barcode sequence unique to each well. For example, two or more wells comprising the same type of analyte-antibody complexes may comprise the same common barcode sequence (e.g., the second barcode sequence). The partitions may be non-well partitions.

The composite barcode molecule, or a derivative thereof, may be sequenced to identify the cell, from the first barcode sequence in the composite barcode sequence, and the analyte, from the second barcode sequence in the composite barcode sequence, and determine that the analyte originated from the cell.

The first barcode molecule may be a nucleic acid molecule, such as an oligonucleotide or polynucleotide. The first barcode sequence may be a nucleic acid sequence. The first

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barcode sequence may uniquely identify the bead that the first barcode molecule is coupled thereto. For example, a first bead and a second bead can be coupled to barcode molecules that comprise different barcode sequences. A plurality of barcode molecules, each coupled to the same bead, may comprise the same barcode sequence. A bead, associated with the unique barcode sequence (e.g., the first barcode sequence), may be co-partitioned with a cell. The unique barcode sequence of the bead may be used to uniquely identify the cell co-partitioned with the bead.

The second barcode molecule may be a nucleic acid molecule, such as an oligonucleotide or polynucleotide. The second barcode sequence may uniquely identify a partition that the second barcode sequence is delivered to. For example, different partitions may each be delivered different barcode sequences. The partition may comprise analytes of a first common type. The unique barcode sequence (e.g., the second barcode sequence) of the partition may be used to uniquely identify the first common type of analytes co-partitioned in the partition. Alternatively, the second barcode sequence may uniquely identify a type of analyte. For example, where the first common type of analytes is partitioned in two or more partitions (e.g., half in a first partition, the other half in a second partition), the second barcode sequence may be delivered to each of the two or more partitions, and different unique barcode sequences may be delivered to other partitions comprising different types of analytes.

The composite barcode molecule may be a nucleic acid molecule, such as an oligonucleotide or polynucleotide. The composite barcode sequence may comprise the first barcode sequence and the second barcode sequence. In some instances, the composite barcode sequence may comprise one or more additional barcode sequences (e.g., unique molecular identifier, anchoring element, etc.) that are described elsewhere herein.

In some instances, the composite barcode molecule may be sequenced to determine a probability that the analyte originated from the cell. For example, in some instances, two or more beads of a plurality of beads may have the identical barcode sequence (e.g., first barcode sequence), which may or may not be intentional. In such cases, identifying the first barcode sequence may be indicative that the analyte originated from one of two or more cells (each co-partitioned with a different bead). For example, where five beads have the identical first barcode sequence, identifying the first barcode sequence may be indicative that the analyte originated from one of five cells (each co-partitioned with a different bead). In some instances, the probability that an analyte originated from a cell may be based at least in part on the probability that the different beads in the system comprise unique first barcode sequences (e.g., if there is substantially a 100% probability that a first bead has a unique first barcode sequence among a plurality of other beads, identifying the first barcode sequence may be indicative that there is substantially a 100% probability the analyte originated from the cell co-partitioned with the first bead). In some instances, a probability that the analyte originated from a cell may be at least about 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100%. Alternatively, the probability may be less than about 50%.

The analyte can be an internal protein. In another example, the analyte can be a metabolite. The analyte can be any constituent of a cell, such as any small molecule, large molecule, or macromolecule (e.g., macromolecular constituent). In yet another example, the analyte can be from a class,

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set, or subset of analytes (e.g., proteins, metabolites, small molecules, etc.) sharing a structural similarity or homology (e.g., moiety, functional group, etc.). The analyte may be a targeted analyte or from a targeted class of analytes, such as by employing antibodies having binding specificity to the targeted analyte or targeted class of analytes. The analyte may be a non-targeted analyte.

FIG. 10 illustrates another method for identifying and measuring an analyte in a cell. A cell **1002** and a bead **1004** may be co-partitioned in a first partition **1006**. The first partition **1006** may be a droplet in an emulsion. The bead **1004** may be delivered to the first partition **1006** prior to, during, or subsequent to partitioning of the cell **1002** in the first partition **1006**. Systems and methods for partitioning are described elsewhere herein. The bead **1004** may be coupled to a plurality of antibodies **1010**, which may or may not have binding specificity to an analyte or a class of analytes in, or suspected to be in, the cell **1002**. Each antibody of the plurality of antibodies **1010** may comprise a common first barcode sequence **1008**. The bead **1004** may be coupled to the plurality of antibodies **1010** via one or more barcode molecules (e.g., comprising the common first barcode sequence **1008**). The bead **1004** may be releasably coupled to the one or more barcode molecules, as described elsewhere herein. For example, FIG. 8 illustrates a bead coupled to a plurality of barcode molecules. The first barcode sequence **1008** may be unique to the bead **1004**.

The cell **1002** may be lysed to release one or more analytes in the first partition **1006**, such as a first analyte **1012**, a second analyte **1014**, and a third analyte **1016**, each a different type of analyte (e.g., different type of protein). The plurality of antibodies **1010** may be released from the bead **1004**, such as described elsewhere herein (e.g., applying a stimulus or stimuli). In the first partition **1006**, a given analyte (e.g., **1012**, **1014**, **1016**) from the cell **1002** may bind with (or otherwise couple to) a given antibody (e.g., **1010**) to form a given analyte-antibody complex, thereby tagging the given analyte with the first barcode sequence **1008**. Each analyte-antibody complex formed in the first partition **1006** may be tagged with the first barcode sequence **1008**, independent of the type of analyte. For example, each of the first analyte **1012**, second analyte **1014**, and the third analyte **1016**, which are a different type of analyte, may be tagged with the first barcode sequence **1008**.

The given analyte-antibody complex may be pooled with other analyte-antibody complexes, such as from the same partition (the first partition **1006**) and/or different partitions (e.g., different droplets). The pool of analyte-antibody complexes may be sorted by type of analyte in each analyte-antibody complex into different partitions. For example, the partitions may be wells. For example, a first analyte-antibody complex comprising the first analyte **1012** may be partitioned into a first well along with other analyte-antibody complexes from the first partition **1006** or other partitions that comprise the same type of analyte as the first analyte **1012**. A second analyte-antibody complex comprising the second analyte **1014** may be partitioned into a second well along with other analyte-antibody complexes from the first partition **1006** or other partitions that comprise the same type of analyte as the second analyte **1014**. A third analyte-antibody complex comprising the third analyte **1016** may be partitioned into a third well along with other analyte-antibody complexes from the first partition **1006** or other partitions that comprise the same type of analyte as the third analyte **1016**.

In FIG. 10, for example, the first well comprises the first analyte-antibody complex from the first partition **1006**, and

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another analyte-antibody complex of the same type as the first analyte **1012** from a first other partition which co-partitioned a second cell and a second bead comprising a second barcode sequence **1018**, thereby tagged with the second barcode sequence **1018**. The second well comprises the second analyte-antibody complex from the first partition **1006**, and another analyte-antibody complex of the same type as the second analyte **1014** from the first other partition, thereby tagged with the second barcode sequence **1018**, and another analyte-antibody complex of the same type as the second analyte **1014** from a second other partition which co-partitioned a third cell and a third bead comprising a third barcode sequence **1020**, thereby tagged with the third barcode sequence **1020**. The third well comprises the third analyte-antibody complex from the first partition **1006**.

In some instances, a type of analyte may be sorted into more than one well. For example, a plurality of analyte-antibody complexes each comprising the same first type of analyte may be distributed across two or more wells, provided that each well comprises only one type of analyte.

A barcode molecule comprising a barcode sequence unique to each well may be delivered to each well. For example, a barcode molecule comprising a fourth barcode sequence **1022** may be delivered to the first well, a barcode molecule comprising a fifth barcode sequence **1024** may be delivered to the second well, and a barcode molecule comprising a sixth barcode sequence **1026** may be delivered to the third well. In some instances, the barcode molecule may be delivered via a bead to each well. In some instances, the barcode molecule may be delivered as part of a plurality of barcode molecules that each comprises the same unique barcode sequence. For example, the fourth barcode sequence **1022** may be unique to the first well, the fifth barcode sequence **1024** may be unique to the second well, and the sixth barcode sequence **1026** may be unique to the third well.

Alternatively or in addition to, a barcode molecule comprising a barcode sequence unique to a type of analyte may be delivered to each well. For example, where two or more wells comprise the same type of analyte, the same barcode sequence may be delivered to the two or more wells.

The wells and their constituents may be subject to conditions sufficient to permit the given barcode molecule (e.g., comprising **1022**, **1024**, **1026**) delivered to the wells and a given barcode molecule of the analyte-antibody complexes to generate a given composite barcode molecule comprising a composite barcode sequence. The composite barcode sequence may comprise the barcode sequence of the barcode molecule delivered to the well and the barcode sequence tagged on the analyte-antibody complex.

For example, in FIG. **10**, in the first well, a first composite barcode sequence comprises the first barcode sequence **1008** from the first partition **1006** (unique to the first partition) and the fourth barcode sequence **1022** (unique to the first well), and a second composite barcode sequence comprises the second barcode sequence **1018** from the first other partition (unique to the first other partition) and the fourth barcode sequence **1022** (unique to the first well). In the second well, a third composite barcode sequence comprises the first barcode sequence **1008** from the first partition **1006** (unique to the first partition) and the fifth barcode sequence **1024** (unique to the second well), a fourth composite barcode sequence comprises the second barcode sequence **1018** from the first other partition (unique to the first other partition) and the fifth barcode sequence **1022** (unique to the second well), and a fifth composite barcode sequence comprises the third barcode sequence **1020** from the second other partition

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(unique to the second other partition) and the fifth barcode sequence **1022** (unique to the second well). In the third well, a sixth composite barcode sequence comprises the first barcode sequence **1008** from the first partition **1006** (unique to the first partition) and the sixth barcode sequence **1026** (unique to the third well).

Upon pooling the composite sequences, such as from different partitions (e.g., wells), the composite sequences may be sequenced to measure one or more analytes in a cell by attributing the one or more analytes to the cell.

For example, from the first composite barcode sequence, identification of the first barcode sequence **1008** may be indicative of the cell **1002** and identification of the fourth barcode sequence **1022** may be indicative of the first analyte **1012** type, permitting the determination that the first analyte **1012** type originated from the cell **1002**. In another example, from the fifth composite barcode sequence, identification of the third barcode sequence **1020** may be indicative of the third cell (co-partitioned with the third bead comprising the third barcode sequence **1020** in the second other partition) and identification of the fifth barcode sequence **1022** may be indicative of the second analyte **1014** type, permitting the determination that the second analyte **1014** type originated from the second cell. Alternatively or in addition, such as when there is a probability that two or more beads may share their common barcode sequence (e.g., first barcode sequence **1008**), identification of the first barcode sequence **1008** may be indicative of a probability that a particular type of analyte originated from the cell **1004**. Respective probabilities for different types of analytes may be determined.

While FIG. **10** illustrates the identification of three different types of analytes, there may be any number of analytes identified. While FIG. **10** illustrates the identification of three different types of cells, there may be any number of cells identified.

While the systems and methods described in relation to FIGS. **9** and **10** illustrate the capture of one or more analytes via the use of one or more antibodies, which may or may not have binding specificity, the systems and methods may be applied using other adaptors or molecules (e.g., peptides) configured to couple to one or more analytes, such as specific priming sequences or random primers. For example, with respect to FIG. **10**, the bead may be coupled to a plurality of adaptors or molecules configured to couple to one or more analytes. In some instances, the bead may be coupled to one or more populations of different adaptors or molecules comprising barcode sequences, such as a first population of antibodies and a second population of priming sequences.

Methods of the present disclosure may be used to identify information about analytes from a cell, including identity (e.g., type) of an analyte and location (e.g., surface-bound, internal) of the analyte. Such analytes may include, for example, proteins, metabolites, ribonucleic acid (RNA) molecules, and deoxyribonucleic acid (DNA) molecules. In some examples, proteomic information may be obtained for the cell and combined with genomic (e.g., DNA or genomic variants) and/or transcriptomic (e.g., mRNA) information about the cell. This may permit identifying, for example, genes that are transcribed into mRNA and mRNA that is translated to protein.

A computer system, for example, described with respect to FIG. **11**, may be used to implement the systems and methods of the present disclosure, such as to generate and maintain one or more libraries, as well as perform sequencing, and analyze the results of such sequencing to identify one or more analytes, identify one or more cells, and/or

attribute an identified analyte to an identified cell. Beneficially, the systems and methods provided herein may efficiently and accurately measure one or more analytes at the single cell level.

Systems and Methods for Sample Compartmentalization

In an aspect, the systems and methods described herein provide for the compartmentalization, depositing, or partitioning of one or more particles (e.g., biological particles, macromolecular constituents of biological particles, beads, reagents, etc.) into discrete compartments or partitions (referred to interchangeably herein as partitions), where each partition maintains separation of its own contents from the contents of other partitions. The partition can be a droplet in an emulsion. A partition may comprise one or more other partitions.

A partition may include one or more particles. A partition may include one or more types of particles. For example, a partition of the present disclosure may comprise one or more biological particles and/or macromolecular constituents thereof. A partition may comprise one or more gel beads. A partition may comprise one or more cell beads. A partition may include a single gel bead, a single cell bead, or both a single cell bead and single gel bead. A partition may include one or more reagents. Alternatively, a partition may be unoccupied. For example, a partition may not comprise a bead. A cell bead can be a biological particle and/or one or more of its macromolecular constituents encased inside of a gel or polymer matrix, such as via polymerization of a droplet containing the biological particle and precursors capable of being polymerized or gelled. Unique identifiers, such as barcodes, may be injected into the droplets previous to, subsequent to, or concurrently with droplet generation, such as via a microcapsule (e.g., bead), as described elsewhere herein. Microfluidic channel networks (e.g., on a chip) can be utilized to generate partitions as described herein. Alternative mechanisms may also be employed in the partitioning of individual biological particles, including porous membranes through which aqueous mixtures of cells are extruded into non-aqueous fluids.

The partitions can be flowable within fluid streams. The partitions may comprise, for example, micro-vesicles that have an outer barrier surrounding an inner fluid center or core. In some cases, the partitions may comprise a porous matrix that is capable of entraining and/or retaining materials within its matrix. The partitions can be droplets of a first phase within a second phase, wherein the first and second phases are immiscible. For example, the partitions can be droplets of aqueous fluid within a non-aqueous continuous phase (e.g., oil phase). In another example, the partitions can be droplets of a non-aqueous fluid within an aqueous phase. In some examples, the partitions may be provided in a water-in-oil emulsion or oil-in-water emulsion. A variety of different vessels are described in, for example, U.S. Patent Application Publication No. 2014/0155295, which is entirely incorporated herein by reference for all purposes. Emulsion systems for creating stable droplets in non-aqueous or oil continuous phases are described in, for example, U.S. Patent Application Publication No. 2010/0105112, which is entirely incorporated herein by reference for all purposes.

In the case of droplets in an emulsion, allocating individual particles to discrete partitions may in one non-limiting example be accomplished by introducing a flowing stream of particles in an aqueous fluid into a flowing stream of a non-aqueous fluid, such that droplets are generated at the junction of the two streams. Fluid properties (e.g., fluid flow rates, fluid viscosities, etc.), particle properties (e.g.,

volume fraction, particle size, particle concentration, etc.), microfluidic architectures (e.g., channel geometry, etc.), and other parameters may be adjusted to control the occupancy of the resulting partitions (e.g., number of biological particles per partition, number of beads per partition, etc.). For example, partition occupancy can be controlled by providing the aqueous stream at a certain concentration and/or flow rate of particles. To generate single biological particle partitions, the relative flow rates of the immiscible fluids can be selected such that, on average, the partitions may contain less than one biological particle per partition in order to ensure that those partitions that are occupied are primarily singly occupied. In some cases, partitions among a plurality of partitions may contain at most one biological particle (e.g., bead, DNA, cell or cellular material). In some embodiments, the various parameters (e.g., fluid properties, particle properties, microfluidic architectures, etc.) may be selected or adjusted such that a majority of partitions are occupied, for example, allowing for only a small percentage of unoccupied partitions. The flows and channel architectures can be controlled as to ensure a given number of singly occupied partitions, less than a certain level of unoccupied partitions and/or less than a certain level of multiply occupied partitions.

FIG. 1 shows an example of a microfluidic channel structure **100** for partitioning individual biological particles. The channel structure **100** can include channel segments **102**, **104**, **106** and **108** communicating at a channel junction **110**. In operation, a first aqueous fluid **112** that includes suspended biological particles (or cells) **114** may be transported along channel segment **102** into junction **110**, while a second fluid **116** that is immiscible with the aqueous fluid **112** is delivered to the junction **110** from each of channel segments **104** and **106** to create discrete droplets **118**, **120** of the first aqueous fluid **112** flowing into channel segment **108**, and flowing away from junction **110**. The channel segment **108** may be fluidically coupled to an outlet reservoir where the discrete droplets can be stored and/or harvested. A discrete droplet generated may include an individual biological particle **114** (such as droplets **118**). A discrete droplet generated may include more than one individual biological particle **114** (not shown in FIG. 1). A discrete droplet may contain no biological particle **114** (such as droplet **120**). Each discrete partition may maintain separation of its own contents (e.g., individual biological particle **114**) from the contents of other partitions.

The second fluid **116** can comprise an oil, such as a fluorinated oil, that includes a fluorosurfactant for stabilizing the resulting droplets, for example, inhibiting subsequent coalescence of the resulting droplets **118**, **120**. Examples of particularly useful partitioning fluids and fluorosurfactants are described, for example, in U.S. Patent Application Publication No. 2010/0105112, which is entirely incorporated herein by reference for all purposes.

As will be appreciated, the channel segments described herein may be coupled to any of a variety of different fluid sources or receiving components, including reservoirs, tubing, manifolds, or fluidic components of other systems. As will be appreciated, the microfluidic channel structure **100** may have other geometries. For example, a microfluidic channel structure can have more than one channel junction. For example, a microfluidic channel structure can have 2, 3, 4, or 5 channel segments each carrying particles (e.g., biological particles, cell beads, and/or gel beads) that meet at a channel junction. Fluid may be directed to flow along one or more channels or reservoirs via one or more fluid flow units. A fluid flow unit can comprise compressors (e.g.,

providing positive pressure), pumps (e.g., providing negative pressure), actuators, and the like to control flow of the fluid. Fluid may also or otherwise be controlled via applied pressure differentials, centrifugal force, electrokinetic pumping, vacuum, capillary or gravity flow, or the like.

The generated droplets may comprise two subsets of droplets: (1) occupied droplets **118**, containing one or more biological particles **114**, and (2) unoccupied droplets **120**, not containing any biological particles **114**. Occupied droplets **118** may comprise singly occupied droplets (having one biological particle) and multiply occupied droplets (having more than one biological particle). As described elsewhere herein, in some cases, the majority of occupied partitions can include no more than one biological particle per occupied partition and some of the generated partitions can be unoccupied (of any biological particle). In some cases, though, some of the occupied partitions may include more than one biological particle. In some cases, the partitioning process may be controlled such that fewer than about 25% of the occupied partitions contain more than one biological particle, and in many cases, fewer than about 20% of the occupied partitions have more than one biological particle, while in some cases, fewer than about 10% or even fewer than about 5% of the occupied partitions include more than one biological particle per partition.

In some cases, it may be desirable to minimize the creation of excessive numbers of empty partitions, such as to reduce costs and/or increase efficiency. While this minimization may be achieved by providing a sufficient number of biological particles (e.g., biological particles **114**) at the partitioning junction **110**, such as to ensure that at least one biological particle is encapsulated in a partition, the Poissonian distribution may expectedly increase the number of partitions that include multiple biological particles. As such, where singly occupied partitions are to be obtained, at most about 95%, 90%, 85%, 80%, 75%, 70%, 65%, 60%, 55%, 50%, 45%, 40%, 35%, 30%, 25%, 20%, 15%, 10%, 5% or less of the generated partitions can be unoccupied.

In some cases, the flow of one or more of the biological particles (e.g., in channel segment **102**), or other fluids directed into the partitioning junction (e.g., in channel segments **104**, **106**) can be controlled such that, in many cases, no more than about 50% of the generated partitions, no more than about 25% of the generated partitions, or no more than about 10% of the generated partitions are unoccupied. These flows can be controlled so as to present a non-Poissonian distribution of single-occupied partitions while providing lower levels of unoccupied partitions. The above noted ranges of unoccupied partitions can be achieved while still providing any of the single occupancy rates described above. For example, in many cases, the use of the systems and methods described herein can create resulting partitions that have multiple occupancy rates of less than about 25%, less than about 20%, less than about 15%, less than about 10%, and in many cases, less than about 5%, while having unoccupied partitions of less than about 50%, less than about 40%, less than about 30%, less than about 20%, less than about 10%, less than about 5%, or less.

As will be appreciated, the above-described occupancy rates are also applicable to partitions that include both biological particles and additional reagents, including, but not limited to, microcapsules or beads (e.g., gel beads) carrying barcoded nucleic acid molecules (e.g., oligonucleotides) (described in relation to FIG. 2). The occupied partitions (e.g., at least about 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, or 99% of the occupied

partitions) can include both a microcapsule (e.g., bead) comprising barcoded nucleic acid molecules and a biological particle.

In another aspect, in addition to or as an alternative to droplet based partitioning, biological particles may be encapsulated within a microcapsule that comprises an outer shell, layer or porous matrix in which is entrained one or more individual biological particles or small groups of biological particles. The microcapsule may include other reagents. Encapsulation of biological particles may be performed by a variety of processes. Such processes may combine an aqueous fluid containing the biological particles with a polymeric precursor material that may be capable of being formed into a gel or other solid or semi-solid matrix upon application of a particular stimulus to the polymer precursor. Such stimuli can include, for example, thermal stimuli (e.g., either heating or cooling), photo-stimuli (e.g., through photo-curing), chemical stimuli (e.g., through cross-linking, polymerization initiation of the precursor (e.g., through added initiators)), mechanical stimuli, or a combination thereof.

Preparation of microcapsules comprising biological particles may be performed by a variety of methods. For example, air knife droplet or aerosol generators may be used to dispense droplets of precursor fluids into gelling solutions in order to form microcapsules that include individual biological particles or small groups of biological particles. Likewise, membrane based encapsulation systems may be used to generate microcapsules comprising encapsulated biological particles as described herein. Microfluidic systems of the present disclosure, such as that shown in FIG. 1, may be readily used in encapsulating cells as described herein. In particular, and with reference to FIG. 1, the aqueous fluid **112** comprising (i) the biological particles **114** and (ii) the polymer precursor material (not shown) is flowed into channel junction **110**, where it is partitioned into droplets **118**, **120** through the flow of non-aqueous fluid **116**. In the case of encapsulation methods, non-aqueous fluid **116** may also include an initiator (not shown) to cause polymerization and/or crosslinking of the polymer precursor to form the microcapsule that includes the entrained biological particles. Examples of polymer precursor/initiator pairs include those described in U.S. Patent Application Publication No. 2014/0378345, which is entirely incorporated herein by reference for all purposes.

For example, in the case where the polymer precursor material comprises a linear polymer material, such as a linear polyacrylamide, PEG, or other linear polymeric material, the activation agent may comprise a cross-linking agent, or a chemical that activates a cross-linking agent within the formed droplets. Likewise, for polymer precursors that comprise polymerizable monomers, the activation agent may comprise a polymerization initiator. For example, in certain cases, where the polymer precursor comprises a mixture of acrylamide monomer with a N,N'-bis-(acryloyl) cystamine (BAC) comonomer, an agent such as tetraethylmethylenediamine (TEMED) may be provided within the second fluid streams **116** in channel segments **104** and **106**, which can initiate the copolymerization of the acrylamide and BAC into a cross-linked polymer network, or hydrogel.

Upon contact of the second fluid stream **116** with the first fluid stream **112** at junction **110**, during formation of droplets, the TEMED may diffuse from the second fluid **116** into the aqueous fluid **112** comprising the linear polyacrylamide, which will activate the crosslinking of the polyacrylamide within the droplets **118**, **120**, resulting in the formation of gel (e.g., hydrogel) microcapsules, as solid or semi-solid beads

or particles entraining the cells 114. Although described in terms of polyacrylamide encapsulation, other ‘activatable’ encapsulation compositions may also be employed in the context of the methods and compositions described herein. For example, formation of alginate droplets followed by exposure to divalent metal ions (e.g., Ca^{2+} ions), can be used as an encapsulation process using the described processes. Likewise, agarose droplets may also be transformed into capsules through temperature based gelling (e.g., upon cooling, etc.).

In some cases, encapsulated biological particles can be selectively releasable from the microcapsule, such as through passage of time or upon application of a particular stimulus, that degrades the microcapsule sufficiently to allow the biological particles (e.g., cell), or its other contents to be released from the microcapsule, such as into a partition (e.g., droplet). For example, in the case of the polyacrylamide polymer described above, degradation of the microcapsule may be accomplished through the introduction of an appropriate reducing agent, such as DTT or the like, to cleave disulfide bonds that cross-link the polymer matrix. See, for example, U.S. Patent Application Publication No. 2014/0378345, which is entirely incorporated herein by reference for all purposes.

The biological particle can be subjected to other conditions sufficient to polymerize or gel the precursors. The conditions sufficient to polymerize or gel the precursors may comprise exposure to heating, cooling, electromagnetic radiation, and/or light. The conditions sufficient to polymerize or gel the precursors may comprise any conditions sufficient to polymerize or gel the precursors. Following polymerization or gelling, a polymer or gel may be formed around the biological particle. The polymer or gel may be diffusively permeable to chemical or biochemical reagents. The polymer or gel may be diffusively impermeable to macromolecular constituents of the biological particle. In this manner, the polymer or gel may act to allow the biological particle to be subjected to chemical or biochemical operations while spatially confining the macromolecular constituents to a region of the droplet defined by the polymer or gel. The polymer or gel may include one or more of disulfide cross-linked polyacrylamide, agarose, alginate, polyvinyl alcohol, polyethylene glycol (PEG)-diacrylate, PEG-acrylate, PEG-thiol, PEG-azide, PEG-alkyne, other acrylates, chitosan, hyaluronic acid, collagen, fibrin, gelatin, or elastin. The polymer or gel may comprise any other polymer or gel.

The polymer or gel may be functionalized to bind to targeted analytes, such as nucleic acids, proteins, carbohydrates, lipids or other analytes. The polymer or gel may be polymerized or gelled via a passive mechanism. The polymer or gel may be stable in alkaline conditions or at elevated temperature. The polymer or gel may have mechanical properties similar to the mechanical properties of the bead. For instance, the polymer or gel may be of a similar size to the bead. The polymer or gel may have a mechanical strength (e.g. tensile strength) similar to that of the bead. The polymer or gel may be of a lower density than an oil. The polymer or gel may be of a density that is roughly similar to that of a buffer. The polymer or gel may have a tunable pore size. The pore size may be chosen to, for instance, retain denatured nucleic acids. The pore size may be chosen to maintain diffusive permeability to exogenous chemicals such as sodium hydroxide (NaOH) and/or endogenous chemicals such as inhibitors. The polymer or gel may be biocompatible. The polymer or gel may maintain or enhance cell viability. The polymer or gel may be biochemically

compatible. The polymer or gel may be polymerized and/or depolymerized thermally, chemically, enzymatically, and/or optically.

The polymer may comprise poly(acrylamide-co-acrylic acid) crosslinked with disulfide linkages. The preparation of the polymer may comprise a two-step reaction. In the first activation step, poly(acrylamide-co-acrylic acid) may be exposed to an acylating agent to convert carboxylic acids to esters. For instance, the poly(acrylamide-co-acrylic acid) may be exposed to 4-(4,6-dimethoxy-1,3,5-triazin-2-yl)-4-methylmorpholinium chloride (DMTMM). The polyacrylamide-co-acrylic acid may be exposed to other salts of 4-(4,6-dimethoxy-1,3,5-triazin-2-yl)-4-methylmorpholinium. In the second cross-linking step, the ester formed in the first step may be exposed to a disulfide crosslinking agent. For instance, the ester may be exposed to cystamine (2,2'-dithiobis(ethylamine)). Following the two steps, the biological particle may be surrounded by polyacrylamide strands linked together by disulfide bridges. In this manner, the biological particle may be encased inside of or comprise a gel or matrix (e.g., polymer matrix) to form a “cell bead.” A cell bead can contain biological particles (e.g., a cell) or macromolecular constituents (e.g., RNA, DNA, proteins, etc.) of biological particles. A cell bead may include a single cell or multiple cells, or a derivative of the single cell or multiple cells. For example after lysing and washing the cells, inhibitory components from cell lysates can be washed away and the macromolecular constituents can be bound as cell beads. Systems and methods disclosed herein can be applicable to both cell beads (and/or droplets or other partitions) containing biological particles and cell beads (and/or droplets or other partitions) containing macromolecular constituents of biological particles.

Encapsulated biological particles can provide certain potential advantages of being more storable and more portable than droplet-based partitioned biological particles. Furthermore, in some cases, it may be desirable to allow biological particles to incubate for a select period of time before analysis, such as in order to characterize changes in such biological particles over time, either in the presence or absence of different stimuli. In such cases, encapsulation may allow for longer incubation than partitioning in emulsion droplets, although in some cases, droplet partitioned biological particles may also be incubated for different periods of time, e.g., at least 10 seconds, at least 30 seconds, at least 1 minute, at least 5 minutes, at least 10 minutes, at least 30 minutes, at least 1 hour, at least 2 hours, at least 5 hours, or at least 10 hours or more. The encapsulation of biological particles may constitute the partitioning of the biological particles into which other reagents are co-partitioned. Alternatively or in addition, encapsulated biological particles may be readily deposited into other partitions (e.g., droplets) as described above.

Beads

A partition may comprise one or more unique identifiers, such as barcodes. Barcodes may be previously, subsequently or concurrently delivered to the partitions that hold the compartmentalized or partitioned biological particle. For example, barcodes may be injected into droplets previous to, subsequent to, or concurrently with droplet generation. The delivery of the barcodes to a particular partition allows for the later attribution of the characteristics of the individual biological particle to the particular partition. Barcodes may be delivered, for example on a nucleic acid molecule (e.g., an oligonucleotide), to a partition via any suitable mechanism. Barcoded nucleic acid molecules can be delivered to

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a partition via a microcapsule. A microcapsule, in some instances, can comprise a bead. Beads are described in further detail below.

In some cases, barcoded nucleic acid molecules can be initially associated with the microcapsule and then released from the microcapsule. Release of the barcoded nucleic acid molecules can be passive (e.g., by diffusion out of the microcapsule). In addition or alternatively, release from the microcapsule can be upon application of a stimulus which allows the barcoded nucleic acid molecules to dissociate or to be released from the microcapsule. Such stimulus may disrupt the microcapsule, an interaction that couples the barcoded nucleic acid molecules to or within the microcapsule, or both. Such stimulus can include, for example, a thermal stimulus, photo-stimulus, chemical stimulus (e.g., change in pH or use of a reducing agent(s)), a mechanical stimulus, a radiation stimulus; a biological stimulus (e.g., enzyme), or any combination thereof.

FIG. 2 shows an example of a microfluidic channel structure **200** for delivering barcode carrying beads to droplets. The channel structure **200** can include channel segments **201**, **202**, **204**, **206** and **208** communicating at a channel junction **210**. In operation, the channel segment **201** may transport an aqueous fluid **212** that includes a plurality of beads **214** (e.g., with nucleic acid molecules, oligonucleotides, molecular tags) along the channel segment **201** into junction **210**. The plurality of beads **214** may be sourced from a suspension of beads. For example, the channel segment **201** may be connected to a reservoir comprising an aqueous suspension of beads **214**. The channel segment **202** may transport the aqueous fluid **212** that includes a plurality of biological particles **216** along the channel segment **202** into junction **210**. The plurality of biological particles **216** may be sourced from a suspension of biological particles. For example, the channel segment **202** may be connected to a reservoir comprising an aqueous suspension of biological particles **216**. In some instances, the aqueous fluid **212** in either the first channel segment **201** or the second channel segment **202**, or in both segments, can include one or more reagents, as further described below. A second fluid **218** that is immiscible with the aqueous fluid **212** (e.g., oil) can be delivered to the junction **210** from each of channel segments **204** and **206**. Upon meeting of the aqueous fluid **212** from each of channel segments **201** and **202** and the second fluid **218** from each of channel segments **204** and **206** at the channel junction **210**, the aqueous fluid **212** can be partitioned as discrete droplets **220** in the second fluid **218** and flow away from the junction **210** along channel segment **208**. The channel segment **208** may deliver the discrete droplets to an outlet reservoir fluidly coupled to the channel segment **208**, where they may be harvested.

As an alternative, the channel segments **201** and **202** may meet at another junction upstream of the junction **210**. At such junction, beads and biological particles may form a mixture that is directed along another channel to the junction **210** to yield droplets **220**. The mixture may provide the beads and biological particles in an alternating fashion, such that, for example, a droplet comprises a single bead and a single biological particle.

Beads, biological particles and droplets may flow along channels at substantially regular flow profiles (e.g., at regular flow rates). Such regular flow profiles may permit a droplet to include a single bead and a single biological particle. Such regular flow profiles may permit the droplets to have an occupancy (e.g., droplets having beads and biological particles) greater than 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or 95%. Such regular flow

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profiles and devices that may be used to provide such regular flow profiles are provided in, for example, U.S. Patent Publication No. 2015/0292988, which is entirely incorporated herein by reference.

The second fluid **218** can comprise an oil, such as a fluorinated oil, that includes a fluorosurfactant for stabilizing the resulting droplets, for example, inhibiting subsequent coalescence of the resulting droplets **220**.

A discrete droplet that is generated may include an individual biological particle **216**. A discrete droplet that is generated may include a barcode or other reagent carrying bead **214**. A discrete droplet generated may include both an individual biological particle and a barcode carrying bead, such as droplets **220**. In some instances, a discrete droplet may include more than one individual biological particle or no biological particle. In some instances, a discrete droplet may include more than one bead or no bead. A discrete droplet may be unoccupied (e.g., no beads, no biological particles).

Beneficially, a discrete droplet partitioning a biological particle and a barcode carrying bead may effectively allow the attribution of the barcode to macromolecular constituents of the biological particle within the partition. The contents of a partition may remain discrete from the contents of other partitions.

As will be appreciated, the channel segments described herein may be coupled to any of a variety of different fluid sources or receiving components, including reservoirs, tubing, manifolds, or fluidic components of other systems. As will be appreciated, the microfluidic channel structure **200** may have other geometries. For example, a microfluidic channel structure can have more than one channel junctions. For example, a microfluidic channel structure can have 2, 3, 4, or 5 channel segments each carrying beads that meet at a channel junction. Fluid may be directed flow along one or more channels or reservoirs via one or more fluid flow units. A fluid flow unit can comprise compressors (e.g., providing positive pressure), pumps (e.g., providing negative pressure), actuators, and the like to control flow of the fluid. Fluid may also or otherwise be controlled via applied pressure differentials, centrifugal force, electrokinetic pumping, vacuum, capillary or gravity flow, or the like.

A bead may be porous, non-porous, solid, semi-solid, semi-fluidic, fluidic, and/or a combination thereof. In some instances, a bead may be dissolvable, disruptable, and/or degradable. In some cases, a bead may not be degradable. In some cases, the bead may be a gel bead. A gel bead may be a hydrogel bead. A gel bead may be formed from molecular precursors, such as a polymeric or monomeric species. A semi-solid bead may be a liposomal bead. Solid beads may comprise metals including iron oxide, gold, and silver. In some cases, the bead may be a silica bead. In some cases, the bead can be rigid. In other cases, the bead may be flexible and/or compressible.

A bead may be of any suitable shape. Examples of bead shapes include, but are not limited to, spherical, non-spherical, oval, oblong, amorphous, circular, cylindrical, and variations thereof.

Beads may be of uniform size or heterogeneous size. In some cases, the diameter of a bead may be at least about 10 nanometers (nm), 100 nm, 500 nm, 1 micrometer (μm), 5 μm, 10 μm, 20 μm, 30 μm, 40 μm, 50 μm, 60 μm, 70 μm, 80 μm, 90 μm, 100 μm, 250 μm, 500 μm, 1 mm, or greater. In some cases, a bead may have a diameter of less than about 10 nm, 100 nm, 500 nm, 1 μm, 5 μm, 10 μm, 20 μm, 30 μm, 40 μm, 50 μm, 60 μm, 70 μm, 80 μm, 90 μm, 100 μm, 250 μm, 500 μm, 1 mm, or less. In some cases, a bead may have

a diameter in the range of about 40-75 μm , 30-75 μm , 20-75 μm , 40-85 μm , 40-95 μm , 20-100 μm , 10-100 μm , 1-100 μm , 20-250 μm , or 20-500 μm .

In certain aspects, beads can be provided as a population or plurality of beads having a relatively monodisperse size distribution. Where it may be desirable to provide relatively consistent amounts of reagents within partitions, maintaining relatively consistent bead characteristics, such as size, can contribute to the overall consistency. In particular, the beads described herein may have size distributions that have a coefficient of variation in their cross-sectional dimensions of less than 50%, less than 40%, less than 30%, less than 20%, and in some cases less than 15%, less than 10%, less than 5%, or less.

A bead may comprise natural and/or synthetic materials. For example, a bead can comprise a natural polymer, a synthetic polymer or both natural and synthetic polymers. Examples of natural polymers include proteins and sugars such as deoxyribonucleic acid, rubber, cellulose, starch (e.g., amylose, amylopectin), proteins, enzymes, polysaccharides, silks, polyhydroxyalkanoates, chitosan, dextran, collagen, carrageenan, ispaghula, acacia, agar, gelatin, shellac, sterculia gum, xanthan gum, Corn sugar gum, guar gum, gum karaya, agarose, alginic acid, alginate, or natural polymers thereof. Examples of synthetic polymers include acrylics, nylons, silicones, spandex, viscose rayon, polycarboxylic acids, polyvinyl acetate, polyacrylamide, polyacrylate, polyethylene glycol, polyurethanes, polylactic acid, silica, polystyrene, polyacrylonitrile, polybutadiene, polycarbonate, polyethylene, polyethylene terephthalate, poly(chlorotrifluoroethylene), poly(ethylene oxide), poly(ethylene terephthalate), polyethylene, polyisobutylene, poly(methyl methacrylate), poly(oxymethylene), polyformaldehyde, polypropylene, polystyrene, poly(tetrafluoroethylene), poly(vinyl acetate), poly(vinyl alcohol), poly(vinyl chloride), poly(vinylidene dichloride), poly(vinylidene difluoride), poly(vinyl fluoride) and/or combinations (e.g., co-polymers) thereof. Beads may also be formed from materials other than polymers, including lipids, micelles, ceramics, glass-ceramics, material composites, metals, other inorganic materials, and others.

In some instances, the bead may contain molecular precursors (e.g., monomers or polymers), which may form a polymer network via polymerization of the molecular precursors. In some cases, a precursor may be an already polymerized species capable of undergoing further polymerization via, for example, a chemical cross-linkage. In some cases, a precursor can comprise one or more of an acrylamide or a methacrylamide monomer, oligomer, or polymer. In some cases, the bead may comprise prepolymers, which are oligomers capable of further polymerization. For example, polyurethane beads may be prepared using prepolymers. In some cases, the bead may contain individual polymers that may be further polymerized together. In some cases, beads may be generated via polymerization of different precursors, such that they comprise mixed polymers, co-polymers, and/or block co-polymers. In some cases, the bead may comprise covalent or ionic bonds between polymeric precursors (e.g., monomers, oligomers, linear polymers), nucleic acid molecules (e.g., oligonucleotides), primers, and other entities. In some cases, the covalent bonds can be carbon-carbon bonds, thioether bonds, or carbon-heteroatom bonds.

Cross-linking may be permanent or reversible, depending upon the particular cross-linker used. Reversible cross-linking may allow for the polymer to linearize or dissociate under appropriate conditions. In some cases, reversible

cross-linking may also allow for reversible attachment of a material bound to the surface of a bead. In some cases, a cross-linker may form disulfide linkages. In some cases, the chemical cross-linker forming disulfide linkages may be cystamine or a modified cystamine.

In some cases, disulfide linkages can be formed between molecular precursor units (e.g., monomers, oligomers, or linear polymers) or precursors incorporated into a bead and nucleic acid molecules (e.g., oligonucleotides). Cystamine (including modified cystamines), for example, is an organic agent comprising a disulfide bond that may be used as a crosslinker agent between individual monomeric or polymeric precursors of a bead. Polyacrylamide may be polymerized in the presence of cystamine or a species comprising cystamine (e.g., a modified cystamine) to generate polyacrylamide gel beads comprising disulfide linkages (e.g., chemically degradable beads comprising chemically-reducible cross-linkers). The disulfide linkages may permit the bead to be degraded (or dissolved) upon exposure of the bead to a reducing agent.

In some cases, chitosan, a linear polysaccharide polymer, may be crosslinked with glutaraldehyde via hydrophilic chains to form a bead. Crosslinking of chitosan polymers may be achieved by chemical reactions that are initiated by heat, pressure, change in pH, and/or radiation.

In some cases, a bead may comprise an acrydite moiety, which in certain aspects may be used to attach one or more nucleic acid molecules (e.g., barcode sequence, barcoded nucleic acid molecule, barcoded oligonucleotide, primer, or other oligonucleotide) to the bead. In some cases, an acrydite moiety can refer to an acrydite analogue generated from the reaction of acrydite with one or more species, such as, the reaction of acrydite with other monomers and cross-linkers during a polymerization reaction. Acrydite moieties may be modified to form chemical bonds with a species to be attached, such as a nucleic acid molecule (e.g., barcode sequence, barcoded nucleic acid molecule, barcoded oligonucleotide, primer, or other oligonucleotide). Acrydite moieties may be modified with thiol groups capable of forming a disulfide bond or may be modified with groups already comprising a disulfide bond. The thiol or disulfide (via disulfide exchange) may be used as an anchor point for a species to be attached or another part of the acrydite moiety may be used for attachment. In some cases, attachment can be reversible, such that when the disulfide bond is broken (e.g., in the presence of a reducing agent), the attached species is released from the bead. In other cases, an acrydite moiety can comprise a reactive hydroxyl group that may be used for attachment.

Functionalization of beads for attachment of nucleic acid molecules (e.g., oligonucleotides) may be achieved through a wide range of different approaches, including activation of chemical groups within a polymer, incorporation of active or activatable functional groups in the polymer structure, or attachment at the pre-polymer or monomer stage in bead production.

For example, precursors (e.g., monomers, cross-linkers) that are polymerized to form a bead may comprise acrydite moieties, such that when a bead is generated, the bead also comprises acrydite moieties. The acrydite moieties can be attached to a nucleic acid molecule (e.g., oligonucleotide), which may include a priming sequence (e.g., a primer for amplifying target nucleic acids, random primer, primer sequence for messenger RNA) and/or one or more barcode sequences. The one more barcode sequences may include sequences that are the same for all nucleic acid molecules coupled to a given bead and/or sequences that are different

across all nucleic acid molecules coupled to the given bead. The nucleic acid molecule may be incorporated into the bead.

In some cases, the nucleic acid molecule can comprise a functional sequence, for example, for attachment to a sequencing flow cell, such as, for example, a P5 sequence for Illumina® sequencing. In some cases, the nucleic acid molecule or derivative thereof (e.g., oligonucleotide or polynucleotide generated from the nucleic acid molecule) can comprise another functional sequence, such as, for example, a P7 sequence for attachment to a sequencing flow cell for Illumina sequencing. In some cases, the nucleic acid molecule can comprise a barcode sequence. In some cases, the primer can further comprise a unique molecular identifier (UMI). In some cases, the primer can comprise an R1 primer sequence for Illumina sequencing. In some cases, the primer can comprise an R2 primer sequence for Illumina sequencing. Examples of such nucleic acid molecules (e.g., oligonucleotides, polynucleotides, etc.) and uses thereof, as may be used with compositions, devices, methods and systems of the present disclosure, are provided in U.S. Patent Pub. Nos. 2014/0378345 and 2015/0376609, each of which is entirely incorporated herein by reference.

FIG. 8 illustrates an example of a barcode carrying bead. A nucleic acid molecule **802**, such as an oligonucleotide, can be coupled to a bead **804** by a releasable linkage **806**, such as, for example, a disulfide linker. The same bead **804** may be coupled (e.g., via releasable linkage) to one or more other nucleic acid molecules **818**, **820**. The nucleic acid molecule **802** may be or comprise a barcode. As noted elsewhere herein, the structure of the barcode may comprise a number of sequence elements. The nucleic acid molecule **802** may comprise a functional sequence **808** that may be used in subsequent processing. For example, the functional sequence **808** may include one or more of a sequencer specific flow cell attachment sequence (e.g., a P5 sequence for Illumina® sequencing systems) and a sequencing primer sequence (e.g., a R1 primer for Illumina® sequencing systems). The nucleic acid molecule **802** may comprise a barcode sequence **810** for use in barcoding the sample (e.g., DNA, RNA, protein, etc.). In some cases, the barcode sequence **810** can be bead-specific such that the barcode sequence **810** is common to all nucleic acid molecules (e.g., including nucleic acid molecule **802**) coupled to the same bead **804**. Alternatively or in addition, the barcode sequence **810** can be partition-specific such that the barcode sequence **810** is common to all nucleic acid molecules coupled to one or more beads that are partitioned into the same partition. The nucleic acid molecule **802** may comprise a specific priming sequence **812**, such as an mRNA specific priming sequence (e.g., poly-T sequence), a targeted priming sequence, and/or a random priming sequence. The nucleic acid molecule **802** may comprise an anchoring sequence **814** to ensure that the specific priming sequence **812** hybridizes at the sequence end (e.g., of the mRNA). For example, the anchoring sequence **814** can include a random short sequence of nucleotides, such as a 1-mer, 2-mer, 3-mer or longer sequence, which can ensure that a poly-T segment is more likely to hybridize at the sequence end of the poly-A tail of the mRNA.

The nucleic acid molecule **802** may comprise a unique molecular identifying sequence **816** (e.g., unique molecular identifier (UMI)). In some cases, the unique molecular identifying sequence **816** may comprise from about 5 to about 8 nucleotides. Alternatively, the unique molecular identifying sequence **816** may compress less than about 5 or more than about 8 nucleotides. The unique molecular identifying

sequence **816** may be a unique sequence that varies across individual nucleic acid molecules (e.g., **802**, **818**, **820**, etc.) coupled to a single bead (e.g., bead **804**). In some cases, the unique molecular identifying sequence **816** may be a random sequence (e.g., such as a random N-mer sequence). For example, the UMI may provide a unique identifier of the starting mRNA molecule that was captured, in order to allow quantitation of the number of original expressed RNA. As will be appreciated, although FIG. 8 shows three nucleic acid molecules **802**, **818**, **820** coupled to the surface of the bead **804**, an individual bead may be coupled to any number of individual nucleic acid molecules, for example, from one to tens to hundreds of thousands or even millions of individual nucleic acid molecules. The respective barcodes for the individual nucleic acid molecules can comprise both common sequence segments or relatively common sequence segments (e.g., **808**, **810**, **812**, etc.) and variable or unique sequence segments (e.g., **816**) between different individual nucleic acid molecules coupled to the same bead.

In operation, a biological particle (e.g., cell, DNA, RNA, etc.) can be co-partitioned along with a barcode bearing bead **804**. The barcoded nucleic acid molecules **802**, **818**, **820** can be released from the bead **804** in the partition. By way of example, in the context of analyzing sample RNA, the poly-T segment (e.g., **812**) of one of the released nucleic acid molecules (e.g., **802**) can hybridize to the poly-A tail of a mRNA molecule. Reverse transcription may result in a cDNA transcript of the mRNA, but which transcript includes each of the sequence segments **808**, **810**, **816** of the nucleic acid molecule **802**. Because the nucleic acid molecule **802** comprises an anchoring sequence **814**, it will more likely hybridize to and prime reverse transcription at the sequence end of the poly-A tail of the mRNA. Within any given partition, all of the cDNA transcripts of the individual mRNA molecules may include a common barcode sequence segment **810**. However, the transcripts made from the different mRNA molecules within a given partition may vary at the unique molecular identifying sequence **812** segment (e.g., UMI segment). Beneficially, even following any subsequent amplification of the contents of a given partition, the number of different UMIs can be indicative of the quantity of mRNA originating from a given partition, and thus from the biological particle (e.g., cell). As noted above, the transcripts can be amplified, cleaned up and sequenced to identify the sequence of the cDNA transcript of the mRNA, as well as to sequence the barcode segment and the UMI segment. While a poly-T primer sequence is described, other targeted or random priming sequences may also be used in priming the reverse transcription reaction. Likewise, although described as releasing the barcoded oligonucleotides into the partition, in some cases, the nucleic acid molecules bound to the bead (e.g., gel bead) may be used to hybridize and capture the mRNA on the solid phase of the bead, for example, in order to facilitate the separation of the RNA from other cell contents.

In some cases, precursors comprising a functional group that is reactive or capable of being activated such that it becomes reactive can be polymerized with other precursors to generate gel beads comprising the activated or activatable functional group. The functional group may then be used to attach additional species (e.g., disulfide linkers, primers, other oligonucleotides, etc.) to the gel beads. For example, some precursors comprising a carboxylic acid (COOH) group can co-polymerize with other precursors to form a gel bead that also comprises a COOH functional group. In some cases, acrylic acid (a species comprising free COOH

groups), acrylamide, and bis(acryloyl)cystamine can be copolymerized together to generate a gel bead comprising free COOH groups. The COOH groups of the gel bead can be activated (e.g., via 1-Ethyl-3-(3-dimethylaminopropyl)carbodiimide (EDC) and N-Hydroxysuccinimide (NHS) or 4-(4,6-Dimethoxy-1,3,5-triazin-2-yl)-4-methylmorpholinium chloride (DMTMM)) such that they are reactive (e.g., reactive to amine functional groups where EDC/NHS or DMTMM are used for activation). The activated COOH groups can then react with an appropriate species (e.g., a species comprising an amine functional group where the carboxylic acid groups are activated to be reactive with an amine functional group) comprising a moiety to be linked to the bead.

Beads comprising disulfide linkages in their polymeric network may be functionalized with additional species via reduction of some of the disulfide linkages to free thiols. The disulfide linkages may be reduced via, for example, the action of a reducing agent (e.g., DTT, TCEP, etc.) to generate free thiol groups, without dissolution of the bead. Free thiols of the beads can then react with free thiols of a species or a species comprising another disulfide bond (e.g., via thiol-disulfide exchange) such that the species can be linked to the beads (e.g., via a generated disulfide bond). In some cases, free thiols of the beads may react with any other suitable group. For example, free thiols of the beads may react with species comprising an acrydite moiety. The free thiol groups of the beads can react with the acrydite via Michael addition chemistry, such that the species comprising the acrydite is linked to the bead. In some cases, uncontrolled reactions can be prevented by inclusion of a thiol capping agent such as N-ethylmaleimide or iodoacetate.

Activation of disulfide linkages within a bead can be controlled such that only a small number of disulfide linkages are activated. Control may be exerted, for example, by controlling the concentration of a reducing agent used to generate free thiol groups and/or concentration of reagents used to form disulfide bonds in bead polymerization. In some cases, a low concentration (e.g., molecules of reducing agent:gel bead ratios of less than or equal to about 1:100,000,000,000, less than or equal to about 1:10,000,000,000, less than or equal to about 1:1,000,000,000, less than or equal to about 1:100,000,000, less than or equal to about 1:10,000,000, less than or equal to about 1:1,000,000, less than or equal to about 1:100,000, less than or equal to about 1:10,000) of reducing agent may be used for reduction. Controlling the number of disulfide linkages that are reduced to free thiols may be useful in ensuring bead structural integrity during functionalization. In some cases, optically-active agents, such as fluorescent dyes may be coupled to beads via free thiol groups of the beads and used to quantify the number of free thiols present in a bead and/or track a bead.

In some cases, addition of moieties to a gel bead after gel bead formation may be advantageous. For example, addition of an oligonucleotide (e.g., barcoded oligonucleotide) after gel bead formation may avoid loss of the species during chain transfer termination that can occur during polymerization. Moreover, smaller precursors (e.g., monomers or cross linkers that do not comprise side chain groups and linked moieties) may be used for polymerization and can be minimally hindered from growing chain ends due to viscous effects. In some cases, functionalization after gel bead synthesis can minimize exposure of species (e.g., oligonucleotides) to be loaded with potentially damaging agents (e.g., free radicals) and/or chemical environments. In some

cases, the generated gel may possess an upper critical solution temperature (UCST) that can permit temperature driven swelling and collapse of a bead. Such functionality may aid in oligonucleotide (e.g., a primer) infiltration into the bead during subsequent functionalization of the bead with the oligonucleotide. Post-production functionalization may also be useful in controlling loading ratios of species in beads, such that, for example, the variability in loading ratio is minimized. Species loading may also be performed in a batch process such that a plurality of beads can be functionalized with the species in a single batch.

A bead injected or otherwise introduced into a partition may comprise releasably, cleavably, or reversibly attached barcodes. A bead injected or otherwise introduced into a partition may comprise activatable barcodes. A bead injected or otherwise introduced into a partition may be degradable, disruptable, or dissolvable beads.

Barcodes can be releasably, cleavably or reversibly attached to the beads such that barcodes can be released or be releasable through cleavage of a linkage between the barcode molecule and the bead, or released through degradation of the underlying bead itself, allowing the barcodes to be accessed or be accessible by other reagents, or both. In non-limiting examples, cleavage may be achieved through reduction of di-sulfide bonds, use of restriction enzymes, photo-activated cleavage, or cleavage via other types of stimuli (e.g., chemical, thermal, pH, enzymatic, etc.) and/or reactions, such as described elsewhere herein. Releasable barcodes may sometimes be referred to as being activatable, in that they are available for reaction once released. Thus, for example, an activatable barcode may be activated by releasing the barcode from a bead (or other suitable type of partition described herein). Other activatable configurations are also envisioned in the context of the described methods and systems.

In addition to, or as an alternative to the cleavable linkages between the beads and the associated molecules, such as barcode containing nucleic acid molecules (e.g., barcoded oligonucleotides), the beads may be degradable, disruptable, or dissolvable spontaneously or upon exposure to one or more stimuli (e.g., temperature changes, pH changes, exposure to particular chemical species or phase, exposure to light, reducing agent, etc.). In some cases, a bead may be dissolvable, such that material components of the beads are solubilized when exposed to a particular chemical species or an environmental change, such as a change temperature or a change in pH. In some cases, a gel bead can be degraded or dissolved at elevated temperature and/or in basic conditions. In some cases, a bead may be thermally degradable such that when the bead is exposed to an appropriate change in temperature (e.g., heat), the bead degrades. Degradation or dissolution of a bead bound to a species (e.g., a nucleic acid molecule, e.g., barcoded oligonucleotide) may result in release of the species from the bead.

As will be appreciated from the above disclosure, the degradation of a bead may refer to the disassociation of a bound or entrained species from a bead, both with and without structurally degrading the physical bead itself. For example, the degradation of the bead may involve cleavage of a cleavable linkage via one or more species and/or methods described elsewhere herein. In another example, entrained species may be released from beads through osmotic pressure differences due to, for example, changing chemical environments. By way of example, alteration of bead pore sizes due to osmotic pressure differences can generally occur without structural degradation of the bead

itself. In some cases, an increase in pore size due to osmotic swelling of a bead can permit the release of entrained species within the bead. In other cases, osmotic shrinking of a bead may cause a bead to better retain an entrained species due to pore size contraction.

A degradable bead may be introduced into a partition, such as a droplet of an emulsion or a well, such that the bead degrades within the partition and any associated species (e.g., oligonucleotides) are released within the droplet when the appropriate stimulus is applied. The free species (e.g., oligonucleotides, nucleic acid molecules) may interact with other reagents contained in the partition. For example, a polyacrylamide bead comprising cystamine and linked, via a disulfide bond, to a barcode sequence, may be combined with a reducing agent within a droplet of a water-in-oil emulsion. Within the droplet, the reducing agent can break the various disulfide bonds, resulting in bead degradation and release of the barcode sequence into the aqueous, inner environment of the droplet. In another example, heating of a droplet comprising a bead-bound barcode sequence in basic solution may also result in bead degradation and release of the attached barcode sequence into the aqueous, inner environment of the droplet.

Any suitable number of molecular tag molecules (e.g., primer, barcoded oligonucleotide) can be associated with a bead such that, upon release from the bead, the molecular tag molecules (e.g., primer, e.g., barcoded oligonucleotide) are present in the partition at a pre-defined concentration. Such pre-defined concentration may be selected to facilitate certain reactions for generating a sequencing library, e.g., amplification, within the partition. In some cases, the pre-defined concentration of the primer can be limited by the process of producing nucleic acid molecule (e.g., oligonucleotide) bearing beads.

In some cases, beads can be non-covalently loaded with one or more reagents. The beads can be non-covalently loaded by, for instance, subjecting the beads to conditions sufficient to swell the beads, allowing sufficient time for the reagents to diffuse into the interiors of the beads, and subjecting the beads to conditions sufficient to de-swell the beads. The swelling of the beads may be accomplished, for instance, by placing the beads in a thermodynamically favorable solvent, subjecting the beads to a higher or lower temperature, subjecting the beads to a higher or lower ion concentration, and/or subjecting the beads to an electric field. The swelling of the beads may be accomplished by various swelling methods. The de-swelling of the beads may be accomplished, for instance, by transferring the beads in a thermodynamically unfavorable solvent, subjecting the beads to lower or high temperatures, subjecting the beads to a lower or higher ion concentration, and/or removing an electric field. The de-swelling of the beads may be accomplished by various de-swelling methods. Transferring the beads may cause pores in the bead to shrink. The shrinking may then hinder reagents within the beads from diffusing out of the interiors of the beads. The hindrance may be due to steric interactions between the reagents and the interiors of the beads. The transfer may be accomplished microfluidically. For instance, the transfer may be achieved by moving the beads from one co-flowing solvent stream to a different co-flowing solvent stream. The swellability and/or pore size of the beads may be adjusted by changing the polymer composition of the bead.

In some cases, an acrydite moiety linked to a precursor, another species linked to a precursor, or a precursor itself can comprise a labile bond, such as chemically, thermally, or photo-sensitive bond e.g., disulfide bond, UV sensitive

bond, or the like. Once acrydite moieties or other moieties comprising a labile bond are incorporated into a bead, the bead may also comprise the labile bond. The labile bond may be, for example, useful in reversibly linking (e.g., covalently linking) species (e.g., barcodes, primers, etc.) to a bead. In some cases, a thermally labile bond may include a nucleic acid hybridization based attachment, e.g., where an oligonucleotide is hybridized to a complementary sequence that is attached to the bead, such that thermal melting of the hybrid releases the oligonucleotide, e.g., a barcode containing sequence, from the bead or microcapsule.

The addition of multiple types of labile bonds to a gel bead may result in the generation of a bead capable of responding to varied stimuli. Each type of labile bond may be sensitive to an associated stimulus (e.g., chemical stimulus, light, temperature, enzymatic, etc.) such that release of species attached to a bead via each labile bond may be controlled by the application of the appropriate stimulus. Such functionality may be useful in controlled release of species from a gel bead. In some cases, another species comprising a labile bond may be linked to a gel bead after gel bead formation via, for example, an activated functional group of the gel bead as described above. As will be appreciated, barcodes that are releasably, cleavably or reversibly attached to the beads described herein include barcodes that are released or releasable through cleavage of a linkage between the barcode molecule and the bead, or that are released through degradation of the underlying bead itself, allowing the barcodes to be accessed or accessible by other reagents, or both.

The barcodes that are releasable as described herein may sometimes be referred to as being activatable, in that they are available for reaction once released. Thus, for example, an activatable barcode may be activated by releasing the barcode from a bead (or other suitable type of partition described herein). Other activatable configurations are also envisioned in the context of the described methods and systems.

In addition to thermally cleavable bonds, disulfide bonds and UV sensitive bonds, other non-limiting examples of labile bonds that may be coupled to a precursor or bead include an ester linkage (e.g., cleavable with an acid, a base, or hydroxylamine), a vicinal diol linkage (e.g., cleavable via sodium periodate), a Diels-Alder linkage (e.g., cleavable via heat), a sulfone linkage (e.g., cleavable via a base), a silyl ether linkage (e.g., cleavable via an acid), a glycosidic linkage (e.g., cleavable via an amylase), a peptide linkage (e.g., cleavable via a protease), or a phosphodiester linkage (e.g., cleavable via a nuclease (e.g., DNAase)). A bond may be cleavable via other nucleic acid molecule targeting enzymes, such as restriction enzymes (e.g., restriction endonucleases), as described further below.

Species may be encapsulated in beads during bead generation (e.g., during polymerization of precursors). Such species may or may not participate in polymerization. Such species may be entered into polymerization reaction mixtures such that generated beads comprise the species upon bead formation. In some cases, such species may be added to the gel beads after formation. Such species may include, for example, nucleic acid molecules (e.g., oligonucleotides), reagents for a nucleic acid amplification reaction (e.g., primers, polymerases, dNTPs, co-factors (e.g., ionic co-factors), buffers) including those described herein, reagents for enzymatic reactions (e.g., enzymes, co-factors, substrates, buffers), reagents for nucleic acid modification reactions such as polymerization, ligation, or digestion, and/or reagents for template preparation (e.g., tagmentation) for

one or more sequencing platforms (e.g., Nextera® for Illumina®). Such species may include one or more enzymes described herein, including without limitation, polymerase, reverse transcriptase, restriction enzymes (e.g., endonuclease), transposase, ligase, proteinase K, DNase, etc. Such species may include one or more reagents described elsewhere herein (e.g., lysis agents, inhibitors, inactivating agents, chelating agents, stimulus). Trapping of such species may be controlled by the polymer network density generated during polymerization of precursors, control of ionic charge within the gel bead (e.g., via ionic species linked to polymerized species), or by the release of other species. Encapsulated species may be released from a bead upon bead degradation and/or by application of a stimulus capable of releasing the species from the bead. Alternatively or in addition, species may be partitioned in a partition (e.g., droplet) during or subsequent to partition formation. Such species may include, without limitation, the abovementioned species that may also be encapsulated in a bead.

A degradable bead may comprise one or more species with a labile bond such that, when the bead/species is exposed to the appropriate stimuli, the bond is broken and the bead degrades. The labile bond may be a chemical bond (e.g., covalent bond, ionic bond) or may be another type of physical interaction (e.g., van der Waals interactions, dipole-dipole interactions, etc.). In some cases, a crosslinker used to generate a bead may comprise a labile bond. Upon exposure to the appropriate conditions, the labile bond can be broken and the bead degraded. For example, upon exposure of a polyacrylamide gel bead comprising cystamine crosslinkers to a reducing agent, the disulfide bonds of the cystamine can be broken and the bead degraded.

A degradable bead may be useful in more quickly releasing an attached species (e.g., a nucleic acid molecule, a barcode sequence, a primer, etc) from the bead when the appropriate stimulus is applied to the bead as compared to a bead that does not degrade. For example, for a species bound to an inner surface of a porous bead or in the case of an encapsulated species, the species may have greater mobility and accessibility to other species in solution upon degradation of the bead. In some cases, a species may also be attached to a degradable bead via a degradable linker (e.g., disulfide linker). The degradable linker may respond to the same stimuli as the degradable bead or the two degradable species may respond to different stimuli. For example, a barcode sequence may be attached, via a disulfide bond, to a polyacrylamide bead comprising cystamine. Upon exposure of the barcoded-bead to a reducing agent, the bead degrades and the barcode sequence is released upon breakage of both the disulfide linkage between the barcode sequence and the bead and the disulfide linkages of the cystamine in the bead.

As will be appreciated from the above disclosure, while referred to as degradation of a bead, in many instances as noted above, that degradation may refer to the disassociation of a bound or entrained species from a bead, both with and without structurally degrading the physical bead itself. For example, entrained species may be released from beads through osmotic pressure differences due to, for example, changing chemical environments. By way of example, alteration of bead pore sizes due to osmotic pressure differences can generally occur without structural degradation of the bead itself. In some cases, an increase in pore size due to osmotic swelling of a bead can permit the release of entrained species within the bead. In other cases, osmotic shrinking of a bead may cause a bead to better retain an entrained species due to pore size contraction.

Where degradable beads are provided, it may be beneficial to avoid exposing such beads to the stimulus or stimuli that cause such degradation prior to a given time, in order to, for example, avoid premature bead degradation and issues that arise from such degradation, including for example poor flow characteristics and aggregation. By way of example, where beads comprise reducible cross-linking groups, such as disulfide groups, it will be desirable to avoid contacting such beads with reducing agents, e.g., DTT or other disulfide cleaving reagents. In such cases, treatment to the beads described herein will, in some cases be provided free of reducing agents, such as DTT. Because reducing agents are often provided in commercial enzyme preparations, it may be desirable to provide reducing agent free (or DTT free) enzyme preparations in treating the beads described herein. Examples of such enzymes include, e.g., polymerase enzyme preparations, reverse transcriptase enzyme preparations, ligase enzyme preparations, as well as many other enzyme preparations that may be used to treat the beads described herein. The terms “reducing agent free” or “DTT free” preparations can refer to a preparation having less than about $1/10$, less than about $1/50$, or even less than about $1/100$ of the lower ranges for such materials used in degrading the beads. For example, for DTT, the reducing agent free preparation can have less than about 0.01 millimolar (mM), 0.005 mM, 0.001 mM DTT, 0.0005 mM DTT, or even less than about 0.0001 mM DTT. In many cases, the amount of DTT can be undetectable.

Numerous chemical triggers may be used to trigger the degradation of beads. Examples of these chemical changes may include, but are not limited to pH-mediated changes to the integrity of a component within the bead, degradation of a component of a bead via cleavage of cross-linked bonds, and depolymerization of a component of a bead.

In some embodiments, a bead may be formed from materials that comprise degradable chemical crosslinkers, such as BAC or cystamine. Degradation of such degradable crosslinkers may be accomplished through a number of mechanisms. In some examples, a bead may be contacted with a chemical degrading agent that may induce oxidation, reduction or other chemical changes. For example, a chemical degrading agent may be a reducing agent, such as dithiothreitol (DTT). Additional examples of reducing agents may include β -mercaptoethanol, (2S)-2-amino-1,4-dimercaptobutane (dithiobutylamine or DTBA), tris(2-carboxyethyl) phosphine (TCEP), or combinations thereof. A reducing agent may degrade the disulfide bonds formed between gel precursors forming the bead, and thus, degrade the bead. In other cases, a change in pH of a solution, such as an increase in pH, may trigger degradation of a bead. In other cases, exposure to an aqueous solution, such as water, may trigger hydrolytic degradation, and thus degradation of the bead. In some cases, any combination of stimuli may trigger degradation of a bead. For example, a change in pH may enable a chemical agent (e.g., DTT) to become an effective reducing agent.

Beads may also be induced to release their contents upon the application of a thermal stimulus. A change in temperature can cause a variety of changes to a bead. For example, heat can cause a solid bead to liquefy. A change in heat may cause melting of a bead such that a portion of the bead degrades. In other cases, heat may increase the internal pressure of the bead components such that the bead ruptures or explodes. Heat may also act upon heat-sensitive polymers used as materials to construct beads.

Any suitable agent may degrade beads. In some embodiments, changes in temperature or pH may be used to degrade

thermo-sensitive or pH-sensitive bonds within beads. In some embodiments, chemical degrading agents may be used to degrade chemical bonds within beads by oxidation, reduction or other chemical changes. For example, a chemical degrading agent may be a reducing agent, such as DTT, wherein DTT may degrade the disulfide bonds formed between a crosslinker and gel precursors, thus degrading the bead. In some embodiments, a reducing agent may be added to degrade the bead, which may or may not cause the bead to release its contents. Examples of reducing agents may include dithiothreitol (DTT), β -mercaptoethanol, (2S)-2-amino-1,4-dimercaptobutane (dithiobutylamine or DTBA), tris(2-carboxyethyl) phosphine (TCEP), or combinations thereof. The reducing agent may be present at a concentration of about 0.1 mM, 0.5 mM, 1 mM, 5 mM, 10 mM. The reducing agent may be present at a concentration of at least about 0.1 mM, 0.5 mM, 1 mM, 5 mM, 10 mM, or greater than 10 mM. The reducing agent may be present at concentration of at most about 10 mM, 5 mM, 1 mM, 0.5 mM, 0.1 mM, or less.

Any suitable number of molecular tag molecules (e.g., primer, barcoded oligonucleotide) can be associated with a bead such that, upon release from the bead, the molecular tag molecules (e.g., primer, e.g., barcoded oligonucleotide) are present in the partition at a pre-defined concentration. Such pre-defined concentration may be selected to facilitate certain reactions for generating a sequencing library, e.g., amplification, within the partition. In some cases, the pre-defined concentration of the primer can be limited by the process of producing oligonucleotide bearing beads.

Although FIG. 1 and FIG. 2 have been described in terms of providing substantially singly occupied partitions, above, in certain cases, it may be desirable to provide multiply occupied partitions, e.g., containing two, three, four or more cells and/or microcapsules (e.g., beads) comprising barcoded nucleic acid molecules (e.g., oligonucleotides) within a single partition. Accordingly, as noted above, the flow characteristics of the biological particle and/or bead containing fluids and partitioning fluids may be controlled to provide for such multiply occupied partitions. In particular, the flow parameters may be controlled to provide a given occupancy rate at greater than about 50% of the partitions, greater than about 75%, and in some cases greater than about 80%, 90%, 95%, or higher.

In some cases, additional microcapsules can be used to deliver additional reagents to a partition. In such cases, it may be advantageous to introduce different beads into a common channel or droplet generation junction, from different bead sources (e.g., containing different associated reagents) through different channel inlets into such common channel or droplet generation junction (e.g., junction 210). In such cases, the flow and frequency of the different beads into the channel or junction may be controlled to provide for a certain ratio of microcapsules from each source, while ensuring a given pairing or combination of such beads into a partition with a given number of biological particles (e.g., one biological particle and one bead per partition).

The partitions described herein may comprise small volumes, for example, less than about 10 microliters (μ L), 5 μ L, 1 μ L, 900 picoliters (pL), 800 pL, 700 pL, 600 pL, 500 pL, 400 pL, 300 pL, 200 pL, 100 pL, 50 pL, 20 pL, 10 pL, 1 pL, 500 nanoliters (nL), 100 nL, 50 nL, or less.

For example, in the case of droplet based partitions, the droplets may have overall volumes that are less than about 1000 pL, 900 pL, 800 pL, 700 pL, 600 pL, 500 pL, 400 pL, 300 pL, 200 pL, 100 pL, 50 pL, 20 pL, 10 pL, 1 pL, or less. Where co-partitioned with microcapsules, it will be appre-

ciated that the sample fluid volume, e.g., including co-partitioned biological particles and/or beads, within the partitions may be less than about 90% of the above described volumes, less than about 80%, less than about 70%, less than about 60%, less than about 50%, less than about 40%, less than about 30%, less than about 20%, or less than about 10% of the above described volumes.

As is described elsewhere herein, partitioning species may generate a population or plurality of partitions. In such cases, any suitable number of partitions can be generated or otherwise provided. For example, at least about 1,000 partitions, at least about 5,000 partitions, at least about 10,000 partitions, at least about 50,000 partitions, at least about 100,000 partitions, at least about 500,000 partitions, at least about 1,000,000 partitions, at least about 5,000,000 partitions, at least about 10,000,000 partitions, at least about 50,000,000 partitions, at least about 100,000,000 partitions, at least about 500,000,000 partitions, at least about 1,000,000,000 partitions, or more partitions can be generated or otherwise provided. Moreover, the plurality of partitions may comprise both unoccupied partitions (e.g., empty partitions) and occupied partitions.

Reagents

In accordance with certain aspects, biological particles may be partitioned along with lysis reagents in order to release the contents of the biological particles within the partition. In such cases, the lysis agents can be contacted with the biological particle suspension concurrently with, or immediately prior to, the introduction of the biological particles into the partitioning junction/droplet generation zone (e.g., junction 210), such as through an additional channel or channels upstream of the channel junction. In accordance with other aspects, additionally or alternatively, biological particles may be partitioned along with other reagents, as will be described further below.

FIG. 3 shows an example of a microfluidic channel structure 300 for co-partitioning biological particles and reagents. The channel structure 300 can include channel segments 301, 302, 304, 306 and 308. Channel segments 301 and 302 communicate at a first channel junction 309. Channel segments 302, 304, 306, and 308 communicate at a second channel junction 310.

In an example operation, the channel segment 301 may transport an aqueous fluid 312 that includes a plurality of biological particles 314 along the channel segment 301 into the second junction 310. As an alternative or in addition to, channel segment 301 may transport beads (e.g., gel beads). The beads may comprise barcode molecules.

For example, the channel segment 301 may be connected to a reservoir comprising an aqueous suspension of biological particles 314. Upstream of, and immediately prior to reaching, the second junction 310, the channel segment 301 may meet the channel segment 302 at the first junction 309. The channel segment 302 may transport a plurality of reagents 315 (e.g., lysis agents) suspended in the aqueous fluid 312 along the channel segment 302 into the first junction 309. For example, the channel segment 302 may be connected to a reservoir comprising the reagents 315. After the first junction 309, the aqueous fluid 312 in the channel segment 301 can carry both the biological particles 314 and the reagents 315 towards the second junction 310. In some instances, the aqueous fluid 312 in the channel segment 301 can include one or more reagents, which can be the same or different reagents as the reagents 315. A second fluid 316 that is immiscible with the aqueous fluid 312 (e.g., oil) can be delivered to the second junction 310 from each of channel segments 304 and 306. Upon meeting of the aqueous fluid

312 from the channel segment 301 and the second fluid 316 from each of channel segments 304 and 306 at the second channel junction 310, the aqueous fluid 312 can be partitioned as discrete droplets 318 in the second fluid 316 and flow away from the second junction 310 along channel segment 308. The channel segment 308 may deliver the discrete droplets 318 to an outlet reservoir fluidly coupled to the channel segment 308, where they may be harvested.

The second fluid 316 can comprise an oil, such as a fluorinated oil, that includes a fluorosurfactant for stabilizing the resulting droplets, for example, inhibiting subsequent coalescence of the resulting droplets 318.

A discrete droplet generated may include an individual biological particle 314 and/or one or more reagents 315. In some instances, a discrete droplet generated may include a barcode carrying bead (not shown), such as via other microfluidics structures described elsewhere herein. In some instances, a discrete droplet may be unoccupied (e.g., no reagents, no biological particles).

Beneficially, when lysis reagents and biological particles are co-partitioned, the lysis reagents can facilitate the release of the contents of the biological particles within the partition. The contents released in a partition may remain discrete from the contents of other partitions.

As will be appreciated, the channel segments described herein may be coupled to any of a variety of different fluid sources or receiving components, including reservoirs, tubing, manifolds, or fluidic components of other systems. As will be appreciated, the microfluidic channel structure 300 may have other geometries. For example, a microfluidic channel structure can have more than two channel junctions. For example, a microfluidic channel structure can have 2, 3, 4, 5 channel segments or more each carrying the same or different types of beads, reagents, and/or biological particles that meet at a channel junction. Fluid flow in each channel segment may be controlled to control the partitioning of the different elements into droplets. Fluid may be directed flow along one or more channels or reservoirs via one or more fluid flow units. A fluid flow unit can comprise compressors (e.g., providing positive pressure), pumps (e.g., providing negative pressure), actuators, and the like to control flow of the fluid. Fluid may also or otherwise be controlled via applied pressure differentials, centrifugal force, electrokinetic pumping, vacuum, capillary or gravity flow, or the like.

Examples of lysis agents include bioactive reagents, such as lysis enzymes that are used for lysis of different cell types, e.g., gram positive or negative bacteria, plants, yeast, mammalian, etc., such as lysozymes, achromopeptidase, lyso-staphin, labiase, kitalase, lyticase, and a variety of other lysis enzymes available from, e.g., Sigma-Aldrich, Inc. (St Louis, MO), as well as other commercially available lysis enzymes. Other lysis agents may additionally or alternatively be co-partitioned with the biological particles to cause the release of the biological particles's contents into the partitions. For example, in some cases, surfactant-based lysis solutions may be used to lyse cells, although these may be less desirable for emulsion based systems where the surfactants can interfere with stable emulsions. In some cases, lysis solutions may include non-ionic surfactants such as, for example, TritonX-100 and Tween 20. In some cases, lysis solutions may include ionic surfactants such as, for example, sarcosyl and sodium dodecyl sulfate (SDS). Electroporation, thermal, acoustic or mechanical cellular disruption may also be used in certain cases, e.g., non-emulsion based partitioning such as encapsulation of biological particles that may be in addition to or in place of droplet partitioning, where any

pore size of the encapsulate is sufficiently small to retain nucleic acid fragments of a given size, following cellular disruption.

Alternatively or in addition to the lysis agents co-partitioned with the biological particles described above, other reagents can also be co-partitioned with the biological particles, including, for example, DNase and RNase inactivating agents or inhibitors, such as proteinase K, chelating agents, such as EDTA, and other reagents employed in removing or otherwise reducing negative activity or impact of different cell lysate components on subsequent processing of nucleic acids. In addition, in the case of encapsulated biological particles, the biological particles may be exposed to an appropriate stimulus to release the biological particles or their contents from a co-partitioned microcapsule. For example, in some cases, a chemical stimulus may be co-partitioned along with an encapsulated biological particle to allow for the degradation of the microcapsule and release of the cell or its contents into the larger partition. In some cases, this stimulus may be the same as the stimulus described elsewhere herein for release of nucleic acid molecules (e.g., oligonucleotides) from their respective microcapsule (e.g., bead). In alternative aspects, this may be a different and non-overlapping stimulus, in order to allow an encapsulated biological particle to be released into a partition at a different time from the release of nucleic acid molecules into the same partition.

Additional reagents may also be co-partitioned with the biological particles, such as endonucleases to fragment a biological particle's DNA, DNA polymerase enzymes and dNTPs used to amplify the biological particle's nucleic acid fragments and to attach the barcode molecular tags to the amplified fragments. Other enzymes may be co-partitioned, including without limitation, polymerase, transposase, ligase, proteinase K, DNase, etc. Additional reagents may also include reverse transcriptase enzymes, including enzymes with terminal transferase activity, primers and oligonucleotides, and switch oligonucleotides (also referred to herein as "switch oligos" or "template switching oligonucleotides") which can be used for template switching. In some cases, template switching can be used to increase the length of a cDNA. In some cases, template switching can be used to append a predefined nucleic acid sequence to the cDNA. In an example of template switching, cDNA can be generated from reverse transcription of a template, e.g., cellular mRNA, where a reverse transcriptase with terminal transferase activity can add additional nucleotides, e.g., polyC, to the cDNA in a template independent manner. Switch oligos can include sequences complementary to the additional nucleotides, e.g., polyG. The additional nucleotides (e.g., polyC) on the cDNA can hybridize to the additional nucleotides (e.g., polyG) on the switch oligo, whereby the switch oligo can be used by the reverse transcriptase as template to further extend the cDNA. Template switching oligonucleotides may comprise a hybridization region and a template region. The hybridization region can comprise any sequence capable of hybridizing to the target. In some cases, as previously described, the hybridization region comprises a series of G bases to complement the overhanging C bases at the 3' end of a cDNA molecule. The series of G bases may comprise 1 G base, 2 G bases, 3 G bases, 4 G bases, 5 G bases or more than 5 G bases. The template sequence can comprise any sequence to be incorporated into the cDNA. In some cases, the template region comprises at least 1 (e.g., at least 2, 3, 4, 5 or more) tag sequences and/or functional sequences. Switch oligos may comprise deoxyribonucleic acids; ribonucleic acids; modi-

fied nucleic acids including 2-Aminopurine, 2,6-Diaminopurine (2-Amino-dA), inverted dT, 5-Methyl dC, 2'-deoxyInosine, Super T (5-hydroxybutynl-2'-deoxyuridine), Super G (8-aza-7-deazaguanosine), locked nucleic acids (LNAs), unlocked nucleic acids (UNAs, e.g., UNA-A, UNA-U, UNA-C, UNA-G), Iso-dG, Iso-dC, 2' Fluoro bases (e.g., Fluoro C, Fluoro U, Fluoro A, and Fluoro G), or any combination.

In some cases, the length of a switch oligo may be at least about 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249 or 250 nucleotides or longer.

In some cases, the length of a switch oligo may be at most about 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249 or 250 nucleotides.

Once the contents of the cells are released into their respective partitions, the macromolecular components (e.g., macromolecular constituents of biological particles, such as RNA, DNA, or proteins) contained therein may be further processed within the partitions. In accordance with the methods and systems described herein, the macromolecular component contents of individual biological particles can be provided with unique identifiers such that, upon characterization of those macromolecular components they may be attributed as having been derived from the same biological particle or particles. The ability to attribute characteristics to individual biological particles or groups of biological particles is provided by the assignment of unique identifiers specifically to an individual biological particle or groups of biological particles. Unique identifiers, e.g., in the form of nucleic acid barcodes can be assigned or associated with individual biological particles or populations of biological particles, in order to tag or label the biological particle's

macromolecular components (and as a result, its characteristics) with the unique identifiers. These unique identifiers can then be used to attribute the biological particle's components and characteristics to an individual biological particle or group of biological particles.

In some aspects, this is performed by co-partitioning the individual biological particle or groups of biological particles with the unique identifiers, such as described above (with reference to FIG. 2). In some aspects, the unique identifiers are provided in the form of nucleic acid molecules (e.g., oligonucleotides) that comprise nucleic acid barcode sequences that may be attached to or otherwise associated with the nucleic acid contents of individual biological particle, or to other components of the biological particle, and particularly to fragments of those nucleic acids. The nucleic acid molecules are partitioned such that as between nucleic acid molecules in a given partition, the nucleic acid barcode sequences contained therein are the same, but as between different partitions, the nucleic acid molecule can, and do have differing barcode sequences, or at least represent a large number of different barcode sequences across all of the partitions in a given analysis. In some aspects, only one nucleic acid barcode sequence can be associated with a given partition, although in some cases, two or more different barcode sequences may be present.

The nucleic acid barcode sequences can include from about 6 to about 20 or more nucleotides within the sequence of the nucleic acid molecules (e.g., oligonucleotides). The nucleic acid barcode sequences can include from about 6 to about 20, 30, 40, 50, 60, 70, 80, 90, 100 or more nucleotides. In some cases, the length of a barcode sequence may be about 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 nucleotides or longer. In some cases, the length of a barcode sequence may be at least about 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 nucleotides or longer. In some cases, the length of a barcode sequence may be at most about 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 nucleotides or shorter. These nucleotides may be completely contiguous, i.e., in a single stretch of adjacent nucleotides, or they may be separated into two or more separate subsequences that are separated by 1 or more nucleotides. In some cases, separated barcode subsequences can be from about 4 to about 16 nucleotides in length. In some cases, the barcode subsequence may be about 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 nucleotides or longer. In some cases, the barcode subsequence may be at least about 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 nucleotides or longer. In some cases, the barcode subsequence may be at most about 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 nucleotides or shorter.

The co-partitioned nucleic acid molecules can also comprise other functional sequences useful in the processing of the nucleic acids from the co-partitioned biological particles. These sequences include, e.g., targeted or random/universal amplification primer sequences for amplifying the genomic DNA from the individual biological particles within the partitions while attaching the associated barcode sequences, sequencing primers or primer recognition sites, hybridization or probing sequences, e.g., for identification of presence of the sequences or for pulling down barcoded nucleic acids, or any of a number of other potential functional sequences. Other mechanisms of co-partitioning oligonucleotides may also be employed, including, e.g., coalescence of two or more droplets, where one droplet contains oligonucleotides, or microdispensing of oligonucleotides into partitions, e.g., droplets within microfluidic systems.

In an example, microcapsules, such as beads, are provided that each include large numbers of the above described

barcoded nucleic acid molecules (e.g., barcoded oligonucleotides) releasably attached to the beads, where all of the nucleic acid molecules attached to a particular bead will include the same nucleic acid barcode sequence, but where a large number of diverse barcode sequences are represented across the population of beads used. In some embodiments, hydrogel beads, e.g., comprising polyacrylamide polymer matrices, are used as a solid support and delivery vehicle for the nucleic acid molecules into the partitions, as they are capable of carrying large numbers of nucleic acid molecules, and may be configured to release those nucleic acid molecules upon exposure to a particular stimulus, as described elsewhere herein. In some cases, the population of beads provides a diverse barcode sequence library that includes at least about 1,000 different barcode sequences, at least about 5,000 different barcode sequences, at least about 10,000 different barcode sequences, at least about 50,000 different barcode sequences, at least about 100,000 different barcode sequences, at least about 1,000,000 different barcode sequences, at least about 5,000,000 different barcode sequences, or at least about 10,000,000 different barcode sequences, or more. Additionally, each bead can be provided with large numbers of nucleic acid (e.g., oligonucleotide) molecules attached. In particular, the number of molecules of nucleic acid molecules including the barcode sequence on an individual bead can be at least about 1,000 nucleic acid molecules, at least about 5,000 nucleic acid molecules, at least about 10,000 nucleic acid molecules, at least about 50,000 nucleic acid molecules, at least about 100,000 nucleic acid molecules, at least about 500,000 nucleic acids, at least about 1,000,000 nucleic acid molecules, at least about 5,000,000 nucleic acid molecules, at least about 10,000,000 nucleic acid molecules, at least about 50,000,000 nucleic acid molecules, at least about 100,000,000 nucleic acid molecules, at least about 250,000,000 nucleic acid molecules and in some cases at least about 1 billion nucleic acid molecules, or more. Nucleic acid molecules of a given bead can include identical (or common) barcode sequences, different barcode sequences, or a combination of both. Nucleic acid molecules of a given bead can include multiple sets of nucleic acid molecules. Nucleic acid molecules of a given set can include identical barcode sequences. The identical barcode sequences can be different from barcode sequences of nucleic acid molecules of another set.

Moreover, when the population of beads is partitioned, the resulting population of partitions can also include a diverse barcode library that includes at least about 1,000 different barcode sequences, at least about 5,000 different barcode sequences, at least about 10,000 different barcode sequences, at least about 50,000 different barcode sequences, at least about 100,000 different barcode sequences, at least about 1,000,000 different barcode sequences, at least about 5,000,000 different barcode sequences, or at least about 10,000,000 different barcode sequences. Additionally, each partition of the population can include at least about 1,000 nucleic acid molecules, at least about 5,000 nucleic acid molecules, at least about 10,000 nucleic acid molecules, at least about 50,000 nucleic acid molecules, at least about 100,000 nucleic acid molecules, at least about 500,000 nucleic acids, at least about 1,000,000 nucleic acid molecules, at least about 5,000,000 nucleic acid molecules, at least about 10,000,000 nucleic acid molecules, at least about 50,000,000 nucleic acid molecules, at least about 100,000,000 nucleic acid molecules, at least about 250,000,000 nucleic acid molecules and in some cases at least about 1 billion nucleic acid molecules.

In some cases, it may be desirable to incorporate multiple different barcodes within a given partition, either attached to a single or multiple beads within the partition. For example, in some cases, a mixed, but known set of barcode sequences may provide greater assurance of identification in the subsequent processing, e.g., by providing a stronger address or attribution of the barcodes to a given partition, as a duplicate or independent confirmation of the output from a given partition.

The nucleic acid molecules (e.g., oligonucleotides) are releasable from the beads upon the application of a particular stimulus to the beads. In some cases, the stimulus may be a photo-stimulus, e.g., through cleavage of a photo-labile linkage that releases the nucleic acid molecules. In other cases, a thermal stimulus may be used, where elevation of the temperature of the beads environment will result in cleavage of a linkage or other release of the nucleic acid molecules from the beads. In still other cases, a chemical stimulus can be used that cleaves a linkage of the nucleic acid molecules to the beads, or otherwise results in release of the nucleic acid molecules from the beads. In one case, such compositions include the polyacrylamide matrices described above for encapsulation of biological particles, and may be degraded for release of the attached nucleic acid molecules through exposure to a reducing agent, such as DTT.

In some aspects, provided are systems and methods for controlled partitioning. Droplet size may be controlled by adjusting certain geometric features in channel architecture (e.g., microfluidics channel architecture). For example, an expansion angle, width, and/or length of a channel may be adjusted to control droplet size.

FIG. 4 shows an example of a microfluidic channel structure for the controlled partitioning of beads into discrete droplets. A channel structure 400 can include a channel segment 402 communicating at a channel junction 406 (or intersection) with a reservoir 404. The reservoir 404 can be a chamber. Any reference to "reservoir," as used herein, can also refer to a "chamber." In operation, an aqueous fluid 408 that includes suspended beads 412 may be transported along the channel segment 402 into the junction 406 to meet a second fluid 410 that is immiscible with the aqueous fluid 408 in the reservoir 404 to create droplets 416, 418 of the aqueous fluid 408 flowing into the reservoir 404. At the junction 406 where the aqueous fluid 408 and the second fluid 410 meet, droplets can form based on factors such as the hydrodynamic forces at the junction 406, flow rates of the two fluids 408, 410, fluid properties, and certain geometric parameters (e.g., w , h_0 , a , etc.) of the channel structure 400. A plurality of droplets can be collected in the reservoir 404 by continuously injecting the aqueous fluid 408 from the channel segment 402 through the junction 406.

A discrete droplet generated may include a bead (e.g., as in occupied droplets 416). Alternatively, a discrete droplet generated may include more than one bead. Alternatively, a discrete droplet generated may not include any beads (e.g., as in unoccupied droplet 418). In some instances, a discrete droplet generated may contain one or more biological particles, as described elsewhere herein. In some instances, a discrete droplet generated may comprise one or more reagents, as described elsewhere herein.

In some instances, the aqueous fluid 408 can have a substantially uniform concentration or frequency of beads 412. The beads 412 can be introduced into the channel segment 402 from a separate channel (not shown in FIG. 4). The frequency of beads 412 in the channel segment 402 may be controlled by controlling the frequency in which the

beads **412** are introduced into the channel segment **402** and/or the relative flow rates of the fluids in the channel segment **402** and the separate channel. In some instances, the beads can be introduced into the channel segment **402** from a plurality of different channels, and the frequency controlled accordingly.

In some instances, the aqueous fluid **408** in the channel segment **402** can comprise biological particles (e.g., described with reference to FIGS. 1 and 2). In some instances, the aqueous fluid **408** can have a substantially uniform concentration or frequency of biological particles. As with the beads, the biological particles can be introduced into the channel segment **402** from a separate channel. The frequency or concentration of the biological particles in the aqueous fluid **408** in the channel segment **402** may be controlled by controlling the frequency in which the biological particles are introduced into the channel segment **402** and/or the relative flow rates of the fluids in the channel segment **402** and the separate channel. In some instances, the biological particles can be introduced into the channel segment **402** from a plurality of different channels, and the frequency controlled accordingly. In some instances, a first separate channel can introduce beads and a second separate channel can introduce biological particles into the channel segment **402**. The first separate channel introducing the beads may be upstream or downstream of the second separate channel introducing the biological particles.

The second fluid **410** can comprise an oil, such as a fluorinated oil, that includes a fluorosurfactant for stabilizing the resulting droplets, for example, inhibiting subsequent coalescence of the resulting droplets.

In some instances, the second fluid **410** may not be subjected to and/or directed to any flow in or out of the reservoir **404**. For example, the second fluid **410** may be substantially stationary in the reservoir **404**. In some instances, the second fluid **410** may be subjected to flow within the reservoir **404**, but not in or out of the reservoir **404**, such as via application of pressure to the reservoir **404** and/or as affected by the incoming flow of the aqueous fluid **408** at the junction **406**. Alternatively, the second fluid **410** may be subjected and/or directed to flow in or out of the reservoir **404**. For example, the reservoir **404** can be a channel directing the second fluid **410** from upstream to downstream, transporting the generated droplets.

The channel structure **400** at or near the junction **406** may have certain geometric features that at least partly determine the sizes of the droplets formed by the channel structure **400**. The channel segment **402** can have a height, h_0 and width, w , at or near the junction **406**. By way of example, the channel segment **402** can comprise a rectangular cross-section that leads to a reservoir **404** having a wider cross-section (such as in width or diameter). Alternatively, the cross-section of the channel segment **402** can be other shapes, such as a circular shape, trapezoidal shape, polygonal shape, or any other shapes. The top and bottom walls of the reservoir **404** at or near the junction **406** can be inclined at an expansion angle, α . The expansion angle, α , allows the tongue (portion of the aqueous fluid **408** leaving channel segment **402** at junction **406** and entering the reservoir **404** before droplet formation) to increase in depth and facilitate decrease in curvature of the intermediately formed droplet. Droplet size may decrease with increasing expansion angle. The resulting droplet radius, R_d , may be predicted by the following equation for the aforementioned geometric parameters of h_0 , w , and α :

$$R_d \approx 0.44 \left(1 + 2.2 \sqrt{\tan \alpha} \frac{w}{h_0} \right) \frac{h_0}{\sqrt{\tan \alpha}}$$

By way of example, for a channel structure with $w=21 \mu\text{m}$, $h=21 \mu\text{m}$, and $\alpha=3^\circ$, the predicted droplet size is $121 \mu\text{m}$. In another example, for a channel structure with $w=25 \mu\text{m}$, $h=25 \mu\text{m}$, and $\alpha=5^\circ$, the predicted droplet size is $123 \mu\text{m}$. In another example, for a channel structure with $w=28 \mu\text{m}$, $h=28 \mu\text{m}$, and $\alpha=7^\circ$, the predicted droplet size is $124 \mu\text{m}$.

In some instances, the expansion angle, α , may be between a range of from about 0.5° to about 4° , from about 0.1° to about 10° , or from about 0° to about 90° . For example, the expansion angle can be at least about 0.01° , 0.1° , 0.2° , 0.3° , 0.4° , 0.5° , 0.6° , 0.7° , 0.8° , 0.9° , 1° , 2° , 3° , 4° , 5° , 6° , 7° , 8° , 9° , 10° , 15° , 20° , 25° , 30° , 35° , 40° , 45° , 50° , 55° , 60° , 65° , 70° , 75° , 80° , 85° , or higher. In some instances, the expansion angle can be at most about 89° , 88° , 87° , 86° , 85° , 84° , 83° , 82° , 81° , 80° , 75° , 70° , 65° , 60° , 55° , 50° , 45° , 40° , 35° , 30° , 25° , 20° , 15° , 10° , 9° , 8° , 7° , 6° , 5° , 4° , 3° , 2° , 1° , 0.1° , 0.01° , or less. In some instances, the width, w , can be between a range of from about $100 \mu\text{m}$ (micrometers) to about $500 \mu\text{m}$. In some instances, the width, w , can be between a range of from about $10 \mu\text{m}$ to about $200 \mu\text{m}$. Alternatively, the width can be less than about $10 \mu\text{m}$. Alternatively, the width can be greater than about $500 \mu\text{m}$. In some instances, the flow rate of the aqueous fluid **408** entering the junction **406** can be between about $0.04 \mu\text{L}/\text{minute}$ ($\mu\text{L}/\text{min}$) and about $40 \mu\text{L}/\text{min}$. In some instances, the flow rate of the aqueous fluid **408** entering the junction **406** can be between about $0.01 \mu\text{L}/\text{minute}$ ($\mu\text{L}/\text{min}$) and about $100 \mu\text{L}/\text{min}$. Alternatively, the flow rate of the aqueous fluid **408** entering the junction **406** can be less than about $0.01 \mu\text{L}/\text{min}$. Alternatively, the flow rate of the aqueous fluid **408** entering the junction **406** can be greater than about $40 \mu\text{L}/\text{min}$, such as $45 \mu\text{L}/\text{min}$, $50 \mu\text{L}/\text{min}$, $55 \mu\text{L}/\text{min}$, $60 \mu\text{L}/\text{min}$, $65 \mu\text{L}/\text{min}$, $70 \mu\text{L}/\text{min}$, $75 \mu\text{L}/\text{min}$, $80 \mu\text{L}/\text{min}$, $85 \mu\text{L}/\text{min}$, $90 \mu\text{L}/\text{min}$, $95 \mu\text{L}/\text{min}$, $100 \mu\text{L}/\text{min}$, $110 \mu\text{L}/\text{min}$, $120 \mu\text{L}/\text{min}$, $130 \mu\text{L}/\text{min}$, $140 \mu\text{L}/\text{min}$, $150 \mu\text{L}/\text{min}$, or greater. At lower flow rates, such as flow rates of about less than or equal to $10 \mu\text{L}/\text{minute}$, the droplet radius may not be dependent on the flow rate of the aqueous fluid **408** entering the junction **406**.

In some instances, at least about 50% of the droplets generated can have uniform size. In some instances, at least about 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or greater of the droplets generated can have uniform size. Alternatively, less than about 50% of the droplets generated can have uniform size.

The throughput of droplet generation can be increased by increasing the points of generation, such as increasing the number of junctions (e.g., junction **406**) between aqueous fluid **408** channel segments (e.g., channel segment **402**) and the reservoir **404**. Alternatively or in addition, the throughput of droplet generation can be increased by increasing the flow rate of the aqueous fluid **408** in the channel segment **402**.

FIG. 5 shows an example of a microfluidic channel structure for increased droplet generation throughput. A microfluidic channel structure **500** can comprise a plurality of channel segments **502** and a reservoir **504**. Each of the plurality of channel segments **502** may be in fluid communication with the reservoir **504**. The channel structure **500** can comprise a plurality of channel junctions **506** between the plurality of channel segments **502** and the reservoir **504**. Each channel junction can be a point of droplet generation.

The channel segment **402** from the channel structure **400** in FIG. **4** and any description to the components thereof may correspond to a given channel segment of the plurality of channel segments **502** in channel structure **500** and any description to the corresponding components thereof. The reservoir **404** from the channel structure **400** and any description to the components thereof may correspond to the reservoir **504** from the channel structure **500** and any description to the corresponding components thereof.

Each channel segment of the plurality of channel segments **502** may comprise an aqueous fluid **508** that includes suspended beads **512**. The reservoir **504** may comprise a second fluid **510** that is immiscible with the aqueous fluid **508**. In some instances, the second fluid **510** may not be subjected to and/or directed to any flow in or out of the reservoir **504**. For example, the second fluid **510** may be substantially stationary in the reservoir **504**. In some instances, the second fluid **510** may be subjected to flow within the reservoir **504**, but not in or out of the reservoir **504**, such as via application of pressure to the reservoir **504** and/or as affected by the incoming flow of the aqueous fluid **508** at the junctions. Alternatively, the second fluid **510** may be subjected and/or directed to flow in or out of the reservoir **504**. For example, the reservoir **504** can be a channel directing the second fluid **510** from upstream to downstream, transporting the generated droplets.

In operation, the aqueous fluid **508** that includes suspended beads **512** may be transported along the plurality of channel segments **502** into the plurality of junctions **506** to meet the second fluid **510** in the reservoir **504** to create droplets **516**, **518**. A droplet may form from each channel segment at each corresponding junction with the reservoir **504**. At the junction where the aqueous fluid **508** and the second fluid **510** meet, droplets can form based on factors such as the hydrodynamic forces at the junction, flow rates of the two fluids **508**, **510**, fluid properties, and certain geometric parameters (e.g., w , h_0 , α , etc.) of the channel structure **500**, as described elsewhere herein. A plurality of droplets can be collected in the reservoir **504** by continuously injecting the aqueous fluid **508** from the plurality of channel segments **502** through the plurality of junctions **506**. Throughput may significantly increase with the parallel channel configuration of channel structure **500**. For example, a channel structure having five inlet channel segments comprising the aqueous fluid **508** may generate droplets five times as frequently than a channel structure having one inlet channel segment, provided that the fluid flow rate in the channel segments are substantially the same. The fluid flow rate in the different inlet channel segments may or may not be substantially the same. A channel structure may have as many parallel channel segments as is practical and allowed for the size of the reservoir. For example, the channel structure may have at least about 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 150, 500, 250, 300, 350, 400, 450, 500, 600, 700, 800, 900, 1000, 1500, 5000 or more parallel or substantially parallel channel segments.

The geometric parameters, w , h_0 , and α , may or may not be uniform for each of the channel segments in the plurality of channel segments **502**. For example, each channel segment may have the same or different widths at or near its respective channel junction with the reservoir **504**. For example, each channel segment may have the same or different height at or near its respective channel junction with the reservoir **504**. In another example, the reservoir **504** may have the same or different expansion angle at the different channel junctions with the plurality of channel

segments **502**. When the geometric parameters are uniform, beneficially, droplet size may also be controlled to be uniform even with the increased throughput. In some instances, when it is desirable to have a different distribution of droplet sizes, the geometric parameters for the plurality of channel segments **502** may be varied accordingly.

In some instances, at least about 50% of the droplets generated can have uniform size. In some instances, at least about 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or greater of the droplets generated can have uniform size. Alternatively, less than about 50% of the droplets generated can have uniform size.

FIG. **6** shows another example of a microfluidic channel structure for increased droplet generation throughput. A microfluidic channel structure **600** can comprise a plurality of channel segments **602** arranged generally circularly around the perimeter of a reservoir **604**. Each of the plurality of channel segments **602** may be in fluid communication with the reservoir **604**. The channel structure **600** can comprise a plurality of channel junctions **606** between the plurality of channel segments **602** and the reservoir **604**. Each channel junction can be a point of droplet generation. The channel segment **402** from the channel structure **400** in FIG. **2** and any description to the components thereof may correspond to a given channel segment of the plurality of channel segments **602** in channel structure **600** and any description to the corresponding components thereof. The reservoir **404** from the channel structure **400** and any description to the components thereof may correspond to the reservoir **604** from the channel structure **600** and any description to the corresponding components thereof.

Each channel segment of the plurality of channel segments **602** may comprise an aqueous fluid **608** that includes suspended beads **612**. The reservoir **604** may comprise a second fluid **610** that is immiscible with the aqueous fluid **608**. In some instances, the second fluid **610** may not be subjected to and/or directed to any flow in or out of the reservoir **604**. For example, the second fluid **610** may be substantially stationary in the reservoir **604**. In some instances, the second fluid **610** may be subjected to flow within the reservoir **604**, but not in or out of the reservoir **604**, such as via application of pressure to the reservoir **604** and/or as affected by the incoming flow of the aqueous fluid **608** at the junctions. Alternatively, the second fluid **610** may be subjected and/or directed to flow in or out of the reservoir **604**. For example, the reservoir **604** can be a channel directing the second fluid **610** from upstream to downstream, transporting the generated droplets.

In operation, the aqueous fluid **608** that includes suspended beads **612** may be transported along the plurality of channel segments **602** into the plurality of junctions **606** to meet the second fluid **610** in the reservoir **604** to create a plurality of droplets **616**. A droplet may form from each channel segment at each corresponding junction with the reservoir **604**. At the junction where the aqueous fluid **608** and the second fluid **610** meet, droplets can form based on factors such as the hydrodynamic forces at the junction, flow rates of the two fluids **608**, **610**, fluid properties, and certain geometric parameters (e.g., widths and heights of the channel segments **602**, expansion angle of the reservoir **604**, etc.) of the channel structure **600**, as described elsewhere herein. A plurality of droplets can be collected in the reservoir **604** by continuously injecting the aqueous fluid **608** from the plurality of channel segments **602** through the plurality of junctions **606**. Throughput may significantly increase with the substantially parallel channel configuration of the channel structure **600**. A channel structure may have as many

substantially parallel channel segments as is practical and allowed for by the size of the reservoir. For example, the channel structure may have at least about 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 150, 200, 250, 300, 350, 400, 450, 500, 600, 700, 800, 900, 1000, 1500, 5000 or more parallel or substantially parallel channel segments. The plurality of channel segments may be substantially evenly spaced apart, for example, around an edge or perimeter of the reservoir. Alternatively, the spacing of the plurality of channel segments may be uneven.

The reservoir **604** may have an expansion angle, α (not shown in FIG. **6**) at or near each channel junction. Each channel segment of the plurality of channel segments **602** may have a width, w , and a height, h_0 , at or near the channel junction. The geometric parameters, w , h_0 , and α , may or may not be uniform for each of the channel segments in the plurality of channel segments **602**. For example, each channel segment may have the same or different widths at or near its respective channel junction with the reservoir **604**. For example, each channel segment may have the same or different height at or near its respective channel junction with the reservoir **604**.

The reservoir **604** may have the same or different expansion angle at the different channel junctions with the plurality of channel segments **602**. For example, a circular reservoir (as shown in FIG. **6**) may have a conical, dome-like, or hemispherical ceiling (e.g., top wall) to provide the same or substantially same expansion angle for each channel segments **602** at or near the plurality of channel junctions **606**. When the geometric parameters are uniform, beneficially, resulting droplet size may be controlled to be uniform even with the increased throughput. In some instances, when it is desirable to have a different distribution of droplet sizes, the geometric parameters for the plurality of channel segments **602** may be varied accordingly.

In some instances, at least about 50% of the droplets generated can have uniform size. In some instances, at least about 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or greater of the droplets generated can have uniform size. Alternatively, less than about 50% of the droplets generated can have uniform size. The beads and/or biological particle injected into the droplets may or may not have uniform size.

FIG. **7A** shows a cross-section view of another example of a microfluidic channel structure with a geometric feature for controlled partitioning. A channel structure **700** can include a channel segment **702** communicating at a channel junction **706** (or intersection) with a reservoir **704**. In some instances, the channel structure **700** and one or more of its components can correspond to the channel structure **100** and one or more of its components. FIG. **7B** shows a perspective view of the channel structure **700** of FIG. **7A**.

An aqueous fluid **712** comprising a plurality of particles **716** may be transported along the channel segment **702** into the junction **706** to meet a second fluid **714** (e.g., oil, etc.) that is immiscible with the aqueous fluid **712** in the reservoir **704** to create droplets **720** of the aqueous fluid **712** flowing into the reservoir **704**. At the junction **706** where the aqueous fluid **712** and the second fluid **714** meet, droplets can form based on factors such as the hydrodynamic forces at the junction **706**, relative flow rates of the two fluids **712**, **714**, fluid properties, and certain geometric parameters (e.g., Δh , etc.) of the channel structure **700**. A plurality of droplets can be collected in the reservoir **704** by continuously injecting the aqueous fluid **712** from the channel segment **702** at the junction **706**.

A discrete droplet generated may comprise one or more particles of the plurality of particles **716**. As described elsewhere herein, a particle may be any particle, such as a bead, cell bead, gel bead, biological particle, macromolecular constituents of biological particle, or other particles. Alternatively, a discrete droplet generated may not include any particles.

In some instances, the aqueous fluid **712** can have a substantially uniform concentration or frequency of particles **716**. As described elsewhere herein (e.g., with reference to FIG. **4**), the particles **716** (e.g., beads) can be introduced into the channel segment **702** from a separate channel (not shown in FIG. **7**). The frequency of particles **716** in the channel segment **702** may be controlled by controlling the frequency in which the particles **716** are introduced into the channel segment **702** and/or the relative flow rates of the fluids in the channel segment **702** and the separate channel. In some instances, the particles **716** can be introduced into the channel segment **702** from a plurality of different channels, and the frequency controlled accordingly. In some instances, different particles may be introduced via separate channels. For example, a first separate channel can introduce beads and a second separate channel can introduce biological particles into the channel segment **702**. The first separate channel introducing the beads may be upstream or downstream of the second separate channel introducing the biological particles.

In some instances, the second fluid **714** may not be subjected to and/or directed to any flow in or out of the reservoir **704**. For example, the second fluid **714** may be substantially stationary in the reservoir **704**. In some instances, the second fluid **714** may be subjected to flow within the reservoir **704**, but not in or out of the reservoir **704**, such as via application of pressure to the reservoir **704** and/or as affected by the incoming flow of the aqueous fluid **712** at the junction **706**. Alternatively, the second fluid **714** may be subjected and/or directed to flow in or out of the reservoir **704**. For example, the reservoir **704** can be a channel directing the second fluid **714** from upstream to downstream, transporting the generated droplets.

The channel structure **700** at or near the junction **706** may have certain geometric features that at least partly determine the sizes and/or shapes of the droplets formed by the channel structure **700**. The channel segment **702** can have a first cross-section height, h_1 , and the reservoir **704** can have a second cross-section height, h_2 . The first cross-section height, h_1 , and the second cross-section height, h_2 , may be different, such that at the junction **706**, there is a height difference of Δh . The second cross-section height, h_2 , may be greater than the first cross-section height, h_1 . In some instances, the reservoir may thereafter gradually increase in cross-section height, for example, the more distant it is from the junction **706**. In some instances, the cross-section height of the reservoir may increase in accordance with expansion angle, β , at or near the junction **706**. The height difference, Δh , and/or expansion angle, β , can allow the tongue (portion of the aqueous fluid **712** leaving channel segment **702** at junction **706** and entering the reservoir **704** before droplet formation) to increase in depth and facilitate decrease in curvature of the intermediately formed droplet. For example, droplet size may decrease with increasing height difference and/or increasing expansion angle.

The height difference, Δh , can be at least about 1 μm . Alternatively, the height difference can be at least about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 200, 300, 400, 500 μm or more. Alternatively, the height difference can be

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at most about 500, 400, 300, 200, 100, 90, 80, 70, 60, 50, 45, 40, 35, 30, 25, 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1 μm or less. In some instances, the expansion angle, β , may be between a range of from about 0.5° to about 4°, from about 0.1° to about 10°, or from about 0° to about 90°. For example, the expansion angle can be at least about 0.01°, 0.1°, 0.2°, 0.3°, 0.4°, 0.5°, 0.6°, 0.7°, 0.8°, 0.9°, 1°, 2°, 3°, 4°, 5°, 6°, 7°, 8°, 9°, 10°, 15°, 20°, 25°, 30°, 35°, 40°, 45°, 50°, 55°, 60°, 65°, 70°, 75°, 80°, 85°, or higher. In some instances, the expansion angle can be at most about 89°, 88°, 87°, 86°, 85°, 84°, 83°, 82°, 81°, 80°, 75°, 70°, 65°, 60°, 55°, 50°, 45°, 40°, 35°, 30°, 25°, 20°, 15°, 10°, 9°, 8°, 7°, 6°, 5°, 4°, 3°, 2°, 1°, 0.1°, 0.01°, or less.

In some instances, the flow rate of the aqueous fluid **712** entering the junction **706** can be between about 0.04 microliters (μL)/minute (min) and about 40 $\mu\text{L}/\text{min}$. In some instances, the flow rate of the aqueous fluid **712** entering the junction **706** can be between about 0.01 microliters (μL)/minute (min) and about 100 $\mu\text{L}/\text{min}$. Alternatively, the flow rate of the aqueous fluid **712** entering the junction **706** can be less than about 0.01 $\mu\text{L}/\text{min}$. Alternatively, the flow rate of the aqueous fluid **712** entering the junction **706** can be greater than about 40 $\mu\text{L}/\text{min}$, such as 45 $\mu\text{L}/\text{min}$, 50 $\mu\text{L}/\text{min}$, 55 $\mu\text{L}/\text{min}$, 60 $\mu\text{L}/\text{min}$, 65 $\mu\text{L}/\text{min}$, 70 $\mu\text{L}/\text{min}$, 75 $\mu\text{L}/\text{min}$, 80 $\mu\text{L}/\text{min}$, 85 $\mu\text{L}/\text{min}$, 90 $\mu\text{L}/\text{min}$, 95 $\mu\text{L}/\text{min}$, 100 $\mu\text{L}/\text{min}$, 110 $\mu\text{L}/\text{min}$, 120 $\mu\text{L}/\text{min}$, 130 $\mu\text{L}/\text{min}$, 140 $\mu\text{L}/\text{min}$, 150 $\mu\text{L}/\text{min}$, or greater. At lower flow rates, such as flow rates of about less than or equal to 10 microliters/minute, the droplet radius may not be dependent on the flow rate of the aqueous fluid **712** entering the junction **706**. The second fluid **714** may be stationary, or substantially stationary, in the reservoir **704**. Alternatively, the second fluid **714** may be flowing, such as at the above flow rates described for the aqueous fluid **712**.

In some instances, at least about 50% of the droplets generated can have uniform size. In some instances, at least about 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or greater of the droplets generated can have uniform size. Alternatively, less than about 50% of the droplets generated can have uniform size.

While FIGS. 7A and 7B illustrate the height difference, z_{lh} , being abrupt at the junction **706** (e.g., a step increase), the height difference may increase gradually (e.g., from about 0 μm to a maximum height difference). Alternatively, the height difference may decrease gradually (e.g., taper) from a maximum height difference. A gradual increase or decrease in height difference, as used herein, may refer to a continuous incremental increase or decrease in height difference, wherein an angle between any one differential segment of a height profile and an immediately adjacent differential segment of the height profile is greater than 90°. For example, at the junction **706**, a bottom wall of the channel and a bottom wall of the reservoir can meet at an angle greater than 90°. Alternatively or in addition, a top wall (e.g., ceiling) of the channel and a top wall (e.g., ceiling) of the reservoir can meet at an angle greater than 90°. A gradual increase or decrease may be linear or non-linear (e.g., exponential, sinusoidal, etc.). Alternatively or in addition, the height difference may variably increase and/or decrease linearly or non-linearly. While FIGS. 7A and 7B illustrate the expanding reservoir cross-section height as linear (e.g., constant expansion angle, β), the cross-section height may expand non-linearly. For example, the reservoir may be defined at least partially by a dome-like (e.g., hemispherical) shape having variable expansion angles. The cross-section height may expand in any shape.

The channel networks, e.g., as described above or elsewhere herein, can be fluidly coupled to appropriate fluidic

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components. For example, the inlet channel segments are fluidly coupled to appropriate sources of the materials they are to deliver to a channel junction. These sources may include any of a variety of different fluidic components, from simple reservoirs defined in or connected to a body structure of a microfluidic device, to fluid conduits that deliver fluids from off-device sources, manifolds, fluid flow units (e.g., actuators, pumps, compressors) or the like. Likewise, the outlet channel segment (e.g., channel segment **208**, reservoir **604**, etc.) may be fluidly coupled to a receiving vessel or conduit for the partitioned cells for subsequent processing. Again, this may be a reservoir defined in the body of a microfluidic device, or it may be a fluidic conduit for delivering the partitioned cells to a subsequent process operation, instrument or component.

The methods and systems described herein may be used to greatly increase the efficiency of single cell applications and/or other applications receiving droplet-based input. For example, following the sorting of occupied cells and/or appropriately-sized cells, subsequent operations that can be performed can include generation of amplification products, purification (e.g., via solid phase reversible immobilization (SPRI)), further processing (e.g., shearing, ligation of functional sequences, and subsequent amplification (e.g., via PCR)). These operations may occur in bulk (e.g., outside the partition). In the case where a partition is a droplet in an emulsion, the emulsion can be broken and the contents of the droplet pooled for additional operations. Additional reagents that may be co-partitioned along with the barcode bearing bead may include oligonucleotides to block ribosomal RNA (rRNA) and nucleases to digest genomic DNA from cells. Alternatively, rRNA removal agents may be applied during additional processing operations. The configuration of the constructs generated by such a method can help minimize (or avoid) sequencing of the poly-T sequence during sequencing and/or sequence the 5' end of a polynucleotide sequence. The amplification products, for example, first amplification products and/or second amplification products, may be subject to sequencing for sequence analysis. In some cases, amplification may be performed using the Partial Hairpin Amplification for Sequencing (PHASE) method.

A variety of applications require the evaluation of the presence and quantification of different biological particle or organism types within a population of biological particles, including, for example, microbiome analysis and characterization, environmental testing, food safety testing, epidemiological analysis, e.g., in tracing contamination or the like.

50 Computer Systems

The present disclosure provides computer systems that are programmed to implement methods of the disclosure. FIG. **11** shows a computer system **1101** that is programmed or otherwise configured to (i) control a microfluidics system (e.g., fluid flow), such as for controlled partitioning, sorting (e.g., of an analyte, of a partition, of a cell, etc.), polymerization, application of one or more stimuli, and the like, (ii) generate and maintain a library or a plurality of libraries of barcodes, (iii) perform sequencing, and/or (iv) analyze and/or output results of such sequencing. The computer system **1101** can regulate various aspects of the present disclosure, such as, for example, regulating fluid flow rate in one or more channels in a microfluidic structure, regulating one or more reactions or interactions of an analyte, biological particle, and/or a sample, regulating the sorting of one or more analytes, and/or regulating the generation of one or more libraries. The computer system **1101** can be an elec-

tronic device of a user or a computer system that is remotely located with respect to the electronic device. The electronic device can be a mobile electronic device.

The computer system **1101** includes a central processing unit (CPU, also “processor” and “computer processor” herein) **1105**, which can be a single core or multi core processor, or a plurality of processors for parallel processing. The computer system **1101** also includes memory or memory location **1110** (e.g., random-access memory, read-only memory, flash memory), electronic storage unit **1115** (e.g., hard disk), communication interface **1120** (e.g., network adapter) for communicating with one or more other systems, and peripheral devices **1125**, such as cache, other memory, data storage and/or electronic display adapters. The memory **1110**, storage unit **1115**, interface **1120** and peripheral devices **1125** are in communication with the CPU **1105** through a communication bus (solid lines), such as a motherboard. The storage unit **1115** can be a data storage unit (or data repository) for storing data. The computer system **1101** can be operatively coupled to a computer network (“network”) **1130** with the aid of the communication interface **1120**. The network **1130** can be the Internet, an internet and/or extranet, or an intranet and/or extranet that is in communication with the Internet. The network **1130** in some cases is a telecommunication and/or data network. The network **1130** can include one or more computer servers, which can enable distributed computing, such as cloud computing. The network **1130**, in some cases with the aid of the computer system **1101**, can implement a peer-to-peer network, which may enable devices coupled to the computer system **1101** to behave as a client or a server.

The CPU **1105** can execute a sequence of machine-readable instructions, which can be embodied in a program or software. The instructions may be stored in a memory location, such as the memory **1110**. The instructions can be directed to the CPU **1105**, which can subsequently program or otherwise configure the CPU **1105** to implement methods of the present disclosure. Examples of operations performed by the CPU **1105** can include fetch, decode, execute, and writeback.

The CPU **1105** can be part of a circuit, such as an integrated circuit. One or more other components of the system **1101** can be included in the circuit. In some cases, the circuit is an application specific integrated circuit (ASIC).

The storage unit **1115** can store files, such as drivers, libraries and saved programs. The storage unit **1115** can store user data, e.g., user preferences and user programs. The computer system **1101** in some cases can include one or more additional data storage units that are external to the computer system **1101**, such as located on a remote server that is in communication with the computer system **1101** through an intranet or the Internet.

The computer system **1101** can communicate with one or more remote computer systems through the network **1130**. For instance, the computer system **1101** can communicate with a remote computer system of a user (e.g., operator). Examples of remote computer systems include personal computers (e.g., portable PC), slate or tablet PC’s (e.g., Apple® iPad, Samsung® Galaxy Tab), telephones, Smart phones (e.g., Apple® iPhone, Android-enabled device, Blackberry®), or personal digital assistants. The user can access the computer system **1101** via the network **1130**.

Methods as described herein can be implemented by way of machine (e.g., computer processor) executable code stored on an electronic storage location of the computer system **1101**, such as, for example, on the memory **1110** or

electronic storage unit **1115**. The machine executable or machine readable code can be provided in the form of software. During use, the code can be executed by the processor **1105**. In some cases, the code can be retrieved from the storage unit **1115** and stored on the memory **1110** for ready access by the processor **1105**. In some situations, the electronic storage unit **1115** can be precluded, and machine-executable instructions are stored on memory **1110**.

The code can be pre-compiled and configured for use with a machine having a processor adapted to execute the code, or can be compiled during runtime. The code can be supplied in a programming language that can be selected to enable the code to execute in a pre-compiled or as-compiled fashion.

Aspects of the systems and methods provided herein, such as the computer system **1101**, can be embodied in programming. Various aspects of the technology may be thought of as “products” or “articles of manufacture” typically in the form of machine (or processor) executable code and/or associated data that is carried on or embodied in a type of machine readable medium. Machine-executable code can be stored on an electronic storage unit, such as memory (e.g., read-only memory, random-access memory, flash memory) or a hard disk. “Storage” type media can include any or all of the tangible memory of the computers, processors or the like, or associated modules thereof, such as various semiconductor memories, tape drives, disk drives and the like, which may provide non-transitory storage at any time for the software programming. All or portions of the software may at times be communicated through the Internet or various other telecommunication networks. Such communications, for example, may enable loading of the software from one computer or processor into another, for example, from a management server or host computer into the computer platform of an application server. Thus, another type of media that may bear the software elements includes optical, electrical and electromagnetic waves, such as used across physical interfaces between local devices, through wired and optical landline networks and over various air-links. The physical elements that carry such waves, such as wired or wireless links, optical links or the like, also may be considered as media bearing the software. As used herein, unless restricted to non-transitory, tangible “storage” media, terms such as computer or machine “readable medium” refer to any medium that participates in providing instructions to a processor for execution.

Hence, a machine readable medium, such as computer-executable code, may take many forms, including but not limited to, a tangible storage medium, a carrier wave medium or physical transmission medium. Non-volatile storage media include, for example, optical or magnetic disks, such as any of the storage devices in any computer(s) or the like, such as may be used to implement the databases, etc. shown in the drawings. Volatile storage media include dynamic memory, such as main memory of such a computer platform. Tangible transmission media include coaxial cables; copper wire and fiber optics, including the wires that comprise a bus within a computer system. Carrier-wave transmission media may take the form of electric or electromagnetic signals, or acoustic or light waves such as those generated during radio frequency (RF) and infrared (IR) data communications. Common forms of computer-readable media therefore include for example: a floppy disk, a flexible disk, hard disk, magnetic tape, any other magnetic medium, a CD-ROM, DVD or DVD-ROM, any other optical medium, punch cards paper tape, any other physical storage medium with patterns of holes, a RAM, a ROM, a PROM and EPROM, a FLASH-EPROM, any other memory chip or

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cartridge, a carrier wave transporting data or instructions, cables or links transporting such a carrier wave, or any other medium from which a computer may read programming code and/or data. Many of these forms of computer readable media may be involved in carrying one or more sequences of one or more instructions to a processor for execution.

The computer system 1101 can include or be in communication with an electronic display 1135 that comprises a user interface (UI) 1140 for providing, for example, results of sequencing analysis. Examples of UIs include, without limitation, a graphical user interface (GUI) and web-based user interface.

Methods and systems of the present disclosure can be implemented by way of one or more algorithms. An algorithm can be implemented by way of software upon execution by the central processing unit 1105. The algorithm can, for example, perform sequencing and process libraries, such as by accessing one or more databases.

Devices, systems, compositions and methods of the present disclosure may be used for various applications, such as, for example, processing a single analyte (e.g., RNA, DNA, metabolite, protein, etc.) or multiple analytes (e.g., DNA and RNA, DNA and protein, RNA and protein, or RNA, DNA and protein, etc.) from a single cell. For example, a biological particle (e.g., a cell or cell bead) is partitioned in a partition (e.g., droplet), and multiple analytes from the biological particle are processed for subsequent processing. The multiple analytes may be from the single cell. This may enable, for example, simultaneous proteomic, transcriptomic and genomic analysis of the cell.

While preferred embodiments of the present invention have been shown and described herein, it will be obvious to those skilled in the art that such embodiments are provided by way of example only. It is not intended that the invention be limited by the specific examples provided within the specification. While the invention has been described with reference to the aforementioned specification, the descriptions and illustrations of the embodiments herein are not meant to be construed in a limiting sense. Numerous variations, changes, and substitutions will now occur to those skilled in the art without departing from the invention. Furthermore, it shall be understood that all aspects of the invention are not limited to the specific depictions, configurations or relative proportions set forth herein which depend upon a variety of conditions and variables. It should be understood that various alternatives to the embodiments of the invention described herein may be employed in practicing the invention. It is therefore contemplated that the invention shall also cover any such alternatives, modifications, variations or equivalents. It is intended that the following claims define the scope of the invention and that methods and structures within the scope of these claims and their equivalents be covered thereby.

What is claimed is:

1. A method for analyzing a cell surface-bound analyte and an intracellular analyte from a cell, comprising:

- (a) binding a first antibody to the cell surface-bound analyte of the cell, wherein the first antibody comprises a first oligonucleotide tagged molecule comprising a first nucleic acid sequence that identifies the first antibody or the cell surface-bound analyte;
- (b) after (a), partitioning the cell into a first partition to generate a cell bead from the cell;
- (c) after (b), lysing the cell within the cell bead;
- (d) after (c), contacting the cell bead with a second antibody that binds to the intracellular analyte of the cell, wherein the second antibody comprises a second

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oligonucleotide tagged molecule comprising a second nucleic acid sequence that identifies the second antibody or the intracellular analyte;

- (e) co-partitioning into a second partition: (1) the cell bead comprising the first antibody coupled with the cell surface-bound analyte, the second antibody coupled with the intracellular analyte, and a nucleic acid molecule from the cell, and (2) a plurality of nucleic acid barcode molecules coupled to a support, wherein each nucleic acid barcode molecule of the plurality of nucleic acid barcode molecules comprises a common barcode sequence for barcoding the first oligonucleotide tagged molecule, the second oligonucleotide tagged molecule, and the nucleic acid molecule from the cell; and
- (f) generating (i) a first barcoded nucleic acid molecule comprising the first nucleic acid sequence and the common barcode sequence, (ii) a second barcoded nucleic acid molecule comprising the second nucleic acid sequence and the common barcode sequence, and (iii) a third barcoded nucleic acid molecule comprising a sequence of the nucleic acid molecule of the cell and the common barcode sequence.

2. The method of claim 1, wherein the method further comprises sequencing: (i) the first barcoded nucleic acid molecule, or a derivative thereof, (ii) the second barcoded nucleic acid molecule, or a derivative thereof, and (iii) the third barcoded nucleic acid molecule, or a derivative thereof, to identify the first nucleic acid sequence, the second nucleic acid sequence, the common barcode sequence, and the sequence of the nucleic acid molecule, or derivatives thereof, wherein the first nucleic acid sequence identifies the first antibody or cell surface-bound analyte, the second nucleic acid sequence identifies the second antibody or intracellular analyte, and the common barcode sequence identifies the cell.

3. The method of claim 1, wherein the support is a bead.

4. The method of claim 1, wherein the cell surface-bound analyte and the intracellular analyte are different types of analytes.

5. The method of claim 1, wherein the cell surface-bound analyte and the intracellular analyte are a same type of analyte.

6. The method of claim 1, wherein the cell surface-bound analyte and the intracellular analyte is a protein.

7. The method of claim 1, wherein the intracellular analyte is a protein.

8. The method of claim 1, wherein the cell surface-bound analyte is a protein.

9. The method of claim 1, wherein the nucleic acid molecule comprises deoxyribonucleic acid (DNA).

10. The method of claim 9, wherein the DNA is genomic DNA.

11. The method of claim 9, further comprising amplifying the DNA.

12. The method of claim 1, wherein the nucleic acid molecule comprises ribonucleic acid (RNA).

13. The method of claim 12, wherein the RNA is messenger RNA (mRNA).

14. The method of claim 13, further comprising performing reverse transcription on the mRNA to generate complementary DNA (cDNA).

15. The method of claim 1, wherein the cell is encased within a cell bead.

16. The method of claim 1, further comprising releasing the plurality of nucleic acid barcode molecules from the support.

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17. The method of claim **1**, wherein (d) comprises subjecting the cell bead to conditions sufficient to permit the second antibody to flow into the cell bead.

18. The method of claim **1**, further comprising, after (a), washing the cell under conditions sufficient to remove 5 unbound antibodies of the first antibody.

19. The method of claim **1**, further comprising, after (d), washing the cell bead under conditions sufficient to permit unbound antibodies of the second antibody to flow out of the cell bead. 10

20. The method of claim **19**, wherein the conditions sufficient comprise subjecting the cell bead to osmotic swelling.

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